

CHAPTER

6

USE OF VIBRATION SIGNATURE ANALYSIS TO DIAGNOSE MACHINE PROBLEMS

6.0 USE OF VIBRATION SIGNATURE ANALYSIS

Probably the greatest shortcoming in today's predictive maintenance programs is the ability to diagnose the mechanical and electrical problems within the machine that are evidenced in the vibration signatures if the vibration exceeds preset overall and spectral alarm levels. For example, an ever increasing number of plants have PMP data collectors and software and successfully build large databases and capture vibration measurements at great numbers of points. However, surveys prove that less than 15% of such plants know how to properly set overall and spectral alarm levels. Then, even a smaller minority know how to diagnose the array of potential problems from the vibration spectra and related variables (i.e. spike energy) which are faithfully printed out by the PMP software for those points that exceed alarm. Therefore, the overriding purpose of this section is to begin to acquaint the reader on how to diagnose such problems from this data. Much valuable information is contained within these vibration spectra, but is only of use if the analyst can unlock its "secrets".

Therefore, Table 6.0 has been developed to put many of these "secrets" right at the fingertips of the analyst. Several hundred hours of research have gone into the development of this four page diagnostic chart. Please note that this chart not only provides text elaborating on vibration symptoms for various machine problems, but it also includes illustrations of "typical vibration spectra" for each problem covered. In addition, drawings are included to illustrate how phase reacts when such problems are predominant. Table 6.0 represents the best understanding to date of the author on how these problems are best diagnosed, based on approximately 16 years field experience in vibration signature analysis and research on a wide range of articles which have been written on the subject.

There are several key items included in Table 6.0. First, the plots under "TYPICAL SPECTRUM" column reveal invaluable information as to the source of the problem. When looking at such spectra, the analyst should ask questions similar to the following:

1. Which frequencies are present in spectrum and how do they relate to machine operating speed (that is, are the peaks present equal to 1X, 2X, 3X, 5.78X RPM or what)?
2. What are the amplitudes of each peak?
3. How do the frequency peaks relate to one another? (i.e., "2X RPM is much higher than 1X RPM"; "there is a large peak at 7.43X RPM"; "there are large number of operating speed harmonics present"; "there are high amplitude sidebands around gear mesh frequency"; "there are 7200 CPM sidebands around a large peak at 46X RPM"; etc.).
4. Finally, if there are significant amplitude peaks, what exactly is their source ("is 7.43X RPM a bearing defect frequency"; "is the 46X RPM peak equal to the number of rotor bars RPM"?).

As its column name implies, "TYPICAL SPECTRUM" is meant to be a representative signature for each problem listed in Table 6.0. These spectra are not intended to be all inclusive. For example, referring to "REMARKS" for the Angular Misalignment problem, note that while the typical spectrum for this problem shows high amplitude 1X RPM and 2X RPM peaks in the axial direction, the discussion shows that it is not unusual for either 1X, 2X or 3X RPM to dominate the whole spectrum when angular misalignment is the problem. In addition, it is not unusual for a machine to have two or more problems present at any one time. For example, if a machine simultaneously had both mechanical looseness and rotor unbalance, they each would contribute frequencies to its spectra which might show high 1X RPM in addition to multiple running speed harmonics.

The next column in Table 6.0 is entitled "PHASE RELATIONSHIP". Information on phase is provided for several of the problem sources listed. Amplitude reveals how much something is vibrating. Frequency relates how many cycles occur per unit of time. Phase completes the picture by showing just how the machine is vibrating. Of great importance, phase is a powerful tool in helping to differentiate which of several problem sources are dominant. For example, there are a large number of problems that generate vibration at 1X and 2X RPM. Using phase, one learns how the machine is vibrating, and in the process, helps zero in on just which problem is present. For example, Table 6.0 shows how phase reacts during the following scenarios:

1. **Force (or "static") unbalance** is evidenced by nearly identical phase in the radial direction on each bearing of a machine rotor.
2. **Couple unbalance** shows approximately a 180° out-of-phase relationship when comparing the outboard and inboard horizontal, or the outboard and inboard vertical direction phase on the same machine.
3. **Dynamic unbalance** is indicated when the phase difference is well removed from either 0° or 180°, but importantly is nearly the same in the horizontal and vertical directions. That is, the horizontal phase difference could be almost anything between the outboard and inboard bearings; but, the key point is that the vertical phase difference should then be almost identical to the horizontal phase difference ($\pm 30^\circ$). For, example, if the horizontal phase difference between the outboard and inboard bearings is 60°, and the dominant problem is dynamic unbalance, the vertical phase difference between these two bearings should be about 60° ($\pm 30^\circ$). If the horizontal phase difference varies greatly from the vertical phase difference, this strongly suggests the dominant problem is not unbalance.
4. **Angular misalignment** is indicated by approximately a 180° phase difference across the coupling, with measurements in the axial direction.
5. **Parallel misalignment** causes radial direction phase across the coupling to be approximately 180° out of phase with respect to one another.
6. **Bent shaft** causes axial phase on the same shaft of a machine to approach a 180° difference when comparing measurements on the outboard and inboard bearings of the same machine rotor.
7. **Resonance** is shown by a 90° phase change at the point when the forcing frequency coincides with a natural frequency, and approaches a full 180° phase change when the machine passes through the natural frequency (depending on the amount of damping present).

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8. **Rotor rub** causes significant, instantaneous changes in phase.
 9. **Mechanical looseness/weakness due to base/frame problems or loose hold-down bolts** is indicated by nearly a 180° phase change when one moves his transducer from the machine foot down to its baseplate and then down to its support base.
 10. **Mechanical looseness due to a cracked frame, loose bearing or loose rotor** causes phase to be unsteady with probable widely differing phase measurements from one measurement to the next. The phase measurement may noticeably differ every time you start up the machine, particularly if the rotor itself is loose and rotates on the shaft a few degrees with each startup.

Often, even though phase measurement capability is now offered by most data collectors, users do not use this powerful tool. If not used, this will severely limit the diagnostic capabilities of any program. However, currently it would be impractical to make phase measurements on all machinery during regular PMP surveys. Its greatest use comes into play when performing diagnostics on machines which have developed high vibration at 1X, 2X or 3X RPM, requiring investigation to detect the predominate cause(s) prior to taking corrective actions.

Note that "PHASE RELATIONSHIP" is illustrated in each of the first 8 problems of Table 6.0 since it is primarily with these problems that phase can be used to differentiate which problem(s) dominate. Phase is then discussed in many of the remarks for the remaining problems in Table 6.0, although it is not illustrated.

Finally, a remarks column is included in Table 6.0 to provide further explanatory information on machine problem symptoms and diagnostics. For example, there is a warning under the remarks column for the "bent shaft" problem source to be sure and account for transducer orientation when taking axial phase measurements.

It is hoped that this illustrated chart will help users in diagnosing a wide variety of machine problems. Further information is now being researched and field tested which may soon be added to the diagnostic chart as we constantly learn more and more about how machines react when subjected to a whole series of problems and how we can "read" these reactive responses via diagnostic techniques.

Following on the next pages will be separate discussions on each of the problems outlined in Table 6.0. Later, real-world case histories will be presented giving real-world examples of each of these problems.

TABLE 6.0 ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	PHASE RELATIONSHIP	REMARKS
MASS UNBALANCE A. FORCE UNBALANCE			Force Unbalance will be in-phase and steady. Amplitude due to unbalance will increase by the square of speed below first rotor critical (a 3X speed increase = 9X higher vibration). 1X RPM always present and normally dominates spectrum. Can be corrected by placement of only one balance correction weight in one plane at Rotor center of gravity (CG). Approx. 0° phase difference should exist between OB & IB horizontals, as well as between OB & IB verticals. Also, approx. 90° phase difference between horizontal & vertical readings usually occurs on each bearing of unbalanced rotor (±30°).
B. COUPLE UNBALANCE			Couple Unbalance results in 180° out-of-phase motion on same shaft. 1X RPM always present and normally dominates spectrum. Amplitude varies with square of increasing speed below first rotor critical speed. May cause high axial vibration as well as radial. Correction requires placement of balance weights in at least 2 planes. Note that approx. 180° phase difference should exist between OB & IB horizontals, as well as between OB & IB verticals. Also, approx. a 90° difference between the horizontal & vertical phase readings on each bearing usually occurs (±30°).
C. DYNAMIC UNBALANCE			Dynamic Unbalance is the dominant type of unbalance found and is a combination of both force and couple unbalance. 1X RPM dominates the spectrum, and truly requires 2 plane correction. Here, the radial phase difference between outboard and inboard bearings can range anywhere from 0° to 180°. However, the horizontal phase difference should closely match the vertical phase difference, when comparing outboard and inboard bearing measurements (±30°). Secondly, if unbalance predominates, roughly a 90° phase difference usually results between the horizontal and vertical readings on each bearing (±30°).
D. OVERHUNG ROTOR UNBALANCE			Overhung Rotor Unbalance causes high 1X RPM in both Axial and Radial directions. Axial readings tend to be in-phase whereas radial phase readings might be unsteady. However, the horizontal phase differences will usually match the vertical phase differences on the unbalanced rotor (±30°). Overhung rotors have both force and couple unbalance, each of which will likely require correction. Thus, correction weights will most always have to be placed in 2 planes to counteract both force and couple unbalance.
ECCENTRIC ROTOR			Eccentricity occurs when center of rotation is offset from geometric centerline of a pulley, gear, bearing, motor armature, etc. Largest vibration occurs at 1X RPM of eccentric component in a direction thru centerlines of the two rotors. Comparative horizontal and vertical phase readings usually differ either by 0° or by 180° (each of which indicate straight-line motion). Attempts to balance eccentric rotors often result in reducing vibration in one radial direction, but increasing it in the other radial direction (depending on amount of eccentricity).
BENT SHAFT			Bent shaft problems cause high axial vibration with axial phase differences tending towards 180° on the same machine component. Dominant vibration normally occurs at 1X if bent near shaft center, but at 2X if bent near the coupling. (Be careful to account for transducer orientation for each axial measurement if you reverse probe direction.) Use dial indicators to confirm bent shaft.
MISALIGNMENT A. ANGULAR MISALIGNMENT			Angular Misalignment is characterized by high axial vibration, 180° out-of-phase across the coupling. Typically will have high axial vibration with both 1X and 2X RPM. However, not unusual for either 1X, 2X or 3X to dominate. These symptoms may also indicate coupling problems as well. Severe angular misalignment may excite many 1X RPM harmonics. Unlike Mechanical Looseness Type 3, these multiple harmonics do not typically have a raised noise floor on the spectra.
B. PARALLEL MISALIGNMENT			Offset Misalignment has similar vibration symptoms to Angular, but shows high radial vibration which approaches 180° out-of-phase across coupling. 2X often larger than 1X, but its height relative to 1X is often dictated by coupling type and construction. When either Angular or Radial Misalignment becomes severe, they can generate either high amplitude peaks at much higher harmonics (4X-8X), or even a whole series of high frequency harmonics similar in appearance to mechanical looseness. Coupling type and material will often greatly influence the entire spectrum when misalignment is severe. Does not typically have raised noise floor.
C. MISALIGNED BEARING COCKED ON SHAFT			Cocked Bearing will generate considerable axial vibration. Will cause Twisting Motion with approximately 180° phase shift top to bottom and/or side to side as measured in axial direction on same bearing housing. Attempts to align coupling or balance the rotor will not alleviate problem. Bearing usually must be removed and correctly installed.
RESONANCE			Resonance occurs when a Forcing Frequency coincides with a System Natural Frequency, and can cause dramatic amplitude amplification, which might result in premature, or even catastrophic failure. This may be a natural frequency of the rotor, but can often originate from support frame, foundation, gearbox or even drive belts. If a rotor is at or near resonance, it can be almost impossible to balance due to the great phase shift it experiences (90° at resonance; nearly 180° when passes thru). Often requires changing natural frequency to a higher or lower frequency. Natural Frequencies do not generally change with a change in speed which helps facilitate their identification (unless on a large plain bearing machine or on a rotor which has significant overhang).

TABLE 6.0
ILLUSTRATED VIBRATION DIAGNOSTIC CHART

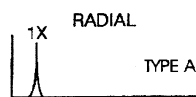
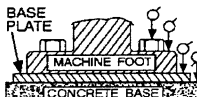
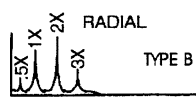
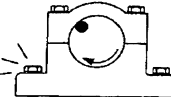
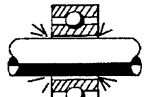
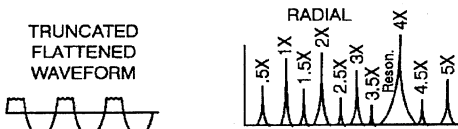
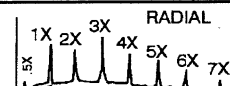
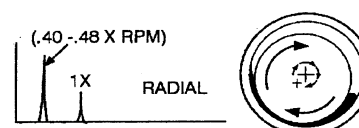
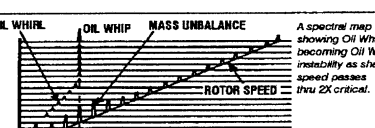
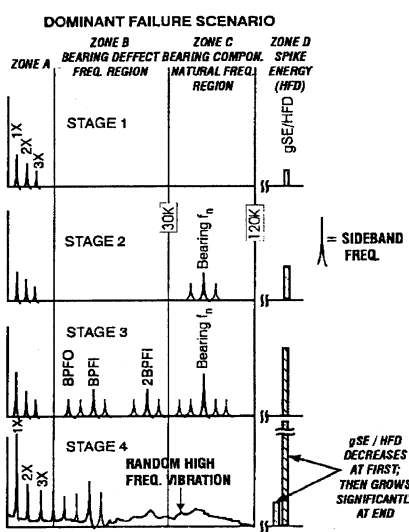
PROBLEM SOURCE	TYPICAL SPECTRUM	PHASE RELATIONSHIP	REMARKS
MECHANICAL LOOSENESS	 1X RADIAL TYPE A		Mechanical Looseness is indicated by either Type A, B or C vibration spectra. Type A is caused by Structural looseness/weakness of machine feet, baseplate or foundation; also by deteriorated grouting, loose hold-down bolts at the base; and distortion of the frame or base (i.e., soft foot). Phase analysis may reveal approx. 90° to 180° phase difference between vertical measurements on bolt, machine foot, baseplate or base itself. Type B is generally caused by loose pillowblock bolts, cracks in frame structure or in bearing pedestal. Type C is normally generated by improper fit between component parts which will cause many harmonics due to nonlinear response of loose parts to dynamic forces from rotor. Causes a truncation of time waveform and a raised noise floor in the spectrum. Type C is often caused by a bearing liner loose in its cap, a bearing loose and turning on its shaft, excessive clearance in either a sleeve or rolling element bearing, a loose impeller on a shaft, etc. Type C Phase is often unstable and may vary widely from one measurement to next, particularly if rotor shifts position on shaft from one startup to next. Mechanical Looseness is often highly directional and may cause very different readings when comparing levels at 30° increments in radial direction all the way around one bearing housing. Also, note that looseness will often cause subharmonic multiples at exactly 1/2 or 1/3X RPM (.5X, 1.5X, 2.5X, etc.).
	 1X 2X 3X 4X 5X 6X 7X 8X 9X 10X RADIAL TYPE B		
	 NOTE RAISED NOISE FLOOR INDICATING LOOSENESS 1X 2X 3X 4X 5X 6X 7X 8X 9X 10X RADIAL TYPE C		
ROTOR RUB	 TRUNCATED FLATTENED WAVEFORM RADIAL .5X 1X 1.5X 2X 2.5X 3X 3.5X 4X 4.5X 5X Reson.		Rotor Rub produces similar spectra to Mechanical Looseness when rotating parts contact stationary components. Rub may be either partial or throughout the entire shaft revolution. Usually generates a series of frequencies, often exciting one or more resonances. Often excites integer fraction subharmonics of running speed (1/2, 1/3, 1/4, 1/5, ..., 1/n), depending on location of rotor natural frequencies. Rotor rub can excite many high frequencies (similar to wide-band noise when chalk is dragged along a blackboard). It can be very serious and of short duration if caused by shaft contacting bearing babbit. A full annular rub throughout an entire shaft revolution can induce "reverse precession" with the rotor whirling at critical speed in a direction opposite shaft rotation (inherently unstable which can lead to catastrophic failure).
JOURNAL BEARINGS			Latter stages of journal bearing wear are normally evidenced by presence of whole series of running speed harmonics (up to 10 or 20). Wiped journal bearings often will allow high vertical amplitudes compared to horizontal, but, may show only one pronounced peak at 1X RPM. Journal bearings with excessive clearance may allow a minor unbalance and/or misalignment to cause high vibration which would be much lower if bearing clearances were set to spec.
A. WEAR/CLEARANCE PROBLEMS	 1X 2X 3X 4X 5X 6X 7X RADIAL NOTE RAISED NOISE FLOOR INDICATING CLEARANCE/LOOSENESS.		
B. OIL WHIRL INSTABILITY	 (.40 - .48 X RPM) 1X RADIAL		
C. OIL WHIP INSTABILITY	 OIL WHIRL OIL WHIP MASS UNBALANCE A spectral map showing Oil Whirl becoming Oil Whip instability as shaft speed passes thru 2X critical. CRITICAL SPEED FREQUENCY		Oil Whirl instability occurs at .40 - .48X RPM and is often quite severe. Considered excessive when amplitude exceeds 40% of bearing clearances. Oil Whirl is an oil film excited vibration where deviations in normal operating conditions (attitude angle and eccentricity ratio) cause oil wedge to "push" shaft around within bearing. Destabilizing force in direction of rotation results in a whirl (or forwards precession). Oil Whirl is unstable since it increases centrifugal forces which increase whirl forces. Can cause oil to no longer support shaft and can become unstable when whirl frequency coincides with a rotor natural frequency. Changes in oil viscosity, lube pressure and external preloads can affect oil whirl. Oil Whip may occur if machine operated at or above 2X rotor critical frequency. When rotor brought up to twice critical speed, whirl will be very close to rotor critical and may cause excessive vibration that oil film may no longer be capable of supporting. Whirl speed will actually "lock onto" rotor critical and this peak will not pass through it even if machine is brought to higher and higher speeds. Produces a lateral forward precessional subharmonic vibration at rotor critical frequency. Inherently unstable which can lead to catastrophic failure.
ROLLING ELEMENT BEARINGS			
(4 Failure Stages)	DOMINANT FAILURE SCENARIO  ZONE A BEARING DEFECT FREQ. REGION ZONE B BEARING DEFECT BEARING COMPON. NATURAL FREQ. REGION ZONE C BEARING DEFECT BEARING COMPON. NATURAL FREQ. REGION ZONE D SPIKE ENERGY (HFD) STAGE 1 STAGE 2 STAGE 3 STAGE 4 BPFI = Inner Race Frequency BPFO = Outer Race Frequency BSF = Ball Spin Frequency FTF = Fund. Train (Cage) Freq. Where: BPFI = Inner Race Frequency BPFO = Outer Race Frequency BSF = Ball Spin Frequency FTF = Fund. Train (Cage) Freq. N_b = Number of Balls or Rollers D_b = Ball/Roller Diameter (in or mm) P_d = Bearing Pitch Diameter (in or mm) α = Contact Angle (degrees)		4 ROLLING ELEMENT BEARING FAILURE STAGES STAGE 1: Earliest indications of bearing problems appear in ultrasonic frequencies ranging from about 250,000 - 350,000 Hz; later, as wear increases, usually drops to approximately 20,000 - 60,000 Hz (1,200,000 - 3,600,000 CPM). These are frequencies evaluated by Spike Energy (gSE), HFD(g) and Shock Pulse (dB). For example, spike energy may first appear at about .25 gSE in Stage 1 (actual value depending on measurement location and machine speed). Acquiring high frequency enveloped spectra confirms whether or not bearing is in Failure Stage 1. STAGE 2: Slight bearing defects begin to "ring" bearing component natural frequencies (f_n) which predominantly occur in 30K - 120K CPM range. Such natural frequencies may also be resonances of bearing support structures. Sideband frequencies appear above and below natural frequency peak at end of Stage 2. Overall spike energy grows (for example, from .25 to .50 gSE). STAGE 3: Bearing defect frequencies and harmonics appear. When wear progresses, more defect frequency harmonics appear and number of sidebands grow, both around these and bearing component natural frequencies. Overall spike energy continues to increase (for example, from .5 to over 1 gSE). Wear is now usually visible and may extend throughout periphery of bearing, particularly when many well formed sidebands accompany bearing defect frequency harmonics. High frequency demodulated and enveloped spectra help confirm Stage III. Replace bearings now! (independent of bearing defect frequency amplitudes in vibration spectra). STAGE 4: Towards the end, amplitude of 1X RPM is even effected. It grows, and normally causes growth of many running speed harmonics. Discrete bearing defect and component natural frequencies actually begin to "disappear" and are replaced by random, broadband high frequency "noise floor". In addition, amplitudes of both high frequency noise floor and spike energy may in fact decrease; but just prior to failure, spike energy and HFD will usually grow to excessive amplitudes.

TABLE 6.0

ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	REMARKS
HYDRAULIC AND AERODYNAMIC FORCES		
A. BLADE PASS & VANE PASS	BPF = # BLADES X RPM	Blade Pass Frequency (BPF) = No. of Blades (or Vanes) X RPM. This frequency is inherent in pumps, fans and compressors, and normally does not present a problem. However, large amplitude BPF (and harmonics) can be generated in pump if gap between rotating vanes and stationary diffusers is not equal all the way around. Also, BPF (or harmonic) sometimes can coincide with a system natural frequency causing high vibration. High BPF can be generated if impeller wear ring seizes on shaft, or if welds fastening diffuser vanes fail. Also, high BPF can be caused by abrupt bends in pipe (or duct), obstructions which disturb flow, damper settings or if pump or fan rotor is positioned eccentrically within housing.
B. FLOW TURBULENCE	BPF = BLADE OR VANE PASS FREQUENCY	Flow Turbulence often occurs in blowers due to variations in pressure or velocity of the air passing thru the fan or connected ductwork. This flow disruption causes turbulence which will generate random, low frequency vibration, typically in the range of 50 to 2000 CPM. If surging occurs within a compressor, random broadband high frequency vibration can occur. Excessive turbulence can also excite broadband high frequency.
C. CAVITATION	120K CPM	Cavitation normally generates random, higher frequency broadband energy which is sometimes superimposed with blade pass frequency harmonics. Normally indicates insufficient suction pressure (starvation). Cavitation can be quite destructive to pump internals if left uncorrected. It can particularly erode impeller vanes. When present, it often sounds as if "gravel" is passing thru pump. Cavitation is usually caused by insufficient inlet flow. Can occur during one survey, and be absent the next survey (if changes in suction valve settings are made).
GEARS		
A. NORMAL SPECTRUM	$GMF = \#T_G \times RPM_G = \#T_P \times RPM_P$	Normal Spectrum shows Gear & Pinion Speeds, along with Gear Mesh Frequency (GMF) and very small GMF harmonics. GMF harmonics commonly will have running speed sidebands around them. All peaks are of low amplitude, and no natural frequencies of gears are excited. F_{max} recommended at 3.25X GMF (minimum) when # teeth are known. If tooth count is not known, set F_{max} at 200X RPM on each shaft.
B. TOOTH WEAR	51T 26T	Key indicator of Tooth Wear is excitation of Gear Natural Frequency (f_n), along with sidebands around it spaced at the running speed of the bad gear. Gear Mesh Frequency (GMF) may or may not change in amplitude, although high amplitude sidebands and number of sidebands surrounding GMF usually occur when wear is noticeable. Sidebands may be better wear indicator than GMF frequencies themselves. Also, high amplitudes commonly occur at either 2XGMF or at 3XGMF (esp. 3XGMF), even when GMF amplitude is acceptable.
C. TOOTH LOAD		Gear Mesh Frequencies are often very sensitive to load. High GMF amplitudes do not necessarily indicate a problem, particularly if sideband frequencies remain low level, and no gear natural frequencies are excited. Each Analysis should be performed with system at maximum operating load for meaningful spectral comparisons.
D. GEAR ECCENTRICITY AND BACKLASH		Fairly high amplitude sidebands around GMF harmonics often suggest gear eccentricity, backlash, or non-parallel shafts which allow the rotation of one gear to "modulate" either the GMF amplitude or the running speed of the other gear. The gear with the problem is indicated by the spacing of the sideband frequencies. Also, 1X RPM level of eccentric gear will normally be high if eccentricity is the dominant problem. Improper backlash normally excites GMF harmonics and Gear Natural Frequency, both of which will be sidebanded at 1X RPM. GMF amplitudes will often decrease with increasing load if backlash is the problem.
E. GEAR MISALIGNMENT		Gear Misalignment almost always excites second order or higher GMF harmonics which are sidebanded at running speed. Often will show only small amplitude 1X GMF, but much higher levels at 2X or 3X GMF. Important to set F_{max} high enough to capture at least 3 GMF harmonics. Also, sidebands around 2XGMF will often be spaced at 2X RPM. Note that sideband amplitudes often are not equal on left and right side of GMF and GMF harmonics due to the tooth misalignment. Causes uneven wear pattern.
F. CRACKED/BROKEN TOOTH	1X GEAR RPM	A Cracked or Broken Tooth will generate a high amplitude at 1X RPM of this gear only in the time waveform, plus it will excite gear natural frequency (f_n) sidebanded at its running speed. It is best detected in Time Waveform which will show a pronounced spike every time the problem tooth tries to mesh with teeth on the mating gear. Time between impacts (Δ) will correspond to 1/RPM of gear with the problem. Amplitudes of Impact Spikes in Time Waveform often will be 10X to 20X higher than that at 1X RPM in the FFT!
G. GEAR ASSEMBLY PHASE PROBLEMS	$GAPF = \frac{GMF}{N_A}$	Gear Assembly Phase Freq. (GAPF) can result in Fractional Gear Mesh Frequencies (if $N_A > 1$). It literally means (T_G/N_A) gear teeth will contact (T_P/N_A) pinion teeth and will generate N_A wear patterns, where N_A is a given tooth combination equals the product of prime factors common to the number of teeth on the gear and pinion (N_A = Assembly Phase Factor). GAPF (or harmonics) can show up right from the beginning if there were manufacturing problems. Also, its sudden appearance in a periodic survey spectrum can indicate damage if contaminate particles pass through the mesh, resulting in damage to the teeth in mesh at the time of ingestion just as they enter and leave meshing or that gears have been reoriented.
H. HUNTING TOOTH PROBLEMS	$f_{HT} = \frac{(GMF)(N_A)}{(T_{GEAR})(T_{PINION})}$	Hunting Tooth Frequency (f_{HT}) occurs when faults are present on both the gear and pinion which might have occurred during the manufacturing process, due to mishandling, or in the field. It can cause quite high vibration, but since it occurs at low frequencies predominately less than 600 CPM, it is often missed. A gear set with this tooth repeat problem normally emits a "growing" sound from the drive. The maximum effect occurs when the faulty pinion and gear teeth both enter mesh at the same time (on some drives, this may occur only 1 of every 10 to 20 revolutions, depending on the f_{HT} formula). Note that T_{GEAR} and T_{PINION} refer to number of teeth on the gear and pinion, respectively. N_A is the Assembly Phase Factor defined above. Will often modulate both GMF and Gear RPM peaks.

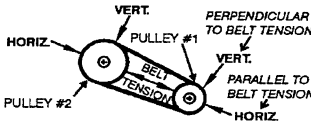
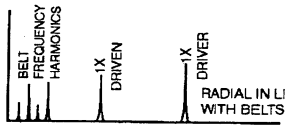
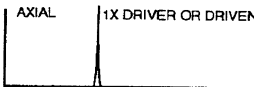
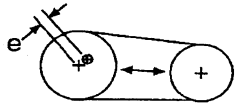
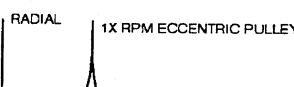
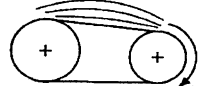
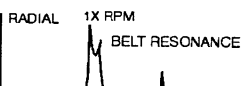
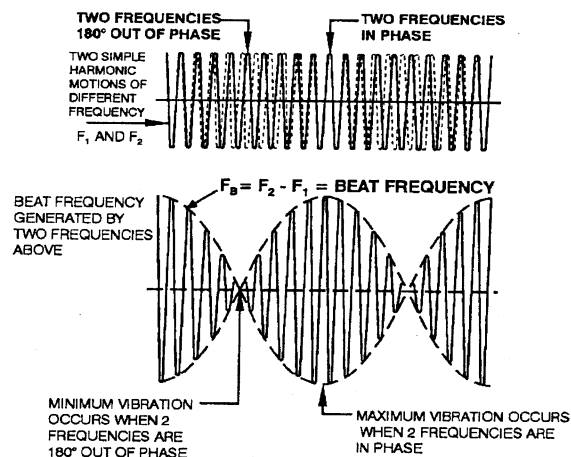
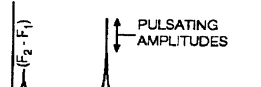
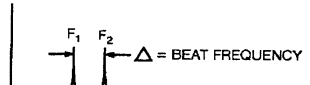
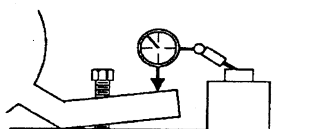
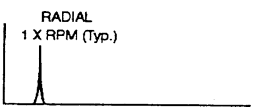
TABLE 6.0

ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	REMARKS
GEARS (CONTINUED)		
I. LOOSE BEARING FIT		Excessive Clearance of bearings supporting the gears can not only excite many running speed harmonics, but will often cause high amplitude response at GMF, 2GMF and/or 3GMF. These high GMF amplitudes are actually a response to, and not the cause of, looseness within the bearings supporting the gearing. Such excessive clearance can be caused either by extensive bearing wear or by improper bearing fit onto the journal during installation. Left uncorrected, it can cause excessive gear wear and damage to other components.
AC INDUCTION MOTORS		
A. STATOR ECCENTRICITY, SHORTED LAMINATIONS OR LOOSE IRON		Stator problems generate high vibration at 2X line frequency (2FL). Stator eccentricity produces uneven stationary air gap between rotor and stator which produces very directional vibration. Differential Air Gap should not exceed 5% for induction motors and 10% for synchronous motors. Soft foot and warped bases can produce an eccentric stator. Loose iron is due to stator support weakness or looseness. Shorted stator laminations can cause uneven, localized heating which can distort the stator itself. This produces thermally-induced vibration which can significantly grow with operating time causing stator distortion and static air gap problems.
B. ECCENTRIC ROTOR (Variable Air Gap) F_e = Electrical Line Freq. N_s = Synch. Speed = $120F_e/P$ F_s = Slip Freq. = $N_s - \text{RPM}$ F_p = Pole Pass Freq. = $F_s \times P$ P = #Poles		Eccentric Rotors produce a rotating variable air gap between the rotor and stator which induces pulsating vibration (normally between 2FL and closest running speed harmonic). Often requires "zoom" spectrum to separate 2FL and running speed harmonic. Eccentric rotors generate 2FL surrounded by Pole Pass frequency sidebands (Fp), as well as Fp sidebands around running speed. Fp appears itself at low frequency (Pole Pass Frequency = Slip Frequency X #Poles). Common values of Fp range from about 20 to 120 CPM (0.3 - 2.0 Hz). Soft foot or misalignment often induces a variable air gap due to distortion (actually a mechanical problem; not electrical).
C. ROTOR PROBLEMS		Broken or Cracked rotor bars or shorting rings; bad joints between rotor bars and shorting rings; or shorted rotor laminations will produce high 1X running speed vibration with pole pass frequency sidebands (Fp). In addition, these problems will often generate Fp sidebands around the second, third, fourth and fifth running speed harmonics. Loose or open rotor bars are indicated by 2X line freq. (2FL) sidebands surrounding Rotor Bar Pass Frequency (RBPF) and/or its harmonics (RBPF = Number of Bars X RPM). Often will cause high levels at 2X RBPF, with only a small amplitude at 1X RBPF. Electrically induced arcing between loose rotor bars and end rings will often show high levels at 2X RBPF (with 2FL sidebands); but little or no increase in amplitudes at 1X RBPF.
D. PHASING PROBLEM (Loose Connector)		Phasing problems due to loose or broken connectors can cause excessive vibration at 2X Line Freq. (2FL) which will have sidebands around it spaced at 1/3 Line Freq. (1/3 FL). Levels at 2FL can exceed 1.0 in/sec if left uncorrected. This is particularly a problem if the defective connector is only sporadically making contact. Loose or broken connectors must be repaired to prevent catastrophic failure.
AC SYNCHRONOUS MOTORS		
(Loose Stator Coils)		Loose stator coils in synchronous motors will generate fairly high vibration at Coil Pass Freq. (CPF) which equals the number of stator coils X RPM (#Stator Coils = #Poles X #Coils/Pole). The Coil Pass Frequency will be surrounded by 1X RPM sidebands. Synchronous motor problems may also be indicated by high amplitude peaks at approx. 60,000 to 90,000 CPM, accompanied by 2FL sidebands. Take at least one spectrum up to 90,000 CPM on each motor bearing housing.
DC MOTORS AND CONTROLS		
A. NORMAL SPECTRUM		Many DC Motor and Control Problems can be detected by vibration analysis. Full-wave rectified, motors (6 SCR's) generate a signal at 6X Line Frequency (6FL = 360 Hz = 21,600 CPM); while half-wave rectified DC motors (3 SCR's) generate 3X Line Freq. (3FL = 180 Hz = 10,800 CPM). The SCR firing Frequency is normally present in a DC Motor Spectrum, but at low amplitude. Note the absence of other peaks at multiples of FL.
B. BROKEN ARMATURE WINDINGS, GROUNDING PROBLEMS OR FAULTY SYSTEM TUNING		When DC Motor spectra are dominated by high levels at SCR or 2X SCR, this normally indicates either Broken Motor Windings or Faulty Tuning of the Electrical Control System. Proper tuning alone can lower vibration at SCR and 2X SCR significantly if control problems predominate. High amplitudes at these frequencies would normally be above approximately .10 in/sec, peak at 1X SCR and about .04 in/sec at 2X SCR Firing Freq.
C. FAULTY FIRING CARD OR BLOWN FUSE		When one firing card fails to fire, then 1/3 of power is lost, and can cause repeated momentary speed changes in the motor. This can lead to high amplitudes at 1/3X and 2/3X SCR Frequency (1/3X SCR Freq. = 1X FL for half-wave rectified, but 2X FL for a full-wave rectified SCR). Caution: Card/SCR configuration should be known before troubleshooting motor (#SCR's, #Firing Cards, etc.).
D. FAULTY SCR, SHORTED CONTROL CARD, LOOSE CONNECTIONS AND/OR BLOWN FUSE		Faulty SCR's, Shorted Control Cards and/or Loose Connections can generate noticeable amplitude peaks at many combinations of line frequency (FL) and SCR firing frequency. Normally, 1 bad SCR can cause high levels at FL and/or 5FL in 6 SCR motors. The point to be made is that neither FL, 2FL, 4FL nor 5FL should be present in DC Motors spectra.
E. FAULTY COMPARATOR CARD		Faulty Comparator Cards cause problems with RPM fluctuation or "hunting". This causes a constant collapsing and regenerating of the magnetic field. These sidebands often approximate the RPM fluctuation and require a high resolution FFT to even detect them. Such sidebands could also be due to generation and regeneration of the magnetic field.
F. ELECTRICAL CURRENT PASSAGE THRU DC MOTOR BEARINGS		Electrically-induced Fluting is normally detected by a series of difference frequencies with the spacing most often at the outer race defect frequency (BPFO), even if such fluting is present on both the outer and inner races. They most often show up in a range centered at about 100,000 to 150,000 CPM. A 180K CPM spectrum with 1600 lines is recommended for detection with measurements on both the OB and IB DC motor bearings.

TABLE 6.0

ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	REMARKS
BELT DRIVE PROBLEMS		
A. WORN, LOOSE OR MISMATCHED BELTS 	 PITCH DIAM. X RPM₁ = PITCH DIAM.₂ X RPM₂	$\text{BELT FREQ.} = \frac{3.142 \times \text{PULLEY RPM} \times \text{PITCH DIAM.}}{\text{BELT LENGTH}}$ $\text{TIMING BELT FREQ.} = \text{BELT FREQ.} \times \# \text{BELT TEETH}$ $= \text{PULLEY RPM} \times \# \text{PULLEY TEETH}$ Belt frequencies are below the RPM of either the motor or the driven machine. When they are worn, loose or mismatched, they normally cause 3 to 4 multiples of belt frequency. Often 2X belt freq. is the dominant peak. Amplitudes are normally unsteady, sometimes pulsing with either driver or driven RPM. On timing belt drives, wear or pulley misalignment is indicated by high amplitudes at the Timing Belt Frequency. Chain drives will indicate problems at Chain Pass Frequency which equals #Sprocket Teeth X RPM.
B. BELT/PULLEY MISALIGNMENT 		Misalignment of pulley produces high vibration at 1X RPM predominantly in the axial direction. The ratio of amplitudes of driver to driven RPM depends on where the data is taken, as well as on relative mass and frame stiffness. Often with pulley misalignment, the highest axial vibration on the motor will be at fan RPM, or vice versa. Can be confirmed by phase measurements by setting Phase Filter at RPM of pulley with highest axial amplitude; then compare phase at this particular frequency on each rotor in the axial direction.
C. ECCENTRIC PULLEYS 		Eccentric pulleys cause high vibration at 1X RPM of the eccentric pulley. The amplitude is normally highest in line with the belts, and should show up on both driver and driven bearings. It is sometimes possible to balance eccentric pulleys by attaching washers to taper-lock bolts. However, even if balanced, the eccentricity will still induce vibration and reversible fatigue stresses in the belt. Pulley eccentricity can be confirmed by phase analysis showing horizontal & vertical phase differences of nearly 0° or 180°.
D. BELT RESONANCE 		Belt Resonance can cause high amplitudes if the belt natural frequency should happen to approach, or coincide with, either the motor or driven RPM. Belt natural frequency can be altered by changing either the belt tension, belt length or cross section. Can be detected by tensioning and then releasing belt while measuring the response on pulleys or bearings. However, when operating, belt natural frequencies will tend to be slightly higher on the tight side and lower on the slack side.
BEAT VIBRATION		
	 WIDEBAND SPECTRUM  ZOOM SPECTRUM	A Beat Frequency is the result of two closely spaced frequencies going into and out of synchronization with one another. The wideband spectrum normally will show one peak pulsating up and down. When you zoom into this peak (lower spectrum below), it actually shows two closely spaced peaks. The difference in these two peaks ($F_2 - F_1$) is the beat frequency which appears itself in the wideband spectrum. The beat frequency is not commonly seen in normal frequency range measurements since it is inherently low frequency, usually ranging from only approximately 5 to 100 CPM. Maximum vibration will result when the time waveform of one frequency (F_1) comes into phase with the waveform of the other frequency (F_2). Minimum vibration occurs when waveforms of these two frequencies line up 180° out of phase.
SOFT FOOT, SPRUNG FOOT AND FOOT-RELATED RESONANCE		
		"Soft Foot" occurs when a machine's foot or frame deflects greatly when a hold-down bolt is loosened to hand tightness, causing the foot to rise more than approximately .002 - .003 inch. This does not always cause a great vibration increase. However, it can do so if the soft foot affects alignment or motor air gap concentricity. "Sprung Foot" can cause great frame distortion, resulting in increased vibration, force and stress in the frame, bearing housing, etc. This can occur when a hold-down bolt is forcefully torqued down on the sprung foot in an attempt to level the foot. "Foot-Related Resonance" can cause dramatic amplitude increases from 5X to 15X or more, as compared with that when the bolt (or combination of bolts) is loosened to hand tightness. When tight, this bolt can notably change the natural frequency of the foot or machine frame itself. Soft Foot, Sprung Foot or Foot-Related Resonance most often affects vibration at 1X RPM, but can also do so at 2X RPM, 3X RPM, 2X line frequency, blade pass frequency, etc. (particularly Foot-Related Resonance).

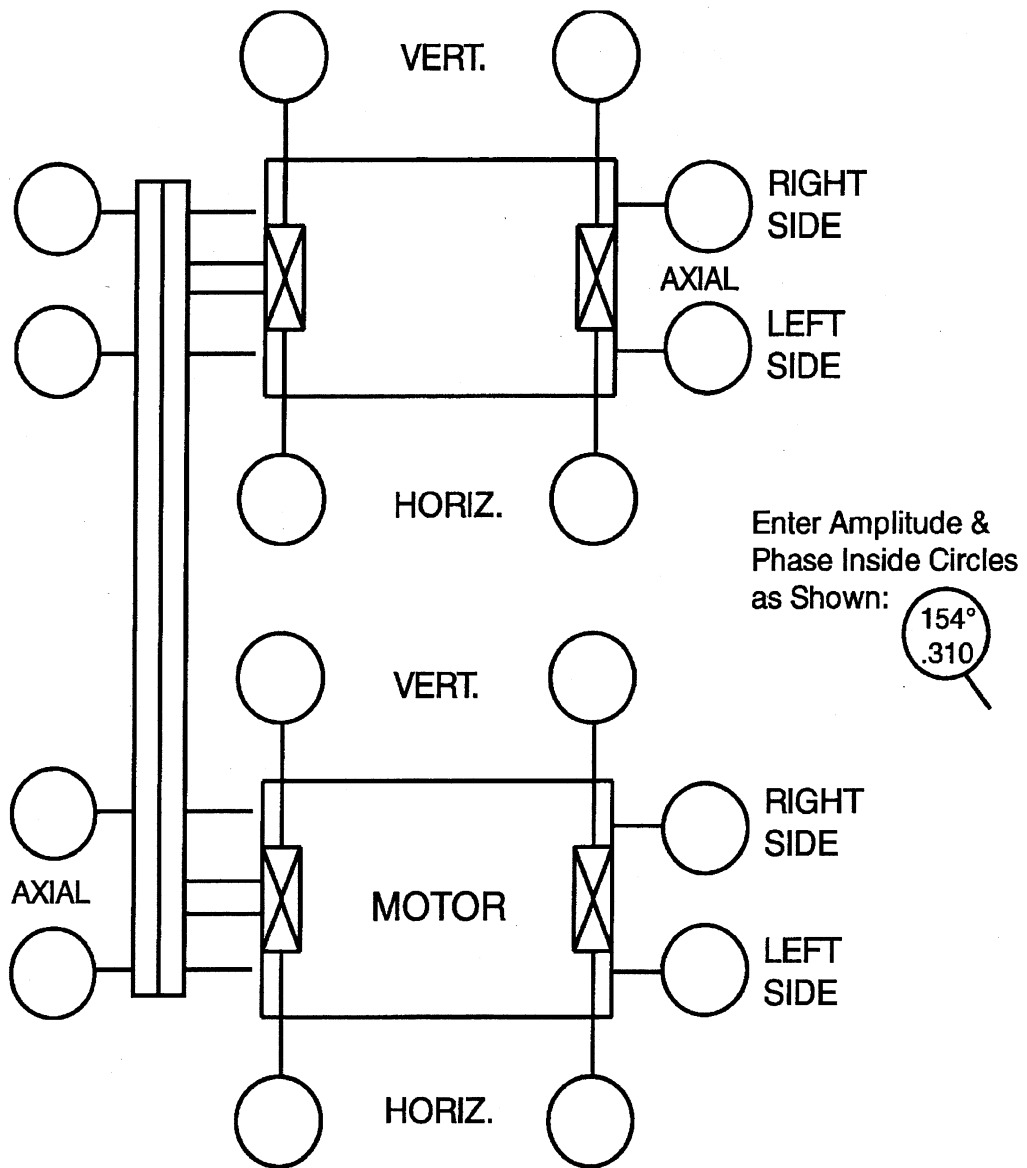
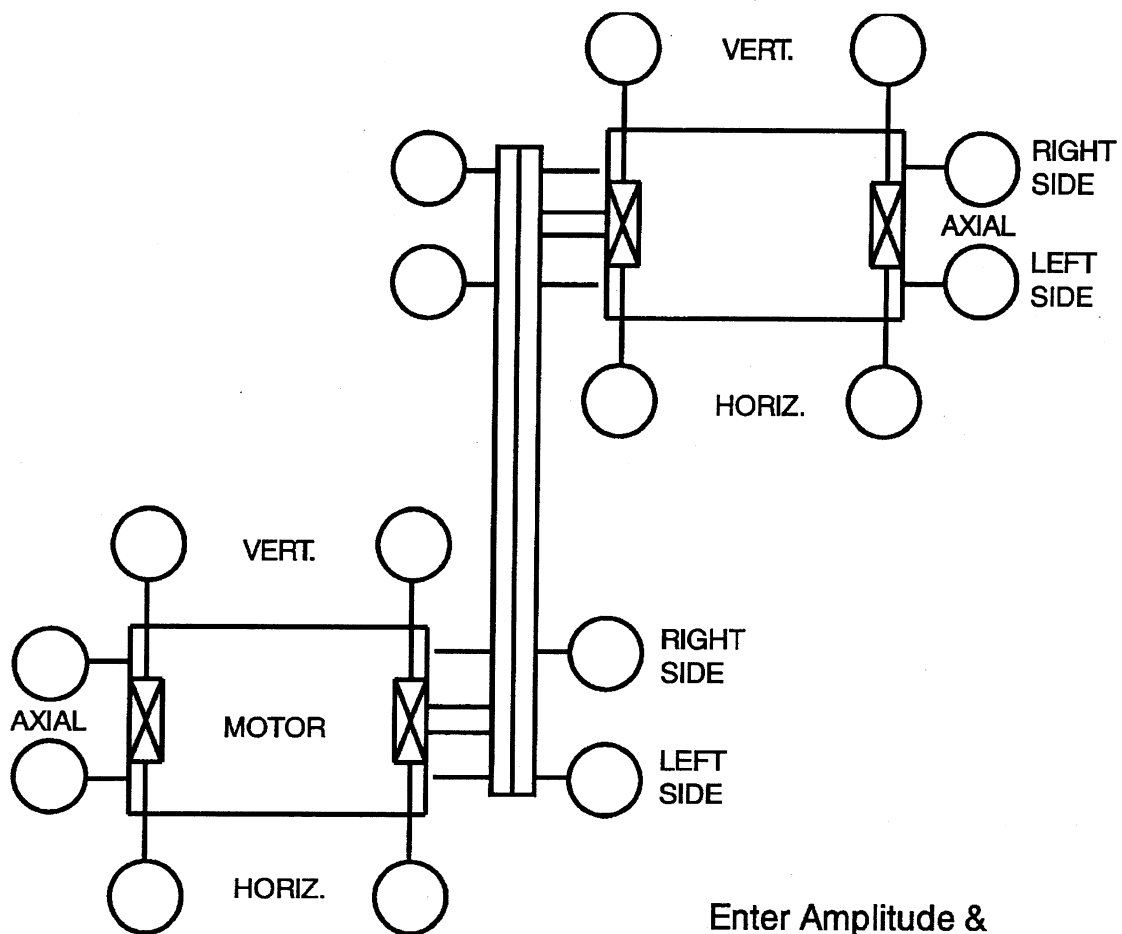


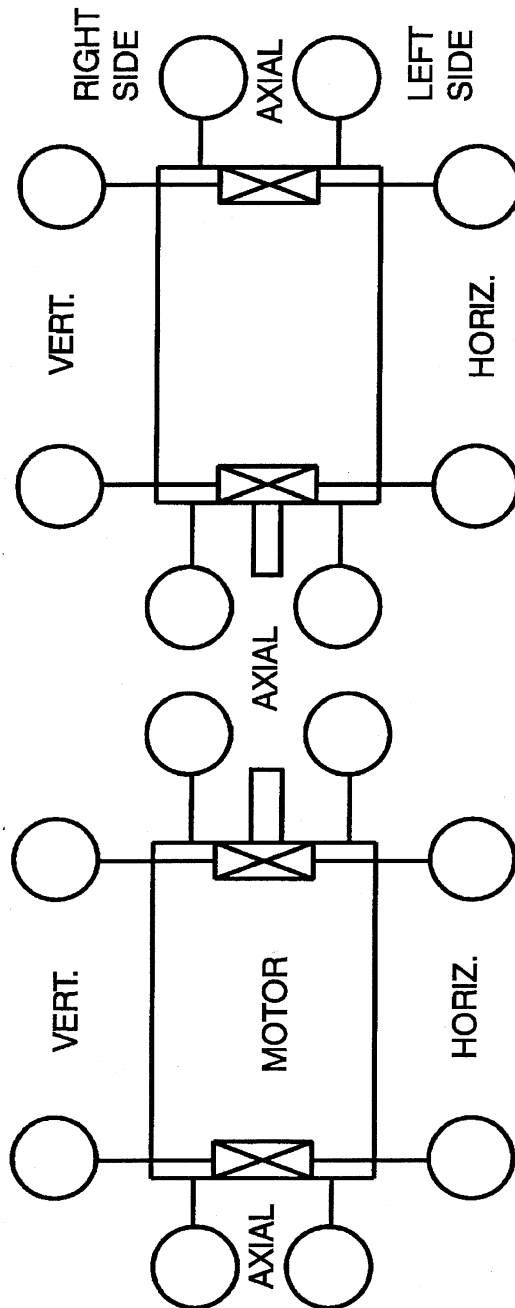
FIGURE 6.0A
PHASE ANALYSIS DIAGRAM BDB-1



Enter Amplitude &
Phase Inside Circles
as Shown:

154°
.310

FIGURE 6.0B
PHASE ANALYSIS DIAGRAM BDB-2



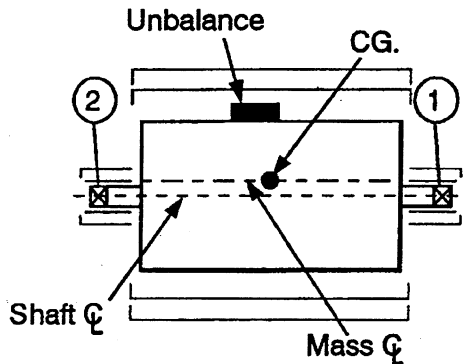
Enter Amplitude &
Phase Inside Circles
as Shown:

154°
.310

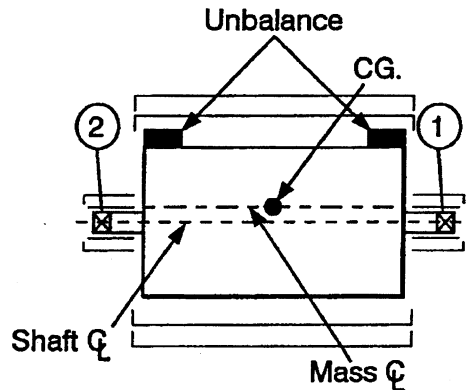
FIGURE 6.0C
PHASE ANALYSIS DIAGRAM DC-1

6.01 MASS UNBALANCE

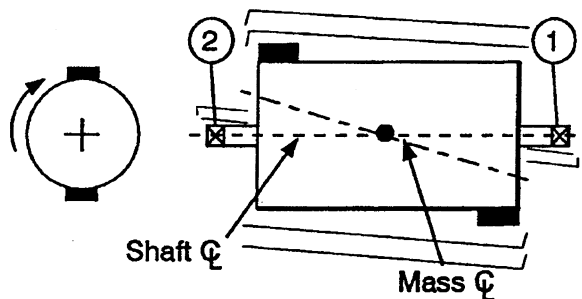
Unbalance occurs when the mass centerline does not coincide with the shaft centerline as shown in Figures 6.01A thru 6.01D. Some degree of unbalance exists in all rotors whether they are a cooling tower fan or a precision grinding wheel. The key is to know how much unbalance is acceptable for the particular type of machine at its specific operating speed which will be discussed in Section 6.015 "Allowable Residual Unbalance".



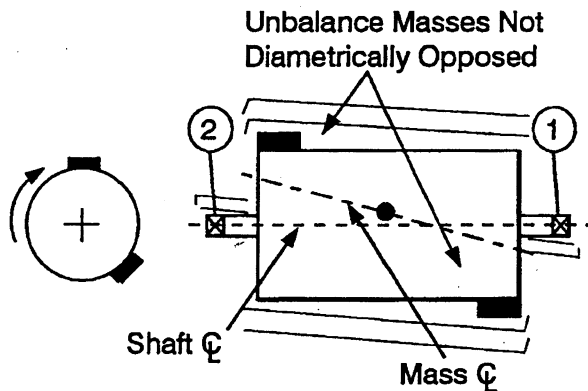
**FIGURE 6.01A
FORCE UNBALANCE**



**FIGURE 6.01B
FORCE UNBALANCE ALSO**



**FIGURE 6.01C
COUPLE UNBALANCE**



**FIGURE 6.01D
DYNAMIC UNBALANCE**

Unbalanced rotors exhibit each of the following characteristics:

1. Unbalance is always indicated by high vibration at 1X RPM of the unbalanced part (but, vibration at 1X RPM is not always unbalance). Normally, this 1X RPM peak will dominate the spectrum.
2. The amplitude at 1X RPM will normally be greater than or equal to 80% of the overall amplitude when the problem is limited to unbalance (may be only 50% to 80% if other problems exist in addition to unbalance).
3. The amplitude of vibration is proportional to how far the mass center is displaced from the shaft center. For example, when operating below the first rotor critical speed, amplitude will vary with the square of RPM (that is, tripling the speed will result in an increase in unbalance vibration by a factor of 9 times).
4. Mass unbalance generates a uniform rotating force which is continually changing direction, but is evenly applied in all radial directions. As a result, the shaft and supporting bearings tend to move in somewhat a circular orbit. However, due to the fact that vertical bearing stiffness is normally higher than that in the horizontal direction, the normal response is a slightly elliptical orbit. Subsequently, horizontal vibration is normally somewhat higher than that in the vertical commonly ranging between 25 to 50% times higher. When the ratio of horizontal to vertical is higher than about 6 to 1, it normally indicates other problems, particularly resonance.
5. When unbalance dominates over other problems, there will normally be about a 90° phase difference between horizontal and vertical directions on a bearing ($\pm 30^\circ$). Therefore, if there is a high vibration at 1X RPM, but this phase difference is either 0° or close to 180°, it normally points to another problem source such as eccentricity.
6. Probably an even greater indicator of unbalance than the approximately 90° phase shift between horizontal and vertical is the fact that when significant unbalance exists, the horizontal phase difference between outboard and inboard bearings should be close to the difference in phase in the vertical direction. That is, instead of comparing horizontal and vertical phase on the same bearing, compare outboard and inboard horizontal phase difference with outboard and inboard vertical phase difference. For example, please refer to Table A of Figure 6.01E which shows a machine having dominant force unbalance. Note that the horizontal phase difference between the #1 and #2 bearings is about 5° (30° minus 25°) compared to a vertical phase difference of about 10° (120° minus 110°). Similarly, over on the pump, the horizontal phase difference (position 3) is about 10° and the vertical phase difference is about 15°. This is the expected phase response with dominant force unbalance.
7. When unbalance is dominant, radial vibration (horizontal and vertical) will normally be quite much higher than that in the axial direction (except for overhung rotors which will be discussed in Section 6.014).
8. Unbalanced rotors normally exhibit steady and repeatable phase in radial directions. When the rotor is trim balanced, the phase can begin to "dwell" back and forth under a strobe light as you achieve a better and better balance, particularly if other problems are present. However, if there is high unbalance, and other problems are not significant, the phase should be steady and repeatable.
9. The effects of unbalance may sometimes be amplified by resonance. This will be discussed in Section 6.05.

10. Unbalance can be a great contributor to high looseness vibration. In fact, on a rotor with unbalance and looseness, if it is possible to balance the rotor, this may substantially reduce the looseness vibration although it will often return when even the least little unbalance component returns. Often, it is not even possible to balance rotors having noticeable looseness.

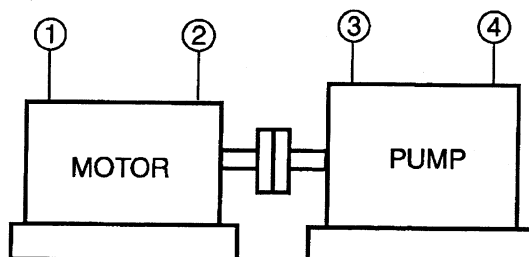


TABLE A.
PHASE AT 1X RPM FOR A MACHINE
WITH DOMINANT FORCE UNBALANCE

DIR.	1	2	3	4
A	60°	*70°	60°	*80°
H	30°	25°	30°	40°
V	120°	110°	120°	135°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

TABLE B.
PHASE AT 1X RPM FOR A MACHINE
WITH DOMINANT COUPLE UNBALANCE

DIR.	1	2	3	4
A	60°	*70°	60°	*80°
H	30°	210°	200°	180°
V	120°	295°	280°	300°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

TABLE C.
PHASE AT 1X RPM FOR A MACHINE
WITH SIGNIFICANT DYNAMIC UNBALANCE

DIR.	1	2	3	4
A	60°	*70°	60°	*80°
H	30°	90°	80°	70°
V	120°	180°	170°	165°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

FIGURE 6.01E
TYPICAL PHASE MEASUREMENTS WHICH WOULD INDICATE EITHER
FORCE, COUPLE OR DYNAMIC UNBALANCE

There are 3 major types of unbalance including force, couple and dynamic unbalance which will be discussed in Sections 6.011, 6.012 and 6.013, respectively, which follow:

6.011 Force Unbalance:

Force unbalance is sometimes known as "static unbalance". Force unbalance is a condition where the mass centerline is displaced from and parallel to the shaft centerline as shown in Figure 6.01A. This is the type of unbalance that has been classically corrected for many years by placing a fan rotor on knife edges or within its bearings and allowing it to "roll to the bottom". That is, when the fan wheel is released, if the heavy spot is angularly displaced from the bottom (6:00 position), it would tend to roll to the bottom hopefully ending up in the 6:00 position, assuming the rotor was sufficiently free within its bearings to rotate. So-called correction of this force unbalance was then accomplished by placing a weight opposite this location (or at about 12:00).

Actually, there are two types of force unbalance as shown in Figure 6.01A and Figure 6.01B. In the case of Figure 6.01A, only one heavy spot exists and is located close to the rotor center of gravity (CG). This is corrected by simply placing an equal weight 180° opposite the angular position of the heavy spot. Figure 6.01B likewise illustrates force unbalance even though it shows heavy spots acting on both the outboard and inboard planes (angularly parallel to one another). In this case, it can either be corrected by placing correction weights either at the CG, or by placing equal and opposite weights at each of the two planes (if corrected at the CG, it would of course require double the correction weight in this case).

Characteristics common to force unbalance can be summarized as follows:

1. Approximately the same unbalance forces at 1X RPM are normally present both on the outboard and inboard rotor bearing housings (however, horizontal and vertical responses may differ somewhat depending on the support stiffness in each direction).
2. With pure force unbalance, the outboard horizontal phase will equal the inboard horizontal phase on the same shaft (that is, if the horizontal phase on the outboard bearing were at 6:00, the inboard reading should likewise be about 6:00 since the two shaft ends are moving together).
3. Likewise, the outboard vertical phase should approximately equal the inboard vertical phase on the same shaft.
4. Force unbalance only requires a single plane correction with the counterweight acting through the rotor CG.
5. The difference in horizontal outboard and inboard phase should approximately equal the phase difference in outboard and inboard vertical phase and the phase change across the coupling should be small (less than 60° to 90°) if force unbalance were dominant.

6.012 Couple Unbalance:

Couple unbalance is a condition where the mass centerline axis intersects the shaft centerline axis at the rotor's center of gravity as shown in Figure 6.01F. Here, a couple is created by equal heavy spots at each end of the rotor, but 180° opposite each other. Significant couple unbalance can introduce severe instability to the rotor causing it to wobble back and forth (like a "seesaw" with the fulcrum at the rotor CG).

Couple unbalance exhibits each of the following characteristics:

1. In pure couple unbalance, the rotor is statically balanced and will not roll to the bottom when the rotor is placed on knife edges. That is, referring to Figure 6.01C, since the heavy spot at position 1 is equal to that at position 2, this meets the requirement for force or static balance. Still, this rotor will also generate considerable vibration at 1X RPM.
2. Couple unbalance generates high amplitude vibration at 1X RPM on both the outboard and inboard bearing housings, but it may be somewhat higher on one bearing than on the other.
3. Substantial couple unbalance can sometimes generate high axial vibration.
4. The horizontal phase difference between the outboard and inboard bearings will approximate 180° (that is, if the outboard horizontal phase were at 6:00, then the inboard horizontal phase will probably be about 12:00 since the two ends are moving opposite each other in a rocking motion).
5. Similarly, the vertical phase difference between outboard and inboard bearings will approximate 180°.
6. Refer to Table B of Figure 6.01E illustrating how phase should react to couple unbalance. Note the 180° phase difference between position 1 and 2 horizontal ($210^\circ - 30^\circ$), and the 175° phase difference between position 1 and 2 vertical ($295^\circ - 120^\circ$). This shows that if the problem is couple unbalance (and not misalignment), both the horizontal and vertical phase differences should roughly be equal to one another - both approximately 180° difference between the outboard and inboard bearings.

6.013 Dynamic Unbalance:

Dynamic unbalance is by far the most common type of unbalance as compared to either purely force or couple unbalance and is defined as "that condition in which the mass centerline is neither parallel to nor intersects the shaft centerline axis". In essence, dynamic unbalance is a combination of both force and couple unbalance. It requires correction in at least 2 planes perpendicular to the shaft centerline axis.

Dynamic unbalance exhibits each of the following characteristics:

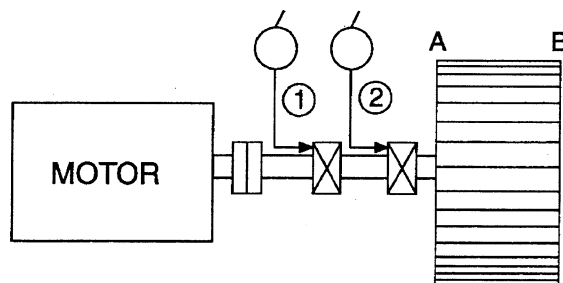
1. Dynamic unbalance generates high vibration at 1X RPM, but the amplitude on the outboard bearing may be somewhat different than that on the inboard bearing housing. Still, they should be within the same order of magnitude, or below about 3 to 1 assuming there are no other significant problems present.
2. Like force and couple unbalance, phase is still steady and repeatable when dynamic unbalance dominates.
3. Although the horizontal phase difference between outboard and inboard bearings could be anything from 0° to 180°, this difference should still approximately equal the vertical phase difference. For example, if the horizontal phase difference was about 60°, the vertical phase difference should likewise be about 60° ($\pm 30^\circ$) as illustrated in Table C of Figure 6.01E. Here, in this example, notice that the phase difference in both horizontal and vertical directions at positions 1 and 2 is about 60° and that the phase difference across the coupling does not approach 180°. Dynamic unbalance requires correction in at least 2 planes.

4. Whether or not force or couple balance dominates, the horizontal phase difference at bearings 1 and 2 should approximately equal the vertical phase difference at these two bearings (if the horizontal phase difference is about 150° showing high couple unbalance, the vertical phase difference will approximate 150° as well) .

6.014 Overhung Rotor Unbalance:

Figure 6.01F shows an overhung rotor. In this case, the driven rotor is placed outboard of bearings 1 & 2 (rotors which are placed between bearings are known as simply supported rotors). Overhung rotors can cause some interesting vibration symptoms and often can present real problems to the analyst when he attempts to balance one. Overhung rotors display the following characteristics:

1. Overhung rotors can generate large axial forces at 1X RPM which can cause axial vibration to be equal to or greater than radial vibration amplitudes.
2. Overhung rotors often generate a high degree of couple unbalance in addition to force unbalance, both of which must be corrected.
3. Referring to Figure 6.01F, for pure unbalance of an overhung rotor, the axial phase at bearing 1 will approximately equal that at bearing 2 ($\pm 30^\circ$). Here again, this phase difference depends on how dominant the unbalance problem is as compared to others such as misalignment, resonance, etc.
4. Normally, overhung rotor unbalance can be corrected by first taking care of the force unbalance component which would leave the remainder as couple unbalance with phase differences approaching 180° . The couple component would then require placement of correction weights in 2 planes 180° opposite one another.



**FIGURE 6.01F
BALANCING OF AN OVERHUNG ROTOR**

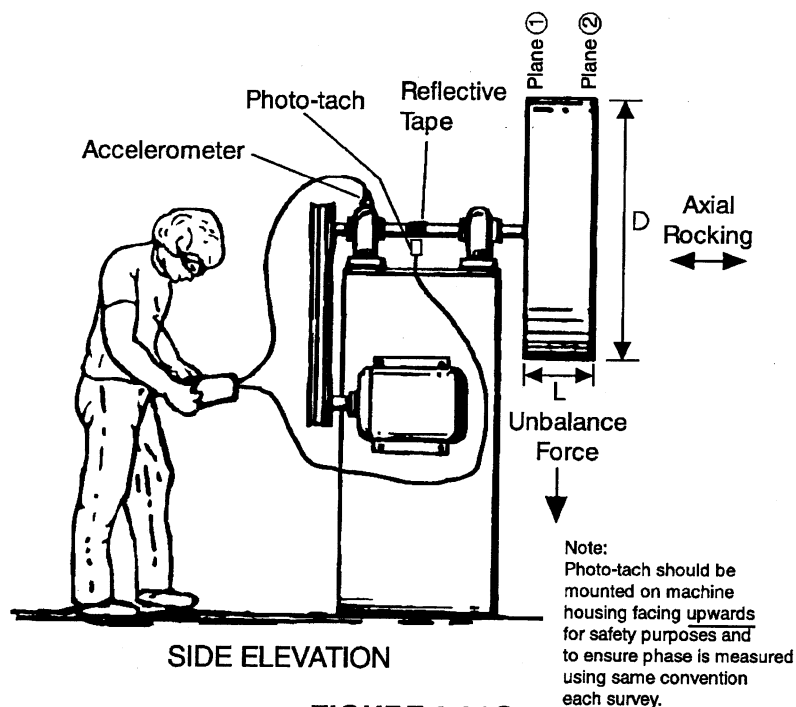
6.0141 Summary of Procedures for Balancing Overhung Rotors

Overhung rotors are machine configurations like that shown in Figure 6.01G where the fan wheel to be balanced is outboard of its two supporting bearings. This configuration is very often found with machines such as blowers, pumps, etc. Because the planes where balance correction weights must be attached are outside the supporting bearings, these rotors will often not respond to standard single and two-plane balancing techniques. In addition, because the unbalance planes are outside the support bearings, even a static unbalance alone will create a couple unbalance proportional to the distance of the unbalance plane from the rotor CG. Therefore, when attempting to balance overhung rotors, the analyst needs to take into account both static and couple unbalance forces, and treat them accordingly.

When balancing an overhung rotor, one of the two following procedures should be taken:

1. Balancing Overhung Rotors by Classic Single-Plane Static-Couple Method:

Figure 6.01G helps explain methods of balancing overhung rotors. Classically, Bearing A is most sensitive to static unbalance whereas the bearing farthest from the fan wheel to be balanced (Bearing B) is most sensitive to couple unbalance. Since Plane 1 is closest to the rotor center of gravity (CG), static corrections should be made in this plane while measuring the response on Bearing A. On the other hand, measurements should be made on Bearing B when making couple corrections in Plane 2. However, placing a trial weight in Plane 2 will destroy the static balance achieved at Bearing A. Therefore, in order to maintain the static balance at Bearing A, a trial weight placement which will generate a couple must be used. Thus, a trial weight of identical size should be placed in Plane 1 at an angle 180° opposite the trial weight location in Plane 2.



Therefore, either the data collector can be used using single-plane balance software or the single-plane graphic technique previously explained can be successfully employed on many overhung rotors, particularly if the ratio of the rotor length-to-diameter (L/D) is less than approximately .50 (where L is length of the rotating component on which correction weights will be placed and D is the diameter of this component - see Figure 6.01G).

Following below will be a description of this classic single-plane balancing technique for overhung rotors:

- a. **Set Up Data Collector and/or Spectrum Analyzer Instruments** - The data collector, phototach, accelerometer and so forth should be set up as previously described under Section D and Figure 6.01G showing the two-plane balancing procedure. Alternatively, the analyst may wish to employ either a swept-filter analyzer which drives a strobe light (like an IRD 350 or IRD 880), or a spectrum analyzer which will fire a phototach for phase measurement.

-
- b. Take Initial Measurements** - Take initial measurements of 1X RPM amplitude, frequency and phase before adding any trial weights. Measurements should be taken on both the outboard and inboard bearings in both vertical and horizontal directions. The radial direction measurement having the highest amplitude will normally be employed for initial balancing (however, after correcting unbalance in the radial direction, measurements will have to be taken in the other radial direction to ensure amplitudes in it are likewise acceptable).
- c. Determine if the Dominant Problem is Either Static or Couple Unbalance** - Looking at the amplitude and phase measurements taken on both bearings in the radial and horizontal directions, determine if the problem is dominated by either static or couple unbalance. If phase differences between the outboard and inboard bearing are approximately 140° or more in both the vertical and horizontal directions, the dominant problem will be couple unbalance. On the other hand, if these differences are both anywhere from 0° to approximately 40° , static unbalance is dominant. Of course, phase differences ranging from approximately 40° to 140° are truly dynamic balance once again with a combination of static and couple. If the problem appears to be mostly couple unbalance, use couple unbalance procedures outlined below. However, if the problem appears to be predominantly static or dynamic unbalance, employ static balance procedures. For now, we will assume that the problem is mostly static.
- d. Make a Single-Plane Static Balance** - Referring to Figure 6.01G, use single-plane techniques taking measurements on Bearing A and placing trial and correction weights in Plane 1.
- e. Determine if Resultant Vibration Amplitudes Meet Required Criteria** - After completing the single-plane static balance using Plane 1, repeat vibration measurements on both the outboard and inboard bearings in each direction (including axial) and ensure that amplitudes now meet allowable criteria.
- f. If Considerable Couple Unbalance Now Remains, Continue With Single-Plane Balance From Bearing B** - Overhung rotors often have large cross-effects which means that single-plane balancing from Plane 1 will often cause high vibration over at Bearing B. Therefore, the analyst will perform another single-plane balance, this time making his measurements from Bearing B farthest from the component to be balanced. When he arrives at the single-plane correction weight solution, he should place this weight in Plane 2; and then place an identical size correction weight over in Plane 1 some 180° away from the weight location in Plane 2.
- g. Determine if Amplitudes Now Meet all Criteria** - After completing the single-plane couple correction, the analyst must again make measurements in horizontal, vertical and axial directions on each bearing and determine that all amplitudes now meet allowable criteria. Often, further balancing must be done at this point beginning with another single-plane balance using Bearing A and Plane 1 which might possibly be followed by another couple balance correction.
- h. If Allowable Criteria Cannot be Met in all Three Directions of Each Bearing, Proceed to Two-Plane Balance Procedure Outlined Below** - Sometimes, this single plane approach will not successfully reduce amplitudes below allowable criteria in all three directions on each bearing, particularly if the L/D ratio is greater than .50 or if the component to be balanced is located far away from the closest bearing. If this happens, two-plane techniques outlined below will have to be taken.

2. Balancing Overhung Rotors by Classic Two-Plane Static-Couple Method :

Due to the significant cross-effects which are often present in overhung rotors, two-plane balance correction techniques often are more successful than those employing single-plane methods. However, one of the problems with two-plane methods is that it can sometimes be a little confusing on deciding which bearing is the left and which is the right bearing; similarly, which plane is the left and which is the right plane? (Some data collectors refer to these as the near and far planes as opposed to left and right; terminology does not matter - only that the analyst remain consistent in his convention.) Referring to Figure 6.01G, when using two-plane techniques, Bearing A will be considered the bearing closest to the overhung rotor while Bearing B will be closest to the pulley. Similarly, Plane 1 will be on the inboard side of the wheel closest to the bearings whereas Plane 2 will be outboard.

Here again, a static/couple solution will be employed when the two-plane correction weight calculations are completed. Since most overhung rotors are so sensitive to static unbalance, only the static correction weight will be placed when this static/couple solution is obtained. Then, after trim balancing, if considerable couple unbalance remains, the analyst will proceed to correct this as well. He should follow the procedure outlined below:

- a. **Set Up Instruments as Outlined in Two-Plane Balance Method in Figure 6.01G** - Here again, this same procedure can be used with either data collectors, swept-filter analyzers or real-time analyzers. However, if using either a swept-filter or real-time analyzer, the analyst should have a two-plane calculator program that is capable of providing static/couple solutions.
- b. **Take Initial Measurements on Both Bearings** - Here again, 1X RPM amplitude, frequency and phase should be measured in horizontal, vertical and axial directions on both the outboard and inboard bearings.
- c. **Complete a Two-Plane Balancing Procedure, But Do Not Yet Place Balance Correction Weights** - A two-plane balance procedure like that outlined in Section D should be employed, but final correction weights not put in place. Instead, when the trial weights sizes and locations are calculated for each plane, the analyst should ask for a static/ couple solution and should initially only make the static correction. For example, if the static solution called for 1 oz in Plane 1 whereas the couple solution called for a 2 oz correction in Planes 1 & 2 180° opposite one another, make only the static correction at this point.
- d. **Determine if Amplitudes Now Meet Allowable Criteria** - After making the static correction in Plane 1, see if amplitudes in all three directions on each bearing are now within compliance with allowable criteria. If not, trim as required. Again, when the two-plane corrections are determined, ask for the static/couple solution and once again, make only the static correction. Most of the time, the problems are resolved at this point. However, if considerable couple unbalance still remains, complete another two-plane procedure again asking for the static/couple solution - this time making the couple correction called for, and not the static correction.
- e. **Determine if Amplitudes Now Meet Allowable Criteria** - After each of the two trials making these static corrections and the single trial making the couple correction, compare amplitudes in horizontal, vertical and axial directions on both the outboard and inboard bearings with allowable criteria. A small percentage of the time, the couple correction will throw the static balance back off. If this is the case, it may require one more static correction before the rotor is successfully balanced.

6.015 Allowable Residual Unbalance and ISO Balance Quality Grade:

When balancing a rotor, one needs to know to what precision he is required to balance the unit. In reality, it is not enough to simply say that it will be acceptable to balance the machine to a level of .10 in/sec or 1.0 mil at a certain speed. While one rotor may satisfactorily be balanced at such levels, another will not. This was recognized back in the 1950's by experts involved in the balance field. They recognized that the residual unbalance is truly proportional to the amount and radius of the remaining rotor eccentricity as well as the weight of the rotor itself and its operating speed. Therefore, they developed a series of balance tolerances known as ISO Standard No. 1940 on "Balance Quality of Rotating Rigid Bodies". Table 6.01A provides the balance quality grades as per these standards for a whole group of rotor types. Then, Table 6.01B provides the numerical standards for each of the ISO balance quality grades (ISO G-1, ISO G-2.5, ISO G-6.3, etc.). Note that the lower the G tolerance, the more precision the balance quality grade. Also note that it is based on the rotor RPM (horizontal axis) as well as the residual unbalance per pound of rotor weight (vertical axis). Table 6.01C is provided showing these same balance quality grades delineated by bands separating one balance quality grade from another. Table 6.01C also shows common nominal RPM's in the United States (1200, 1800 and 3600 RPM).

When balancing a machine, you first refer to the tabulated information in Table 6.01A to determine to what tolerance you should balance it. For example, if balancing an automobile crankshaft, this falls under ISO G-16 quality grade. On the other hand, fans fall under G-6.3, and grinding-machine drives fall under G-1. Note that the quality grade number itself represents the maximum permissible circular velocity of the rotor center of gravity expressed in millimeters per second (mm/sec). For example, a quality grade G-6.3 corresponds to rotor velocity of 6.3 mm/sec RMS which corresponds to an equivalent .248 in/sec RMS (.351 in/sec peak).

It is the experience of the author that this ISO Standard, which was established in 1966, is a little too conservative, possibly because it might be based on the technology available during that day and time. It is recommended that when using the ISO tolerance, that you should use one quality grade better than that specified for the specific machine you wish to balance (that is, if the standard calls for G-6.3, we would recommend using ISO G-2.5). For example, if balancing a fan wheel, note that Table 6.01A calls for an ISO quality grade G-6.3 for fans. In this case, we would recommend ISO grade G-2.5. Following below will be a procedure on how to determine the allowable residual unbalance, the ISO balance quality grade that you have achieved, and the rotor balance sensitivity:

How to Determine Residual Unbalance Remaining in a Rotor After Balancing

When field balancing, one must know when to determine that the job is complete. He will know this not only when he has achieved low vibration levels, but also when he knows he has balanced the rotor within allowable specifications. To know this, he must determine the residual unbalance remaining in the rotor. This can be accomplished by following the procedure below (refer to Figure 6.01H):

- a. Make original measurements of amplitude and phase and graph this to scale on polar coordinate paper. Call this vector the "O" vector.
- b. Attach a trial weight and document the trial weight size (oz) and radius (in) to which it is attached. ($mr = \text{trial weight size} \times \text{trial weight radius}$)
- c. After attaching the trial weight, spin the rotor and measure amplitude and phase. Graph this on the polar coordinate paper as the "O + T" vector.

- d. Draw a vector called "T" from the end of vector "O" to the end of vector "O + T". Vector "T" represents the effect of the trial weight alone. Measure the length of vector "T" to the same scale as that used for vectors "O" and "O + T". Using this scale, determine the equivalent vibration level (mils).
- e. Calculate Rotor Sensitivity as per the following equation:

$$\text{Rotor Sensitivity} = \frac{(\text{Trial Wt. Size})(\text{Trial Wt. Radius})}{\text{Trial Weight Effect}}$$

(oz-in/mil) (mils)

(Eqn. 6.01A)

- f. Calculate Residual Unbalance using Equation 6.01B. If Residual Unbalance not brought within tolerances, trim balance using current correction weight as the trial weight for the trim run. Continue trim balancing until Residual Unbalance is reduced within required balance tolerances:

$$\text{Residual Unbalance} = \text{Rotor Sens.} \times \text{Vib. Ampl. After Bal.}$$

(oz-in) (oz-in/mil) (mils)

(Eqn. 6.01B)

Example (see Figure 6.01H):

Given:

Required ISO Balance Quality = G 2.5
Rotor Weight = 100 lb
Rotor Speed = 800 RPM
Amplitude After Balancing = 2.0 mils

Therefore, Required U_{per} = 1.76 oz-in total (single-plane balance)

- a. Original reading = 10 mils @ 240° = "O" vector.
- b. Trial weight of 3 oz is attached in the balance plane at a 6 inch radius
(mr = 3 oz X 6 in = 18 oz-in)
- c. Trial run reading = 8 mils @ 120° = "O + T" vector
- d. Effect of trial weight alone = T = 15.5 mils (from Figure 6.01H)
- e. Rotor Sensitivity = $\frac{18 \text{ oz-in}}{15.5 \text{ mils}} = 1.16 \frac{\text{oz-in}}{\text{mil}}$
- f. Residual Unbalance = $(1.16 \frac{\text{oz-in}}{\text{mil}})(2.0 \text{ mils}) = \underline{2.32 \text{ oz-in}}$
(not within specs)

Continued balancing and reduced vibration to 1.0 mil

$$\text{Residual Unbalance} = (1.16 \frac{\text{oz-in}}{\text{mil}})(1.0 \text{ mil}) = \underline{1.16 \text{ oz-in}}$$

(in compliance)

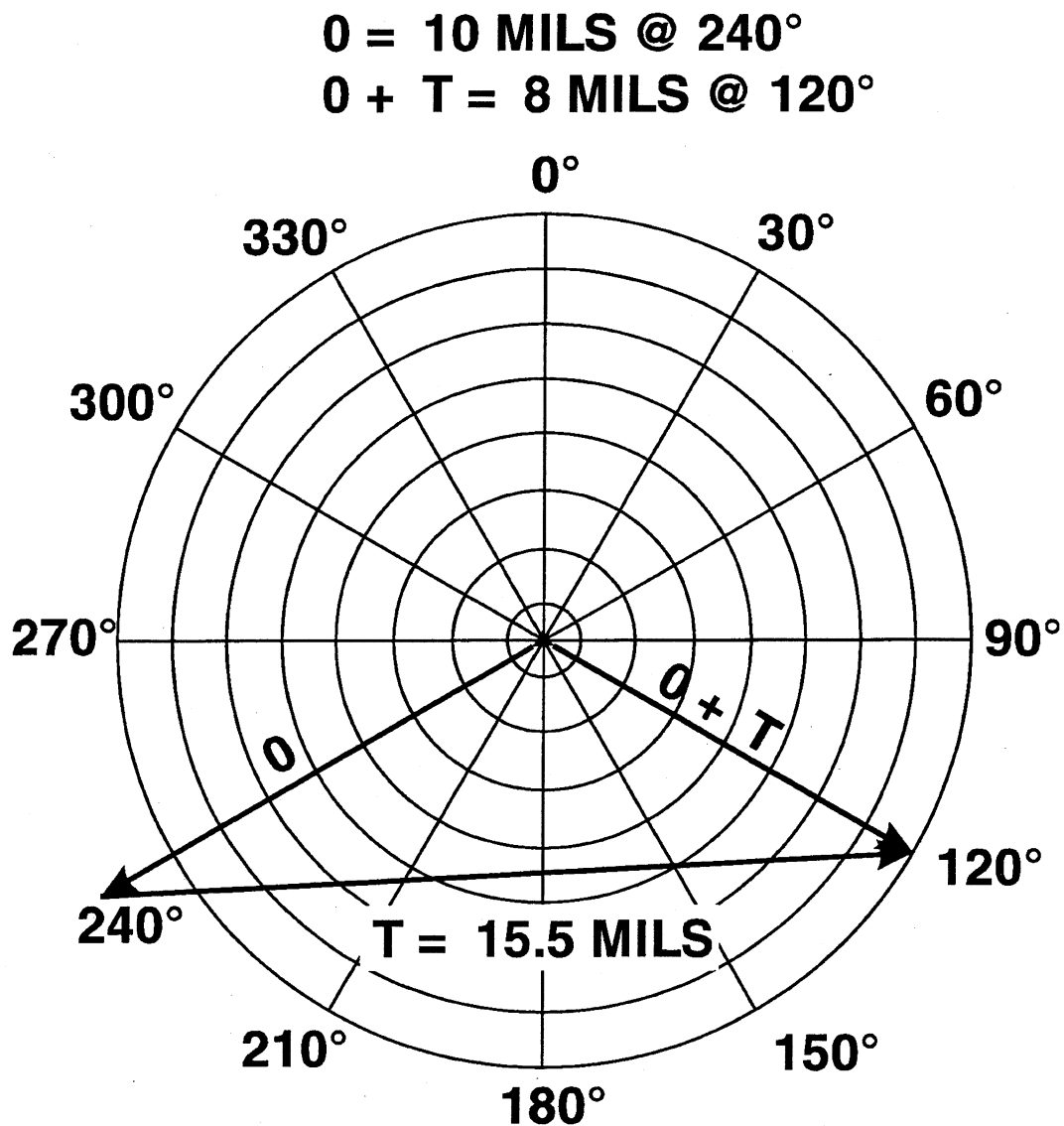


FIGURE 6.01H
STANDARD SINGLE-PLANE VECTOR SOLUTION

TABLE 6.01A*
BALANCE QUALITY GRADES FOR VARIOUS GROUPS OF
REPRESENTATIVE RIGID ROTORS IN ACCORDANCE WITH ISO 1940 AND
ANSI S2.19-1975

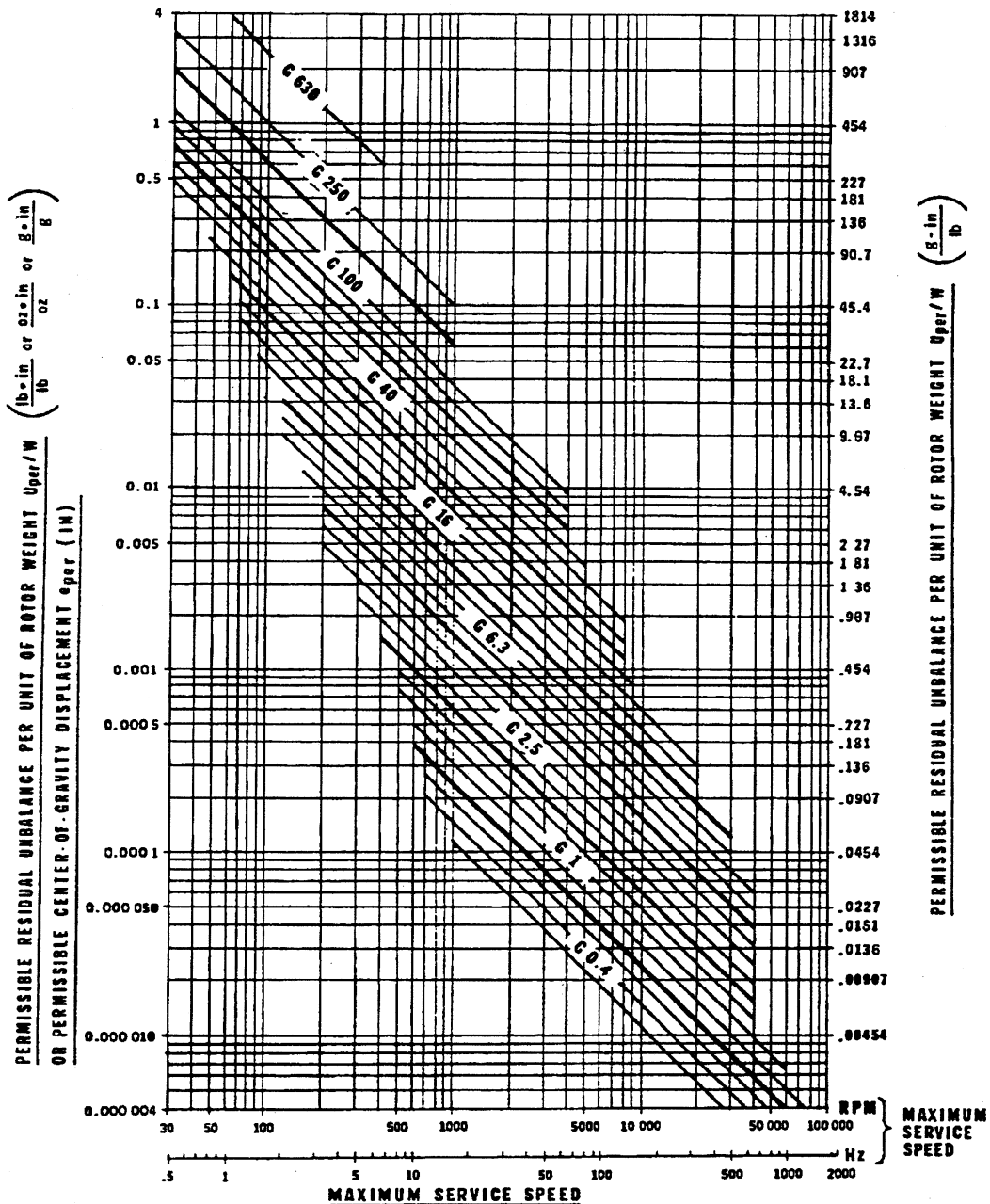
*(Reference 20)

Balance Quality Grades G	EW 1, 2 (mm/sec)	Rotor Types - General Examples
G 4000	4000	Crankshaft drives ³ of rigidly mounted slow marine diesel engines with uneven number of cylinders. ⁴
G 1600	1600	Crankshaft drives of rigidly mounted large two-cycle engines.
G 630	630	Crankshaft drives of rigidly mounted large four-cycle engines. Crankshaft drives of elastically mounted marine diesel engines.
G 250	250	Crankshaft drives of rigidly mounted fast four-cylinder diesel engines. ⁴
G 100	100	Crankshaft drives of fast diesel engines with six or more cylinders. ⁴ Complete engines (gasoline or diesel) for cars, trucks and locomotives. ⁵
G 40	40	Car wheels, wheel rims, wheel sets, drive shafts. Crankshaft drives or elastically mounted fast four-cycle engines (gasoline or diesel) with six or more cylinders. ⁴ Crankshaft drives for engines of cars, trucks or locomotives.
G 16	16	Drive shafts (propeller shafts, cardan shafts) with special requirements. Parts of crushing machinery. Parts of agricultural machinery. Individual components of engines (gasoline or diesel) for cars, trucks, and locomotives. Crankshaft drives of engines with six or more cylinders under special requirements. Slurry or dredge pump impeller.
G 6.3	6.3	Parts or process plant machines. Marine main turbine gears (merchant service). Centrifuge drums. Fans. Assembly aircraft gas turbine rotors. Fly wheels. Pump impellers. Machine-tool and general machinery parts. Normal electrical armatures. Individual components of engines under special requirements.
G 2.5	2.5	Gas and steam turbines, including marine main turbines (merchant service). Rigid turbo-generator rotors. Rotors. Turbo-compressors. Machine-tool drives. Medium and large electrical armatures with special requirements. Small electrical armatures. Turbine-driven pumps.
G 1	1	Tape recorder and phonograph (gramophone) drives. Grinding machine drives. Small electrical armatures with special requirements.
G 0.4	0.4	Spindles, disks, and armatures of precision grinders. Gyroscopes.

1. The quality grade number represents the maximum permissible circular velocity of the center of gravity (CG) in mm/sec.
2. $\omega = 2\pi n/60 \sim n/10$, if n is measured in revolutions per minute and ω in radians per second.
3. In general, for rigid rotors with two correction planes, one-half of the recommended residual unbalance is to be taken for each plane; these values apply usually for any two arbitrarily chosen planes, but the state of unbalance may be improved upon at the bearings. For disk-shaped rotors the full recommended value holds for one plane.
4. A crankshaft-drive is an assembly which includes the crankshaft, a flywheel, clutch, pulley, vibration damper, rotating portion of connecting rod, etc.
5. For purposes of this Standard, slow diesel engines are those with a piston velocity of less than 9 mm/sec; fast diesel engines are those with a piston velocity of greater than 9 m/sec.
6. In complete engines, the rotor mass comprises the sum of all masses belonging to the crankshaft drive described in Note 3 above.

TABLE 6.01B*
MAXIMUM PERMISSIBLE RESIDUAL SPECIFIC UNBALANCE
CORRESPONDING TO VARIOUS BALANCE QUALITY GRADES G,
IN ACCORDANCE WITH ISO 1940

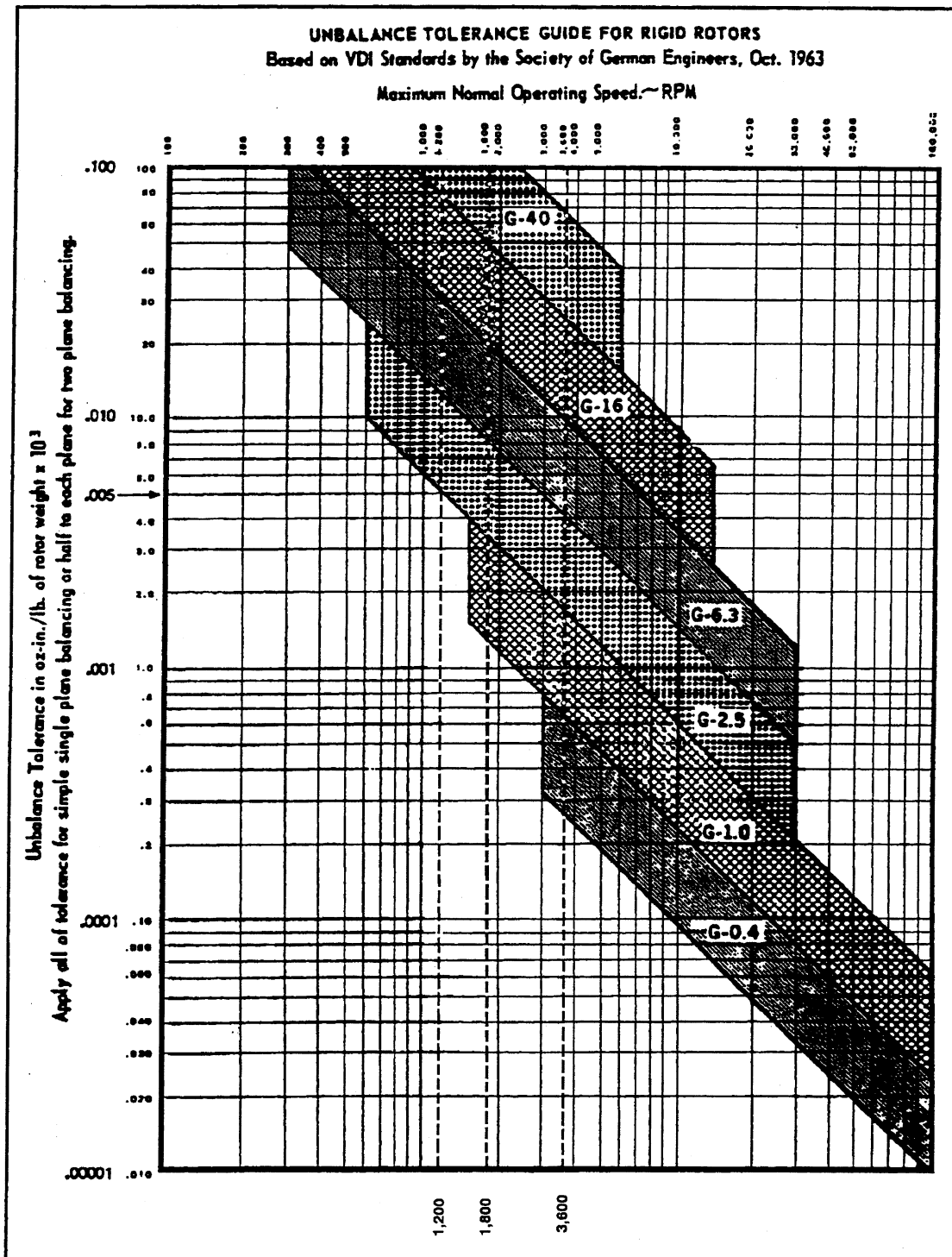
*(Reference 20)



NOTES:

- (1) In general, for rigid rotors with two correction planes, one half of the recommended residual unbalance is allowed for each plane; these values apply usually for any two arbitrarily chosen planes. (See paragraph on Applying Tolerances to Two-plane Rotors). For disc-shaped rotors the full recommended value applies to one plane.
- (2) To obtain acceptable residual unbalance in oz-in per lb of rotor weight multiply lb-in value by 16.
- (3) To avoid need for multiplication use appropriate Balance Tolerance Nomogram on following pages.

TABLE 6.01C*
EQUIVALENT ISO QUALITY GRADES SHOWN IN BANDED REGIONS
 *(Reference 21)

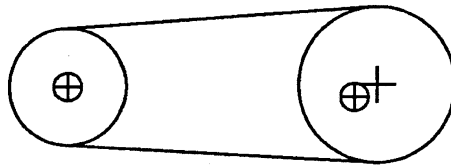


6.02 ECCENTRIC ROTORS

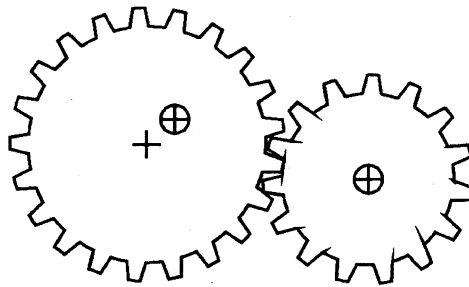
McGraw Hill's "Dictionary of Mechanical and Design Engineering" defines Eccentricity as:

"the distance of the geometric center of a revolving body from the axis of rotation".

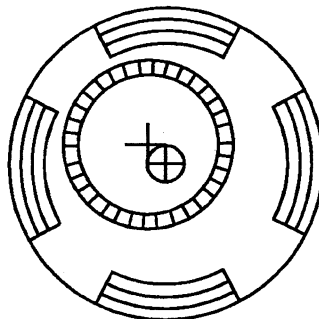
In other words, referring to Figures 6.02A through 6.02C, an eccentric rotor is one in which the shaft centerline does not line up with the rotor centerline. This results in more weight being on one side of the rotating centerline than the other and causes the shaft to wobble in an irregular orbit. This is inherently unstable and can be the source of troublesome vibration. Sometimes, it is possible to "balance out" part of the effect of eccentricity, but much of the displaced motion still remains. In other cases, it is not even possible to perform a good balance on rotors having more eccentricity. Today with the emphasis on higher and higher rotating speeds, it is very important that eccentricity be minimized.



**FIGURE 6.02A
ECCENTRIC SHEAVE**



**FIGURE 6.02B
ECCENTRIC GEAR**



**FIGURE 6.02C
ECCENTRIC MOTOR ARMATURE**

The classic spectrum of an eccentric rotor is shown on Figure 6.02D. Note that, like unbalance, the spectrum is dominated by the frequency at 1X RPM of the eccentric component, particularly when the measurement is taken in the direction through the centers of the two rotors. Looking at Figure 6.02D which shows a motor driving a fan with an eccentric pulley, note that the motor operating speed peak will be much lower than that at fan speed, particularly when taken in line with the belt direction. This eccentricity causes a very highly directional loading so that vibration at 1X RPM can be very much higher in one radial direction than in the other (depending on the amount of eccentricity).

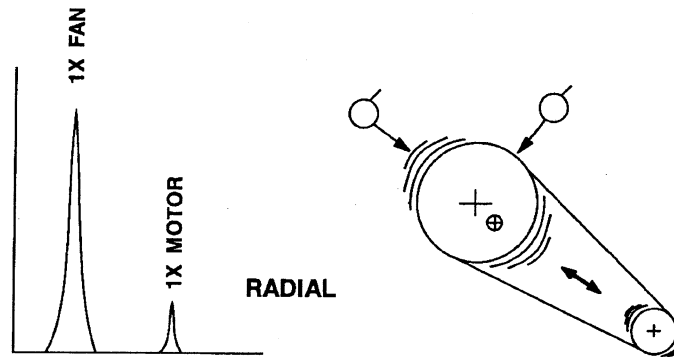


FIGURE 6.02D
TYPICAL SPECTRUM OF AN ECCENTRIC ROTOR

An eccentric rotor exhibits each of the following characteristics:

1. Some of the more common types of eccentric rotors include eccentric pulleys, gears, motor rotors and pump impellers:
 - a. Figure 6.02A shows an eccentric pulley. In these units, the largest vibration most often occurs in the direction of belt tension and at the frequency of 1X RPM of the eccentric pulley. Eccentric pulleys represent one of the most troublesome sources of undesirable vibration in belt drives today. Unfortunately, the industry to date has not sufficiently policed itself to minimize eccentricity in common pulleys. Often, attempts are made to overcome pulley eccentricity after the fact by balancing. Even when this is done, balancing alone will not significantly lower the back-and-forth belt motion which results in continuous belt tension variation, depending on the position of the eccentric pulley at any instant. Plants need to protect themselves by writing eccentricity specifications into their belt drive orders if they want to maximize the life of their machinery and lower their vibration.
 - b. Figure 6.02B shows an eccentric gear in which the largest vibration will occur in a direction in line with the centers of the two gears, and at a frequency of 1X RPM of the eccentric gear. The vibration signature will appear like unbalance of this gear, but it is not. If the eccentricity is significant, it can induce very high dynamic loads on gear teeth as they are forced into and out of a bind with the mating gear. Phase analysis can be used on gears having high 1X RPM vibration to differentiate whether unbalance or eccentricity is the source (See characteristic #3 below). Not only do eccentric gears result in higher 1X RPM vibration, but they also can generate high amplitude gear mesh frequencies and harmonics which will be accompanied by higher than normal amplitude sideband frequencies spaced around the gear mesh frequency at the eccentric gear RPM. Sometimes, these sidebands will be at 2X RPM of the eccentric gear. These sidebands will modulate the amplitude of gear mesh frequencies themselves.

-
- c. Figure 6.02C shows an eccentric motor rotor. Eccentric rotors produce a rotating variable air gap between the rotor and stator which induces pulsating vibration between 2X line frequency (7200 CPM) and its closest running speed harmonic as well as generating pole pass frequency (F_p) sidebands around 2X line frequency (see "Electrical Problem Vibration Symptoms" in Section 6.12). That is, for a 3580 RPM motor, this would be between 2X running speed and 2X line frequency, whereas for a 1780 RPM unit, it would be between 4X RPM and 7200 CPM. Section 6.12 will show that an eccentric motor rotor will also generate pole pass frequency sidebands around 2X line frequency (where pole pass frequency, F_p , equals #Poles times slip frequency). Finally, the eccentric rotor motion itself will cause a variation in the magnetic field between the stator poles and rotor, thereby inducing 1X RPM vibration between the rotor and stator.
- d. Eccentric pump impellers can result in unequal hydraulic forces distributed between the rotating impeller and stationary diffuser vanes. This can result not only in high vibration at pump RPM, but also at vane pass frequency and multiples (# vanes times RPM and multiples) due to a "hydraulic unbalance" induced by the eccentric impeller.
2. Attempts to balance eccentric rotors will often result in reducing vibration in one direction, but increasing it in the other radial direction.
3. Eccentric rotors may cause significantly higher vibration in one radial direction than in the other (as does resonance, wiped bearings and sometimes looseness as well). Phase analysis can be employed as an effective tool to detect whether or not the source of high vibration at 1X RPM is from eccentricity or from another 1X RPM source such as unbalance. Comparative horizontal and vertical phases usually differ by approximately 0° or 180° since the force induced by eccentricity is highly directional (rather than a 90° phase difference in horizontal and vertical as in the case of dominant unbalance problems).

6.03 BENT SHAFT

A bowed or bent shaft can generate excessive vibration in a machine, depending on the amount and location of the bend. Like eccentric shafts, the effects can sometimes be decreased by balancing. However, more often than not, it is not possible to achieve a satisfactory balance in a shaft which has any noticeable bend. Analysts are sometimes successful in removing the bend by various techniques sometimes involving thermal treatments. In these cases, however, one must be careful not to introduce residual stresses which might later lead to shaft fatigue. Bent shafts exhibit the following characteristics:

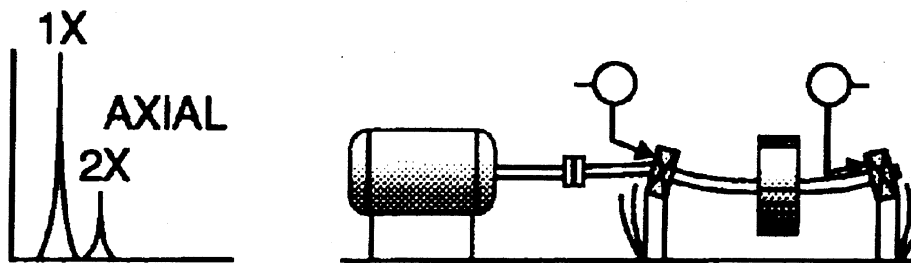
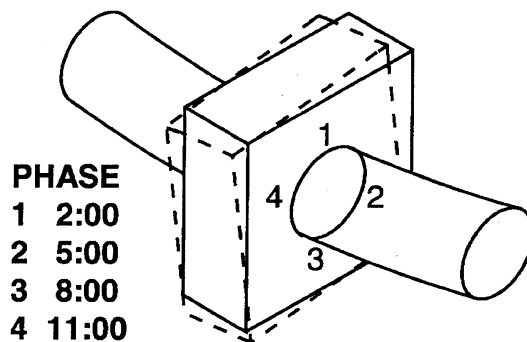
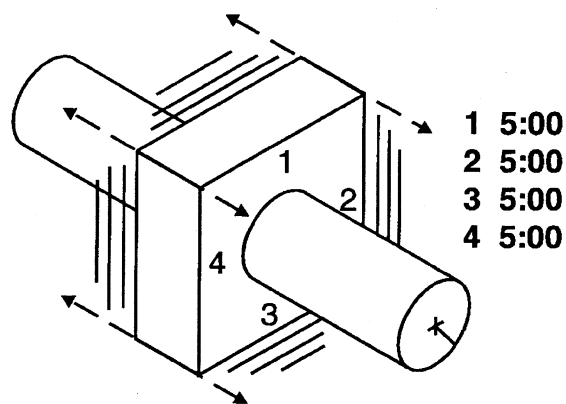


FIGURE 6.03A
BENT SHAFT SPECTRAL AND PHASE RESPONSE

1. Figure 1 shows that high axial vibration is generated by the rocking motion induced by the bent shaft. Dominant vibration normally is at 1X RPM if bent near the shaft center, but a higher than normal 2X RPM component can also be produced, particularly if bent near the coupling.
2. Axial phase change between two bearings on the same component (motor, fan, pump, etc) approaches 180°, dependent on the amount of the bend (as shown in Figure 1). In addition, if one makes several measurements on the same bearing at various points in the axial direction, he will normally find that phase differences approaching 180° occur between that measured on the left and right hand side of the bearing, and also between the upper and lower sides of the same bearing.
3. Amplitudes of 1X RPM and 2X RPM will normally be steady, assuming that 2X RPM is not located close to twice line frequency (7200 CPM) which might induce a beat of the 2X RPM component with 2X line frequency if there is high electromagnetic vibration present.
4. Please note the axial phase measurements on 4 points of a bearing housing pictured in Figure 6.03B. If the shaft is bowed through or very near a bearing, you get a twisting motion by the bearing housing itself which will result in significantly different phase readings on this bearing housing in the axial direction as pictured in Drawing A of Figure 6.03B. Drawing B of this figure shows the axial phase which results from a true, straight shaft.
5. When much run-out is present at the rotating mass, it appears as unbalance. When run-out at the coupling occurs, it appears as misalignment.



A: AXIAL PHASE MEASUREMENTS INDICATING TWISTING MOTION DUE TO A BENT SHAFT



B: AXIAL PHASE MEASUREMENTS INDICATING A TRUE SHAFT IN PROPER MOTION

FIGURE 6.03B

6. In bent shafts, amplitude can vary with the square of speed and preload. If unbalance is more of the problem than bow, vibration will decrease abruptly if operating below the first critical speed. However, if the rotor is brought above its first critical speed, unbalance amplitude will change only a small amount, whereas if the dominant problem is a bent shaft, the amplitude will again drop significantly as the speed is dropped towards the first critical speed.
7. If a rotor is located between bearings and should operate at or close to its fundamental natural frequency, it will appear to be a "bent" shaft and will display these symptoms (see Figure 6.05E in Section 6.05 on "Resonant Vibration"). However, this is only temporary. When the machine is stopped or at another non-resonant speed, it will then "straighten out".
8. When electric motors have problems such as shorted laminations, they will thermally induce a bend as the machine heats up, with the resultant vibration getting higher and higher as the rotor heats. This again will introduce bent shaft symptoms (see Figure 6.09H in Section 6.09 on "Electrical Vibration"). In this case, the shaft again will straighten when allowed to come back to room temperature if the plastic limit of the shaft material has not been exceeded. This will be covered later in the electrical problems Section 6.09.

6.04 MISALIGNMENT

Likely the most widespread mechanical problem in industry today is misalignment. Many plants have begun to resolve a number of their unbalance problems as more and more data collectors having this capability, as well as other analyzers are acquired by plants, and balance actions are taken. However, new instruments are also now becoming available to resolve alignment problems. These include optical as well as the newer laser devices. With these newer instruments, we can now realize that machines have been operating for quite some time with much higher levels of misalignment than had previously been thought. In fact, we are learning that it is not uncommon at all to have 30% to 50% or more of machines in any plant that have high degrees of misalignment.

The trouble with such high levels of misalignment is that it induces high vibration levels leading to premature failure of expensive machine components and increased energy demands as well. Misalignment is now probably one of the leading causes of bearing failures as well.

Although vibration responds to the degree of misalignment, there is not a direct 1-for-1 relationship between the amount of misalignment (angularity and offset) and the amount of vibration. As John Mitchell states on page 182 of Reference 2:

"the vibration characteristics associated with misaligned flexible couplings are not a direct measure of the amount of misalignment but of the coupled system's ability to accommodate misalignment. Thus, the external symptoms of misalignment, in addition to being a function of the offset between shafts, are also affected by speed, torque, or any other condition such as corrosion or sludging which may alter the coupling's stiffness and hence its ability to accommodate a given offset."

The first page of the Vibration Diagnostic Chart (Table 6.0) shows there are 3 types of alignment concerns including angular misalignment, parallel misalignment and a misaligned bearing cocked on a shaft. Each of these will be covered separately along with a section on coupling problems later. First, since misalignment problems are so prevalent today, a number of key facts should be considered about it including what effect it has on component lives; where it directs its potentially harmful forces; what are its spectral characteristics (harmonic content); what are its directional characteristics; what are its phase characteristics; and finally, what should be done to monitor alignment:

1. Component Failures Due To Misalignment - Misalignment can of course cause the coupling to fail, but other machine components as well. For example, if the coupling is stronger than the adjacent bearing, it can subject the bearing to excessive forces with little or no damage to the coupling. Similarly, such misalignment can detrimentally affect other components including gears, belts, sheaves, blading, etc.
2. Reaction On Free (or Outboard) End - It is possible for the highest reaction to misalignment to occur not on the bearing closest to the coupling, but on the free or outboard machine end. In these cases, incoming forces from the coupling may be strong enough to stabilize this system adjacent to the coupling and suppress the symptoms on this end.
3. Axial Vibration - Misalignment normally causes both high axial and radial vibration (as opposed to unbalance which acts mostly in the radial direction with the exception of overhung rotors).

-
4. Other Sources Of High Axial Vibration - Again, while misalignment is probably the most common source of high axial vibration, there are several other sources which can generate it including:
 - a. Bent Shafts;
 - b. Shafts in Resonant Whirl;
 - c. Misaligned Bearings Cocked on the Shaft (See Section 6.043);
 - d. Resonance of Some Component in the Axial Direction;
 - e. Worn Thrust Bearings;
 - f. Worn Helical or Bevel Gears;
 - g. A Sleeve Bearing Motor Hunting for its Magnetic Center;
 - h. Couple Component of a Dynamic Unbalance.

Therefore, when high axial vibration occurs, do not quickly jump to the conclusion that the problem is misalignment. Instead, refer especially to phase; and then to the vibration spectrum.

5. Low Axial Vibration During Misalignment - Although misalignment is classically categorized as having high axial vibration, it does not always occur. For example, the writer has experienced some cases where misalignment was the problem even though axial levels were only about 1/4 of those in the radial direction. This is quite possible for machines with predominately parallel offset versus angular misalignment.
6. Significant Difference in Horizontal And Vertical Amplitudes - Since it has been pointed out that it is possible for a machine to have good horizontal alignment but poor vertical, it is quite possible for misaligned machines to have much higher vibration in one radial direction versus another.
7. Radial Vibration Response To Misalignment - One would think that if driver and driven shafts were horizontally offset, it would cause high horizontal forces. Although this is sometimes the case, Reference 4 states that in most cases, high horizontal amplitudes are primarily the result of vertical misalignment and vice versa.
8. 2X RPM Vibration - Often, misalignment generates a higher than normal 2X RPM vibration which can act not only in the axial direction, but also in the radial. This second operating speed harmonic is caused by asymmetric stiffness in the machine and its supports, or in the coupling. That is, there is often quite a difference in stiffness around the supporting housing, frame, foundation and coupling itself which can allow a "back-and-forth" motion with each revolution, thereby resulting in 2X RPM vibration.
9. Higher Harmonics - Misalignment can also cause large numbers of harmonics which will make the spectrum appear like looseness/excessive clearance problems. The key distinguishing feature still appears to be the high level at 2X RPM in the axial direction. Several tests have been conducted purposely misaligning units and measuring their response (Reference 3). During these referenced tests, multiple harmonics often begin to appear when the misalignment became more and more severe.
10. Phase Is Best Indicator - When high vibration occurs on a machine predominately at 1X RPM and 2X RPM, the best overall indicator of misalignment problems is phase (that is, how the machine is shaking). Phase will differentiate between a number of other potential 1X RPM and 2X RPM vibration sources. Phase behavior in response to misalignment can be summarized as follows:
 - a. Probably the best indicator of misalignment problems is evaluation of phase across the coupling. Here, one is checking how the driver shaft and its coupling half is

reacting relative to the driven shaft with its coupling half. When this phase difference across the coupling approaches 180° ($\pm 40^\circ$ to 50°), misalignment is often indicated, particularly when other misalignment symptoms are present. The higher the severity of misalignment, the closer this difference will approach 180° . Similarly, the less significant are other problems such as unbalance, eccentricity, resonance, etc., the more this difference will approach 180° . Here, it is important that not only vibration spectra, but also phase measurements be captured in horizontal, vertical and axial directions on each of the bearing housings which are accessible.

- b. Since it is possible for shafts to have good horizontal alignment, but poor vertical alignment (or vice versa), it is common in these cases for the horizontal phase difference to be quite different from the vertical phase difference. In fact, this is the case most of the time. In the special case where shafts have good horizontal, but poor vertical alignment, the shaft orbit itself would tend to be greatly elliptical which may make the alignment problem to appear to be eccentricity, resonance or a similar problem. That is, the amplitudes may be greatly different in one radial direction versus another. However, examination of the phase differences throughout the machine will indicate the misalignment problem.
- c. When examining the phase difference on one of the rotors (just the motor, pump, fan, etc), the radial phase differences for significant misalignment will be either 0° or 180° ($\pm 30^\circ$). This is unlike unbalance in which such phase differences could be most anything (i.e., both the horizontal and vertical phase differences might be 70°). The key here is that misalignment phase differences will approach either 0° or 180° .
- d. When comparing horizontal phase differences with vertical phase differences on the same rotor, about 90% of misaligned machines will show a difference approaching 180° between the vertical and horizontal. For example, if the horizontal phase differed about 30° between the outboard and inboard bearings, the vertical phase difference would be about 210° for most misaligned rotors. An unbalanced rotor will not show this phase behavior since whatever phase difference occurs on the horizontal direction will be very close to that in the vertical direction.

In summary, phase data should always be taken if possible on machines having high vibration at 1X and 2X RPM since phase will be the key indicator in differentiating whether the dominant problem source is misalignment as opposed to other problems of similar symptoms. While other symptoms such as high axial vibration and harmonic vibration are also good symptoms, these should not receive as much weight as phase (for example, if phase does indicate misalignment, but axial vibration does not, one should give more weight to the data provided him by phase) if the vibration is high.

- 11. **Effect Of Other Problem Sources** - When other problems such as unbalance, bent shaft, resonance, etc. are present, along with misalignment, this can affect not only the vibration spectrum, but also phase behavior. For example, if both unbalance and misalignment are present, it might show high levels at both 1X RPM and 2X RPM, plus radial phase differences which may or may not approach 150° to 180° , depending on the severity of each problem (in this case, axial phase differences across the coupling will still likely approach 180°). Reference 4 suggests that when several problem sources are all present, each of them will contribute vectorially. That is, if one had polar coordinate graph paper, you might show the contribution of unbalance as a 3 mil level at 30° ; the misalignment a 2 mil level at 60° ; and a simultaneous eccentricity problem contributing 1 mil at 0° . The resultant vector would not show a phase at any of these 3 individual angles, but instead would produce a vector somewhere on the order of 4 mils at about 40° . This would still not be radically different in any of the original phase angles.

However, if each of these 3 were at widely differing phase angles, the resultant phase angle and magnitude could be quite different. In this case, one would first have to solve one problem at a time (in this case, he should likely solve first the eccentricity problem; then the misalignment, and finally balance the rotor). As each problem is solved, the vibration spectra and phase will react accordingly.

12. Monitoring Alignment Change - When aligning especially critical machines, it is often helpful to monitor the alignment and see how it might change. When doing so, it is especially important to monitor phase in all 3 directions at each bearing on each machine involved in the alignment. In most machines that have thermal offsets, if the machine is brought up to speed from room temperature, it should display misalignment symptoms in the beginning which should disappear as the machine comes up to full operating temperature. For example, the phase difference across the coupling should initially be on the order of 150° to 210°, but should drop to close to 0° to 30° in the end. In addition to phase, one should monitor how the vibration spectrum changes as well as other non-vibration related variables such as bearing temperature, temperature of the support legs and oil film pressure. When monitoring alignment change, each of the following should be considered:
- a. 1X RPM - It might be better to monitor vibration at higher harmonics of 2X up to 4X RPM rather than 1X RPM since the first harmonic will be effected by so many other things (unbalance, resonance, eccentricity, bent shaft, etc).
 - b. 2X RPM - The 2X component should be a much better indicator of alignment than 1X assuming that this is not a 3600 RPM nominal motor (if so, 2X RPM will likely be very close to 2X line frequency which will "contaminate" the apparent 2X amplitude unless one is able to separate 2X RPM from 2X line frequency).
 - c. 3X RPM - This 3X component may be the best indicator of alignment change if this particular machine does in fact cause an increase at 3X RPM with an alignment change. This is often the case. In such cases, it is not necessary for the 3X component to be larger than either 1X or 2X RPM, just that it be sensitive itself to alignment change.
 - d. 4X RPM - The same analogy applies to 4X RPM as does 3X RPM with the exception being that this is not an 1800 RPM nominal speed machine (in which case 4X RPM would closely approach 2X line frequency at 7200 CPM).
 - e. Number of Coupling Grids (or Segments) X RPM - Some coupling types include a number of grids or segments which often cause vibration at the number of grids (or segments) on one coupling half times the RPM, particularly when misalignment becomes severe. In these cases, where the coupling components themselves are effected and do respond to misalignment, this frequency will be an excellent choice to monitor since it will be well removed from any effects of unbalance, bent shaft, eccentricity or any other such source other than alignment.

6.041 Angular Misalignment:

Angular misalignment is pictured in Figure 6.04A. Each of the following characteristics are demonstrated by angular misalignment:

- 1. Angular misalignment primarily generates high axial vibration, particularly at 1X and 2X RPM. However, it is not unusual for either one of these peaks (1X, 2X or 3X RPM) to dominate alone.

2. However typically, when the amplitude of either 2X RPM or 3X RPM exceeds approximately 30% to 50% of that at 1X RPM in the axial direction, angular misalignment is indicated. This assumes that there is high vibration (that is, misalignment may be of concern if 1X = .30 ips and 2X = .20 ips; but not if 1X = .03 and 2X = .02 ips).
3. Angular misalignment is best detected by 180° phase change across the coupling in the axial direction as pictured in Figure 6.04A. If each of the bearings on one of the side are moving one way, while those on the other side are moving in the opposite direction, angular alignment is highly suspect.

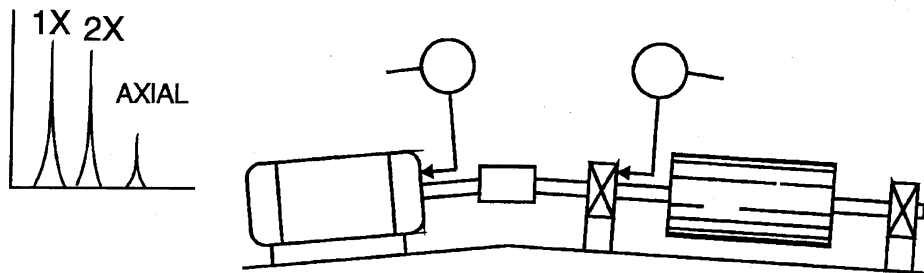


FIGURE 6.04A
ANGULAR MISALIGNMENT SPECTRAL AND PHASE RESPONSE

6.042 Parallel Misalignment (Also known as Radial Offset Misalignment):

Radial misalignment is pictured in Figure 6.04B. It displays each of the following characteristics:

1. Parallel misalignment primarily affects radial vibration as opposed to angular which affects axial.
2. Like angular alignment problems, parallel misalignment causes phase to approach 180° difference across the coupling, but in the radial direction (horizontal or vertical).
3. Radial misalignment is often indicated in a spectrum when 2X RPM exceeds approximately 50% of the amplitude at 1X RPM, but its height relative to 1X RPM is often dictated by the coupling type and construction. It is not uncommon for 2X RPM to exceed that at 1X RPM, particularly when the parallel misalignment becomes severe.
4. When either angular or parallel misalignment becomes severe, each can generate an array of harmonics ranging up to and including the 4th through the 8th harmonic. In this case, the severe misalignment spectrum can appear to be mechanical looseness (see page 1 of Table 6.0 Diagnostics Chart).

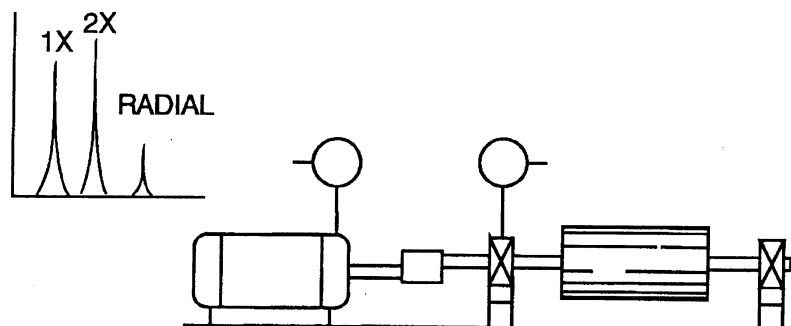
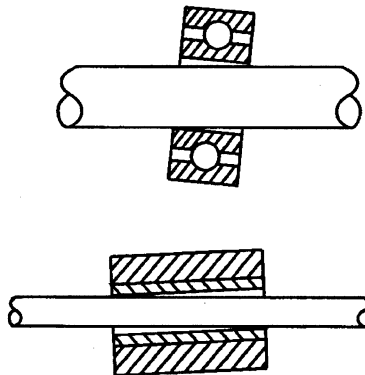


FIGURE 6.04B
PARALLEL MISALIGNMENT SPECTRAL AND PHASE RESPONSE

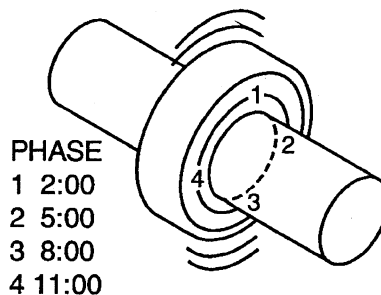
6.043 Misaligned Bearing Cocked on the Shaft:

When either a sleeve or rolling element bearing is misaligned or cocked on the shaft, it can cause high vibration and unusual loading. If it is detected, it should quickly be resolved before causing premature component failures. This problem is pictured in Figure 6.04C. Each of the following characteristics are indicative of a misaligned bearing on a shaft:

1. A cocked bearing will normally generate considerable axial vibration which can affect not only that at 1X RPM, but also 2X RPM as well.
2. If phase is measured in the axial direction at each of 4 points 90° apart from each other as shown in Figure 6.04D, a cocked bearing will be indicated by a 180° phase shift from top to bottom or from side to side.
3. Attempts to align the coupling or balance the rotor will not alleviate the problem. The effected bearing must be removed and correctly installed.



**FIGURE 6.04C
MISALIGNED BEARINGS COCKED ON SHAFT**



**FIGURE 6.04D
AXIAL PHASE MEASUREMENTS INDICATING
A COCKED BEARING ON A SHAFT**

6.044 Coupling Problems:

It is often difficult to tell from vibration signatures or phase analysis whether the problem is misalignment or a coupling problem. Each of the many types of couplings has a different effect on the response of the machinery to which it is coupled. Other factors affecting its response include spacing between shafts, shaft diameter and bearing type. However, problem couplings do display the following characteristics:

1. **3X RPM** will often respond to a coupling problem, particularly for a coupling having a spacer that is too long or too short. In these cases, the radial spectrum will normally indicate a fairly noticeable 3X running speed vibration, and that in the axial direction will show a much higher 3X RPM component. These problems are resolved by either correctly sizing the spacer or repositioning either the driver or driven equipment.
2. Gear Type Couplings can experience coupling lockup where the frictional force developed at gear teeth is greater than the applied force causing the coupling to become a rigid member. Friction welding of teeth can occur at this point, particularly if there is a lack of lubrication. A locked coupling can cause severe problems and may lead to thrust bearing failure if it results in the thrust load of 2 machines being applied to only one thrust bearing. Also, if the teeth do weld together and then break loose, it leaves pit marks on the coupling teeth. Coupling lockup can be broken temporarily either due to a change in load or by striking the coupling with a mallet or a piece of wood. However, this coupling should be closely inspected as soon as possible looking for tooth damage, lubrication problems and alignment problems, replacing if necessary.

Coupling lockup will normally cause an increase in both axial and radial vibration with axial vibration normally being higher. Most of the time, the 1X RPM is most effected. However, certain types of couplings will generate a frequency distribution resembling a "Christmas Tree" effect. In these cases, many harmonics can appear with the vibration dropping approximately 25% from one harmonic to the next (Reference 5). What gives the spectrum a "Christmas Tree" effect is that there is a fairly uniform drop of about 25% all the way from the 2nd through the 5th or 6th harmonic.

3. A loose coupling is likely to cause sidebands around blade pass frequencies (#blades X RPM) and mesh frequencies (#teeth X RPM) as shown in Figure 6.04E (however, sidebanding of blade pass and mesh frequencies does not always indicate a loose coupling). This is caused by the fact that a loose coupling does not drive the rotating equipment at a uniform speed, but rather, pulses at multiples of the shaft speed causing its running speed to modulate these other frequencies. Therefore, a signal similar to that shown in Figure 6.04E with equally spaced sidebands at coupling RPM can mean the coupling is loose (either from a poor fit on the shaft or from worn coupling components).

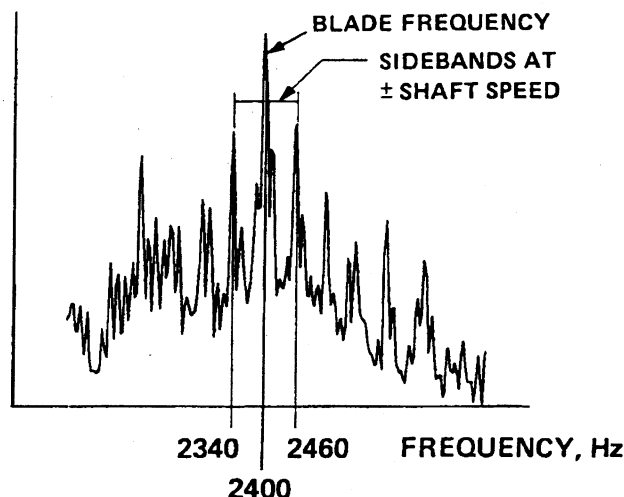


FIGURE 6.04E
LOOSENESS OF COUPLING INDICATED BY COUPLING RUNNING SPEED
SIDEBANDS ABOUT BLADE PASS FREQUENCY

6.05 NATURAL FREQUENCIES AND RESONANCE

Natural frequencies and resonance are concepts that must be covered in any type of vibration course. The two are related; however, natural frequencies exist for every object in the universe while resonance is a special condition where some outside force is exciting a natural frequency. Natural frequencies, by themselves, are not harmful to a machine. Resonance, which does not always occur, can be extremely harmful to an object, such as a machine or support structure, and can determine the rate of failure of a machine's components. Also, a resonant condition can affect the performance of machinery and lower product quality. More often than not, natural frequency and resonance are discussed with an emphasis on mathematical theory rather than from a practical application viewpoint. This section will address these concepts as they pertain to condition monitoring.

6.051 Natural Frequency

Definition

The SHOCK AND VIBRATION HANDBOOK¹ defines Natural Frequency as "the frequency of free vibration of a system". In other words, when an object is standing still and is impacted with one short burst of energy, the object will vibrate at a series of frequencies which are known as its natural frequencies. Every object has an infinite number of natural frequencies and each of these natural frequencies has a corresponding physical shape associated with it called a "mode shape". An example of the first few natural frequencies and corresponding mode shapes for a shaft supported by bearings on each end (pinned-pinned beam) are given in **Figure 6.05A**. **Figure 6.05A** only shows the first 3 modes. In reality, there are an infinite number of natural frequencies and mode shapes, each higher in frequency than the last and with one more "hump" than the last. Note from the graphical representation of the mode shapes that the higher modes have less deflection from the shaft's position at rest. Therefore, less stress is typically induced in the shaft and bearings if a higher mode of vibration is excited.

Directionality

Not only does the shaft in **Figure 6.05A** have an infinite number of natural frequencies in the direction shown (in the plane of the page), it also has an infinite number of natural frequencies in the direction perpendicular to those shown in its horizontal direction (into and out of the page), as well as along the direction of the shaft in its axial direction (shortening and elongating the shaft). Each of these natural frequencies is unique (i.e. - they all occur at different frequencies).

Identification

The natural frequencies of an object can be identified by a variety of methods that are taught in later courses. However, a brief description of each will be given in this section.

- 1) **IMPULSE NATURAL FREQUENCY TEST** - An impulse natural frequency test is conducted by beginning with the machine at rest. An accelerometer is placed on a component of the machine to determine its natural frequencies. The machine is then impacted in the same or opposite direction while a FFT device (Data Collector or RTA) records the response of the accelerometer. Peaks that show up on the frequency spectrum are probable natural frequencies in the direction of the accelerometer. The test is repeated at many different locations and in all three spatial directions (horizontal, vertical and axial) to detect as many natural frequencies as possible.

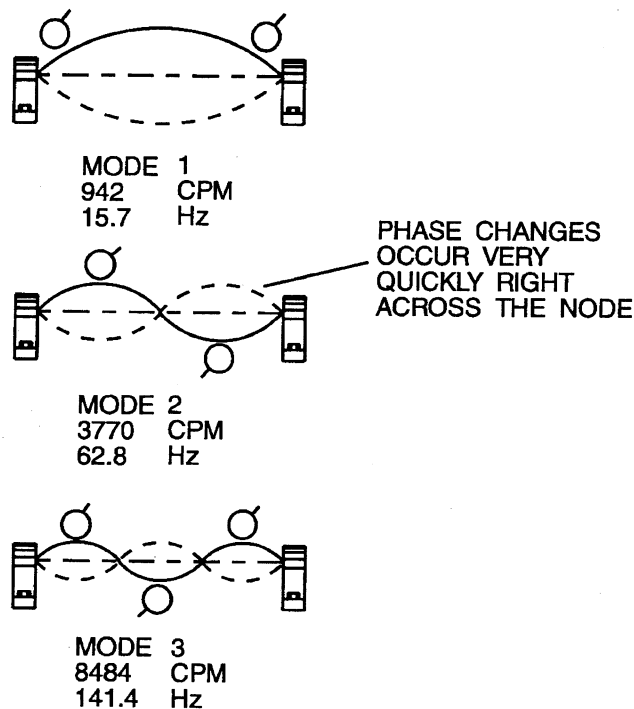


FIGURE 6.05A
MODE SHAPES DURING RESONANCE OF A SIMPLY SUPPORTED BEAM
(SHAFT BETWEEN TWO BEARINGS)

- 2) **COASTDOWN or RUNUP TEST** - A coastdown test is conducted by placing the accelerometer on a machine component while the machine is running and then turning off the machine and letting it coast to a stop. An FFT device (Data Collector or RTA) records a series of spectra as the machine coasts down. Peaks will occur in the spectrum at suspected natural frequencies when the 1X RPM (unbalance force) excites them. Such natural frequencies can also be excited by other forcing frequencies as the machine coasts to a stop (2X RPM, blade pass frequency, gear mesh frequency, etc.). It is important to remember that a peak in the spectrum does not necessarily mean that a natural frequency exists. It could be that the speed of the rotor at that time equals the speed of another machine running nearby and a "beat" is formed which increases the amplitude at that particular frequency. To verify that a natural frequency is encountered, a Bode' plot can be run. A Bode' plot is similar to a coastdown plot except that phase is measured as well as the amplitude. A natural frequency is identified by a 90 degree phase shift as well as an increase in the amplitude. As the machine continues to coastdown, the phase will shift another 90 degrees for a total 180° phase shift when it passes through resonance and the amplitude will drop. **Figure 6.05B** is a plot of a coastdown and **Figure 6.05C** is a Bode' plot of the same coastdown. Note that the peak to the right is not a natural frequency since the Bode' plot does not show a phase shift. The peak to the left is a natural frequency, since the Bode' plot exhibits the 90 degree phase change at the natural frequency, and then continues onto its full 180 degree phase change.

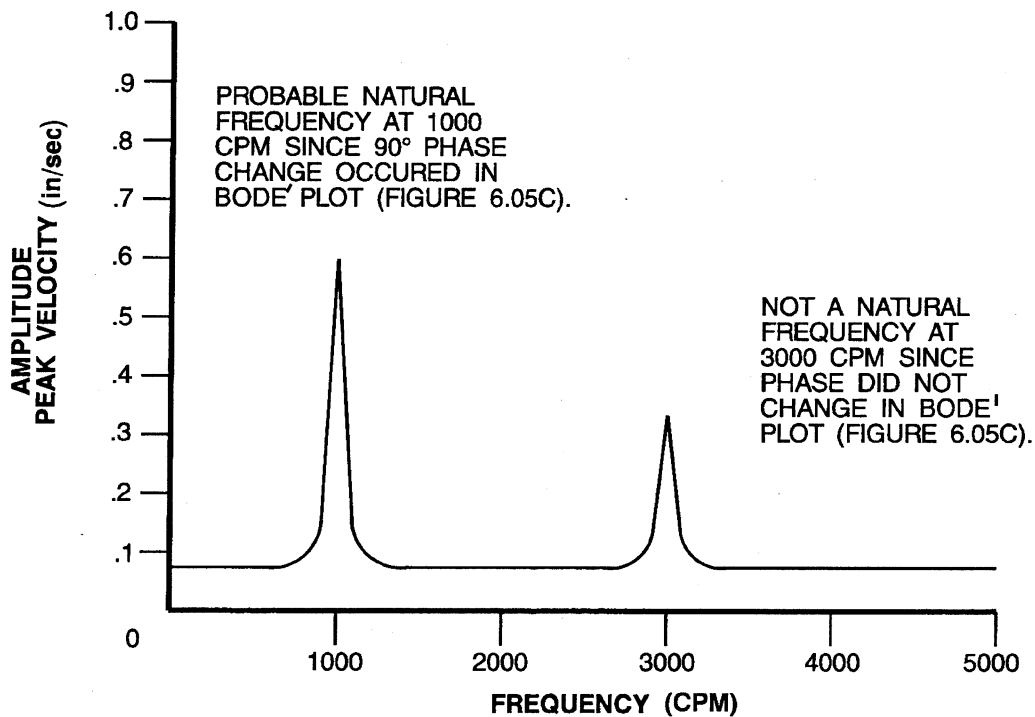


FIGURE 6.05B
COASTDOWN PLOT OF A MACHINE
(BODE' PLOT IS SHOWN IN FIGURE 6.05C)

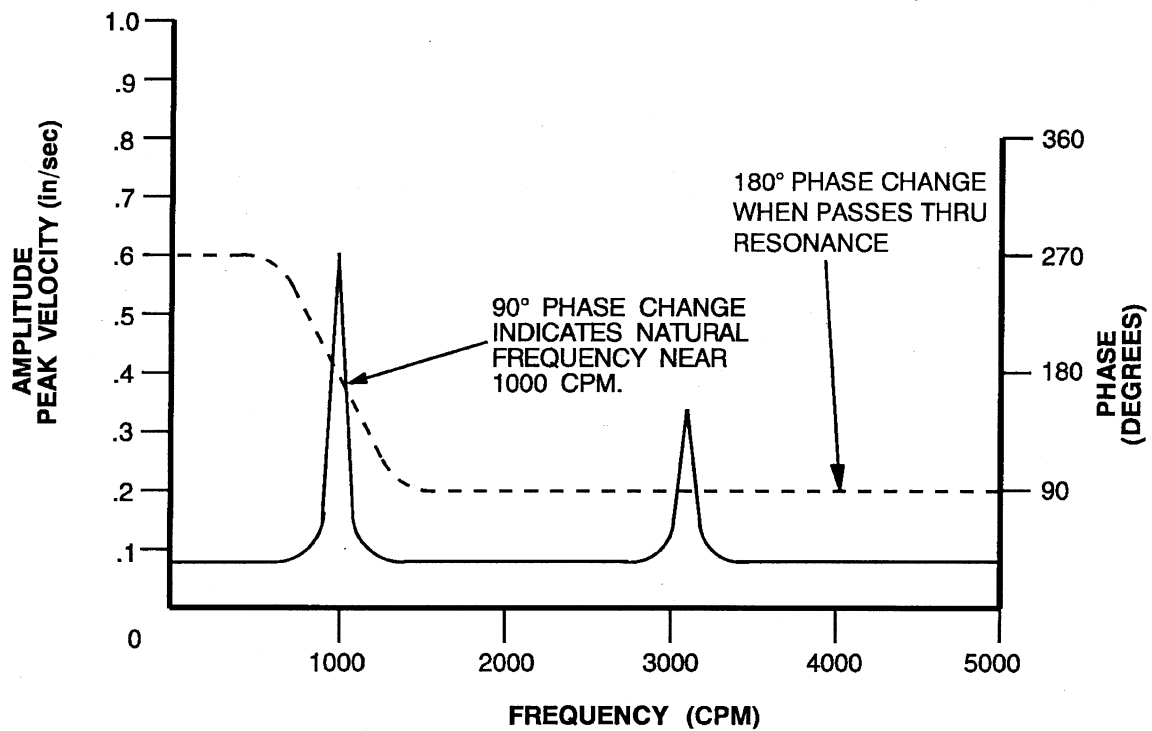


FIGURE 6.05C
BODE' PLOT OF SAME MACHINE IN FIGURE 6.05B

A runup test is identical to the coastdown except that the machine begins at rest and then is brought up to speed. Vibration and phase (Bode) measurements are taken during this runup and are plotted similar to the coastdown to determine the natural frequencies of the system.

- 3) **OTHER METHODS-** Other methods for determining natural frequencies include shaker tests, modal analysis, and finite element modeling. Shaker tests are conducted by attaching a variable speed motor with a preset unbalance to an object. The unbalance will force the object to which it is attached. Vibration response is monitored as the speed of the motor is "swept" through its range of frequencies. A high amplitude peak as well as a 90° phase change continuing on to 180° indicates a natural frequency. Modal analysis utilizes a force hammer (the input force can be recorded) a transducer, and a FFT device for recording the data. A computer graphical representation of the object being tested is created. The force input as well as the objects response to an impact from the hammer are recorded. The modal software computes the mode shapes and natural frequencies and displays them on the computer in an animated fashion. A finite element model consists of purely mathematical information to approximate the object's mode shapes and natural frequencies. Information such as the objects physical properties (mass, stiffness, etc...) and geometry are input into a finite element software program. The program then computes the estimated mode shapes and natural frequencies and displays them in an animated fashion on the computer. Note that the finite element method does not require any "field" information to be collected from the object, therefore, this method is very useful in estimating natural frequencies of objects in a design environment before the object is actually built.

Harmfulness

There is nothing harmful about a natural frequency unless a resonance occurs at a particular natural frequency.

6.052 Resonance

Simply speaking, resonance is a condition that occurs when a force is applied to an object at a particular frequency that is very near or equal to a natural frequency of the object. Also, the force or part of the force must be in the same direction as the natural frequency. The result of a resonance condition is a terrific increase in amplitude (10x - 30x higher) at the frequency that is resonant. The applied force may be from a number of sources. As was learned earlier in **Chapter 6**, unbalance creates a force at 1X RPM, misalignment creates a force at 1X, 2X, and 3X RPM, many other problems such as eccentricity or bent shaft create forces at 1X RPM, bearing problems create forces at noninteger harmonics of 1X RPM (i.e. - 4.64X RPM, 8.33X RPM, etc.), even motor electrical problems cause forces at frequencies such as 7200 CPM or at rotor bar pass frequency. If the frequency of any of these forces is equal to a machine's natural frequency and in the same direction, a resonance condition will be created. The result will be an increase in the vibration amplitude that would normally be generated by the problem on its own. It is important to remember 3 concepts:

- 1) A natural frequency that does not have a forcing frequency near it will not be excited (i.e. - no peak will be found in the frequency spectrum).
- 2) A problem with the machine will show up in the frequency spectrum and will have an amplitude that is based on the amount of force generated by the problem and the stiffness of the machine that is restraining the force provided that no natural frequency is present at that problem frequency.
- 3) If a problem occurs and its frequency is the same as one of the machine's natural frequencies, the amplitude in the frequency spectrum will be 10X to 30X higher than described in number 2 for the particular problem.

6.06 MECHANICAL LOOSENESS

Many texts and seminars today simply refer to a general term called "Mechanical Looseness" and that it either can be detected by high vibration at 2X RPM or multiple running speed harmonics. However, a comprehensive study of a great amount of available literature as well as review of a number of case histories accomplished by Technical Associates actually has revealed that there are at least 3 different types of Mechanical Looseness, each of which has its own characteristic vibration spectra as well as vibration phase behavior. A discussion will follow on each of these which are as follows:

6.061 *Type A* - Structural Frame/Base Looseness (Primarily 1X RPM);

6.062 *Type B* - Looseness Due To Rocking Motion Or Cracked Structure/Bearing Pedestal (Primarily 2X RPM);

6.063 *Type C* - Loose Bearing In Housing or Improper Fit Between Component Parts (Multiple Harmonics Due To Nonlinearity Often Induced By Impulse Events)

One of the important facts about each type of mechanical looseness is that it alone is not a cause of vibration. Instead, looseness is a reaction to other problems which are present such as unbalance, misalignment, eccentricity, bearing problems, etc. Resolution of these other problems often will remove many of the symptoms, and therefore the response of looseness. However, the problem is that only minute amounts of such problems as misalignment or unbalance can cause vibration if a looseness condition exists. Looseness aggravates the situation. Therefore, mechanical looseness allows much more vibration than would otherwise occur from these other problems alone. Resolution of the other problems themselves will often remove most of the symptoms due to looseness. However, this is often virtually impossible in reality because such steps would require extraordinary levels of precision of alignment or balancing. Therefore, in these cases, the looseness condition will first have to be resolved. Then, if remaining vibration is still high, other steps such as alignment and balancing can be accomplished with much greater ease than before the looseness conditions were resolved. Following below is a discussion on each of the 3 types of looseness which were listed previously:

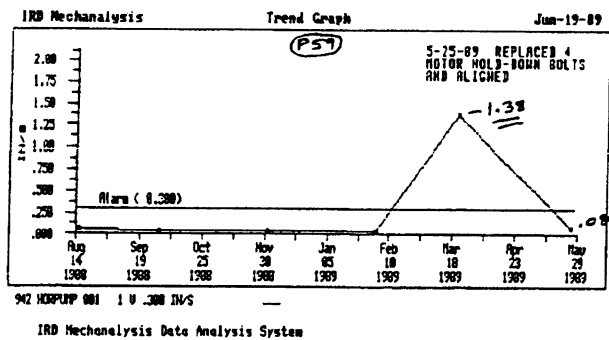
6.061 *Type A* - Structural Frame/Base Looseness (1X RPM):

This type of looseness includes each of the following problems:

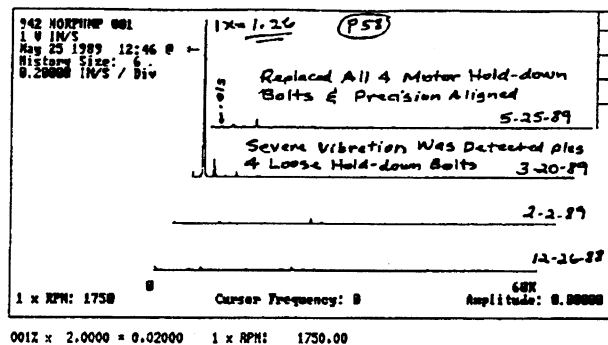
- Structural Looseness/Weakness of Machine Feet, Baseplate & Concrete Base;
- Deteriorated or Crumbled Grouting;
- Distortion of Frame or Base (Soft Foot);
- Loose Hold-Down Bolts.

Type A Looseness Problems are often misdiagnosed as unbalance or misalignment problems since they have almost identical vibration spectra. Therefore, it is important to look beyond vibration spectra and compare relative amplitudes between directions, look closely at phase behavior which departs radically from such things as unbalance, and to examine other characteristics which are listed as follows:

1. Type A Looseness spectra are dominated by high 1X RPM vibration and appear identical to that for an unbalance or eccentric rotor condition. A spectrum illustrating this type signature is shown in Figure 6.06A.



Machine	942 HORPUMP 001	Pos	1	Dir	V	Units	IN/S	Alarm	.300
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
Code	mm/dd/yy			Code	mm/dd/yy				
.062	0	08-14-88	10:01						
.054	0	08-15-88	13:00						
.042	0	09-29-88	11:03						
.038	0	12-01-88	9:49						
.041	0	02-02-89	15:00						
1.38	0	03-20-89	13:43						
.080	0	05-25-89	12:46						



Machine	942 HORPUMP 001	Pos	1	Dir	V	Units	IN/S	Date	May 25 1989 12:46
Frequency	1 x RPM	Amplitude							
3600	2.06	0.02478							
7200	4.11	0.05782							
3X x 2.0000	= 0.06000	1 x RPM:	1800.00						

Machine	942 HORPUMP 001	Pos	1	Dir	V	Units	IN/S	Date	Mar 20 1989 13:43
Frequency	1 x RPM	Amplitude							
1800	1.00	1.26005							
3450	1.72	0.12994							
001X x 2.0000	= 0.02000	1 x RPM:	1750.00						

Machine	942 HORPUMP 001	Pos	1	Dir	V	Units	IN/S	Date	Feb 02 1989 13:00
Frequency	1 x RPM	Amplitude							
21600	12.34	0.03940							
1X x 2.0000	= 0.02000	1 x RPM:	1750.00						

Machine	942 HORPUMP 001	Pos	1	Dir	V	Units	IN/S	Date	Dec 26 1988 10:01
Frequency	1 x RPM	Amplitude							
150	0.09	0.02542							
21600	12.34	0.02478							

FIGURE 6.06A
SEVERE LOOSENESS INTRODUCED BY LOOSE HOLD-DOWN BOLTS

2. Often, high vibration is pretty well confined to only one rotor (that is, the driver or driven component or the gearbox alone). This is unlike unbalance or misalignment in which rather high vibration levels due to these other problems are not confined to just one of the rotors.
3. Two different phase behaviors can occur with Type A Looseness:
 - a. When comparing vertical and horizontal phase on each of the bearing housings, the vibration will sometimes be found to be highly directional with phase differences of either 0° or 180° depending on whether or not the horizontal reading was taken on one side or the other (either a phase difference of 0° or 180° means that the motion is directly up and down or side to side). This does not normally occur with simple unbalance in which horizontal and vertical phase usually differs approximately 90° ($\pm 30^\circ$).
 - b. When this first phase behavior occurs (0° or 180° phase difference in horizontal and vertical), the analyst should not confine his measurements to the bearing housings alone, but move on down to the machine foot, baseplate, concrete base and surrounding floor. This is illustrated in Figure 6.06B. Here, comparative amplitude and phase measurements should show relatively identical amplitude and phase at 1X RPM at each location. If there is a great difference in amplitude and phase, this will suggest relative motion. Using the point where this great phase change occurs, one can locate where the problem exists. For example, the measurements in Figure 6.06B show a problem between the baseplate and concrete base indicated by the great difference in phase (note 180° out of phase with the other two measurements). This indicates structural looseness/weakness allowing relative movement in machine components which may be due to a problem with the grouting between the baseplate and concrete base; or broken or cracked foundations, etc. On the other hand, if a great phase difference occurred between machine foot and baseplate, this might suggest looseness of the mounting bolt and/or possible stripped mounting bolt threads. Either of these two problem conditions can cause a great vibration at 1X RPM like that shown in Figure 6.06A on the machine component where this occurs, particularly on the bearing housing directly above this base location.

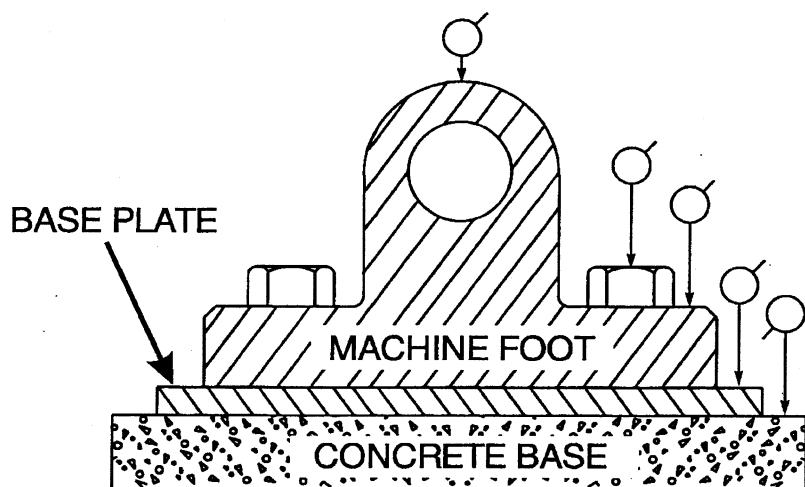


FIGURE 6.06B
PHASE ANALYSIS USED TO PINPOINT LOOSENESS

4. Figure 6.06C illustrates another case involving this high 1X RPM looseness vibration. In this case, bolts that are parallel to the pump shaft are mounted on each pump bearing housing acting in the axial direction. In this case, if these bolts become loose, it will cause a high vibration in the axial direction at 1X RPM which will closely resemble a misalignment problem. However, simply tightening these bolts will greatly reduce the vibration.

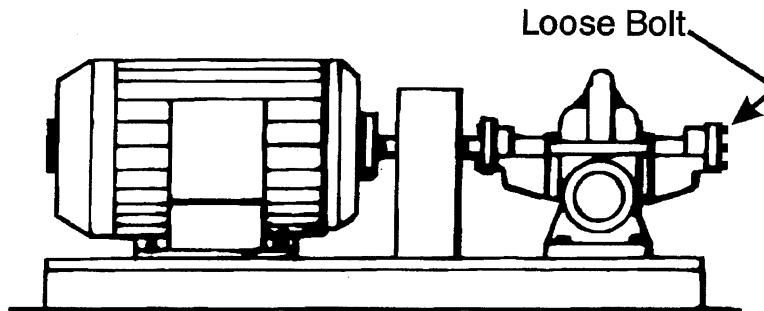


FIGURE 6.06C
EFFECT OF LOOSE BOLT ON PUMP OUTBOARD BEARING HOUSING

5. Distortion induced either by a soft foot or piping strain shows another situation which would result in high 1X RPM vibration signatures looking like unbalance. However, in these cases, when phase readings are taken, they will show highly directional vibration with the difference in horizontal and vertical phase approaching either 0° or $180^\circ (\pm 30^\circ)$ rather than 90° in the case of simple unbalance. If the problem were distortion rather than looseness, amplitude and phase measurements would show the machine foot, baseplate and concrete base pretty much vibrating in the same direction (equal phase readings). However, they may show that the amplitude on either one of the foot bolts alone is much higher than that on any of the other 3 bolts; or, for example, the right front foot and left rear foot may be significantly higher than the left front and right rear feet. In this second case, the motor would be vibrating diagonally. Great amplitude differences on these foot bolts would suggest a soft foot which must be corrected to reduce the high vibration levels. (In fact, the analyst may find that if he backs off slightly on those bolts having high vibration, he may see much lower vibration than before).

6.062 Looseness Due To Rocking Motion or Cracked Structure/Bearing Pedestal (2X RPM):

The 2X RPM "looseness symptom" referenced in many vibration texts seems only to occur for the following looseness problems:

- o Crack in Structure or Bearing Pedestal;
- o Rocking Motion Sometimes Induced By Differential Length Support Legs;
- o Occasionally on Some Loose Bearing Housing Bolts;
- o When Loose Bearing or Improper Component Fit Problems Are Only of Minor Severity (No Impulse or Impact Events).

Figure 6.06D shows a spectrum typically showing these problems. These problems exhibit the following characteristics:

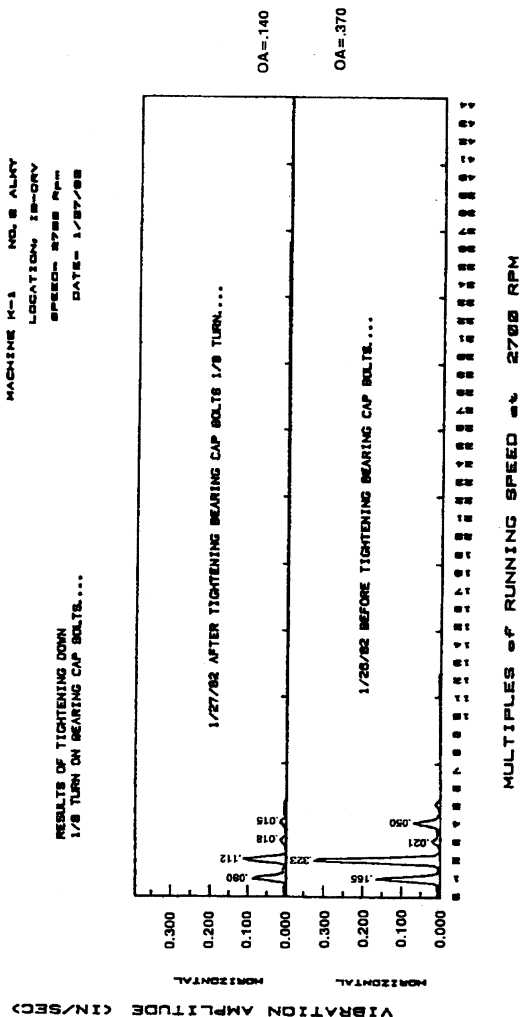
1. Typically, these problems are suggested when the amplitude at 2X RPM exceeds about 50% of that at 1X RPM in the radial direction.
2. Amplitudes are somewhat erratic.

3. If phase readings are taken with a strobe light, it will often show 2 reference marks which are slightly erratic.
4. These looseness symptoms will not normally occur unless there is some other exciting force such as unbalance or misalignment. However, if this looseness condition exists, it will be extremely difficult to balance or align the unit sufficiently to bring down the final vibration sufficiently.
5. If the looseness problem is a bearing loose in the housing or a loose component on the shaft, the vibration will pretty well remain at 1X and 2X RPM until it worsens allowing an impulse or impact event. When this occurs, these impulses cause nonlinearities in the time waveform which will begin exciting many harmonics advancing to Type C Looseness.

CASE VI

This case is on our K-1 refrigeration compressor located at our Alkylation Plant. The turbine and compressor both run between 2,700-3,100 RPM requiring 6,000 HP. In this signature (Figure 13) there is a high two times running speed frequency showing up on the turbine inboard. The vibration in this case was suspected to be caused by a

looseness problem between the bearing and the bearing housing. The bearing housing nuts were found to be slightly loose and they were tightened. The signature also shows the level had dropped significantly after the housing was tightened; a slight misalignment problem might still be indicated.



Vibration signatures of Turbine 1B bearing showing highest level at two times running speed. Bottom signature is before bearing cap is tightened down and top is after.

FIGURE 6.06D
EXAMPLE OF LOOSENESS OF FASTENERS OCCURRING AT 2X RPM
(Ref. 17)

6.063 Loose Bearing In Housing or Improper Fit Between Component Parts (Multiple Harmonics Due To Nonlinearity Often Induced By Impulse Events):

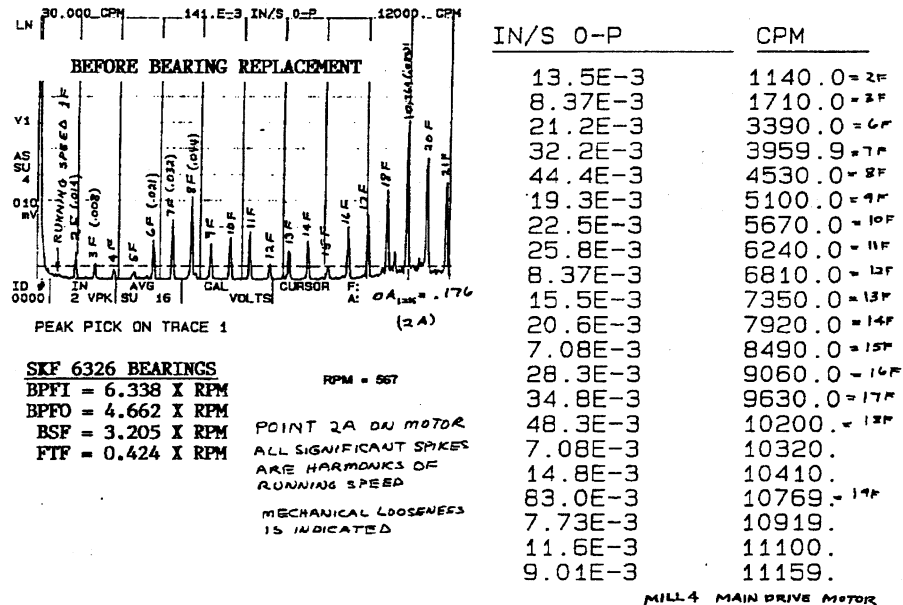
Each of the following problems occur in Type C Looseness:

- Bearing Loose in Housing;
- Excessive Internal Bearing Clearances;
- Bearing Liner Loose In Its Cap;
- Loose Rotor;
- Bearing Loose and Turning on Shaft.

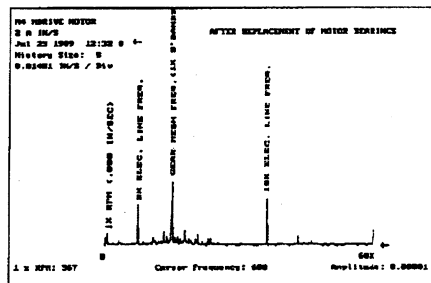
Figures 6.06E thru 6.06G illustrate typical spectra indicating Type C Looseness. Note the presence of multiple running speed harmonics in both spectra. Also note in Figure 6.06G that one of the running speed harmonics lies close to a natural frequency causing a resonant response at this frequency (if the looseness condition is resolved, the resonance condition will likely be as well). **Type C is the most common mechanical looseness problem** and exhibits the following characteristics:

1. Multiple running speed harmonics sometimes up to 10X or even 20X RPM are clearly present in this spectrum. These harmonics are a result of impulses and truncation (limiting) in the machine response. This impulse event causes a nonlinearity in the time waveform. When this occurs, multiple harmonics will appear in the resulting FFT spectrum (see Figure 6.06E).
2. This looseness tends to produce vibration that is directional which differs from unbalance. It normally will be highest in the direction and vicinity of the looseness problem. For example, this may show that the highest vibration is not either horizontal or vertical, but somewhere in between the two.
3. If the amplitude of harmonics becomes significant, this can also generate frequencies spaced at 1/2 times RPM (that is, .50X, 1.50X, 2.50X, etc) or even sometimes at 1/3 times RPM.
4. The analyst is cautioned that amplitudes of these 1/2 times RPM "harmonics" may appear deceptively low when compared with those at 1X RPM and running speed harmonics. However, he should remember that no peaks at 1/2 times RPM intervals should be present whatsoever. If the peaks are clearly evident, they do indicate a more advanced looseness problem (or possibly, presence of a rub).
5. One half times RPM "harmonics" usually are accompanied by other problem sources such as unbalance and misalignment.
6. Phase measurements of Type C looseness problems are normally somewhat erratic, but can approach differences of 0° and 180° between horizontal and vertical directions if the vibration itself becomes highly directional. It normally acts in a radial direction, but can occur in the axial, dependent on the exact type of looseness.
7. In the case of a loose rotor such as a loose pump impeller, phase will vary from one startup to the next. The amplitude itself may be steady for a given run, but likewise will vary from startup to startup. Such a loose rotor is impossible to balance since the heavy spot itself is constantly changing directions. This shift in amplitude and phase is likely caused by center of gravity shifts.

8. **CAUTION:** Vibration spectra simply appearing to be Type C looseness (many 1X RPM harmonics) can, in fact, signal a problem of much greater severity - a bearing loose and turning on a shaft. This can be the case even if amplitude of 1X RPM and its harmonics are fairly low, on the order of .05 in/sec or less. In these cases, the turning of the bearing on the shaft can actually cause great damage, actually removing material from the shaft diameter. In these cases, it can cause catastrophic failure of the machine as the bearing finally locks, and can do so without even generating any bearing defect frequencies. Figure 6.06E below is an example of just such a catastrophic failure where over .25 inch of the shaft diameter was removed from a 3.50 inch shaft before the bearing locked up.



AFTER REPLACEMENT OF MOTOR SHAFT AND BEARINGS
OVERALL = .059 in/sec

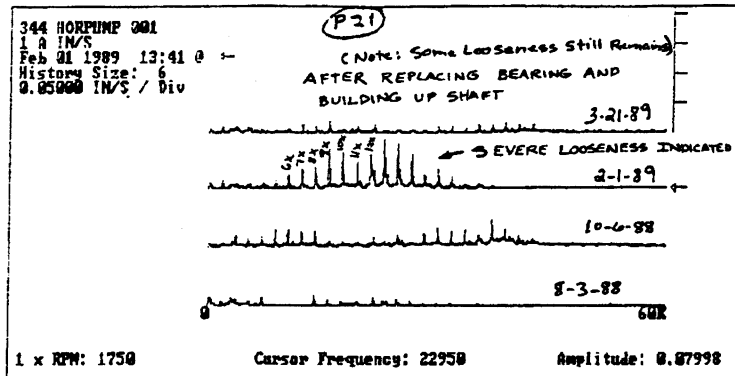


Machine		Pos	Dir	Units	Date	
M4 MAIN DRIVE MOTOR		2	A	IN/S	Jul 25 1989 12:32	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude	
150	0.26	0.00350	36000	82.49	0.03050	
600	1.06	0.00801	43200	76.19	0.00648	
7200	12.70	0.02701	59850	105.56	0.01048	
10500	18.52	0.00800				
10950	19.31	0.00400				
12900	22.73	0.00950				
13500	23.81	0.00648				
14700	25.93	0.04149				
15300	26.98	0.00550				
15900	28.04	0.00448				
16500	29.10	0.00448				
17550	30.95	0.01048				
18450	32.54	0.00448				
19950	35.19	0.00448				
20550	36.24	0.00750				
21800	40.21	0.00499				
23400	41.27	0.00499				

T - Threshold adjust Q - Quit

FIGURE 6.06E

BEFORE & AFTER REPLACEMENT OF A DC MOTOR BEARING WHICH WAS TURNING ON THE SHAFT RESULTING IN CATASTROPHIC FAILURE



$$2\% \times 0.5000 = 0.01000 \quad 1 \times \text{RPM: } 1750.00$$

Machine	Pos	Dir	Units	Date
344 HOPUMP 001	1	A	IN/S	Mar 21 1989 12:57

Frequency	1 x RPM	Amplitude
12300	7.03	0.01247
15900	9.09	0.01843
19500	11.14	0.01048
21600	12.34	0.01096
35250	20.14	0.01144
37050	21.17	0.01191
38850	22.20	0.01493
40650	23.23	0.01295
42450	24.26	0.01096

$$006\% \times 0.5000 = 0.03000 \quad 1 \times \text{RPM: } 1750.00$$

Machine	Pos	Dir	Units	Date
344 HOPUMP 001	1	A	IN/S	Feb 01 1989 13:41

Frequency	1 x RPM	Amplitude
12300	7.03	0.03002
14100	8.06	0.03503
15900	9.09	0.06497
17700	10.11	0.05997
19500	11.14	0.04247
21150	12.09	0.05496
22950	13.11	0.07998
24750	14.14	0.07498
26550	15.17	0.05496
30000	17.14	0.03002

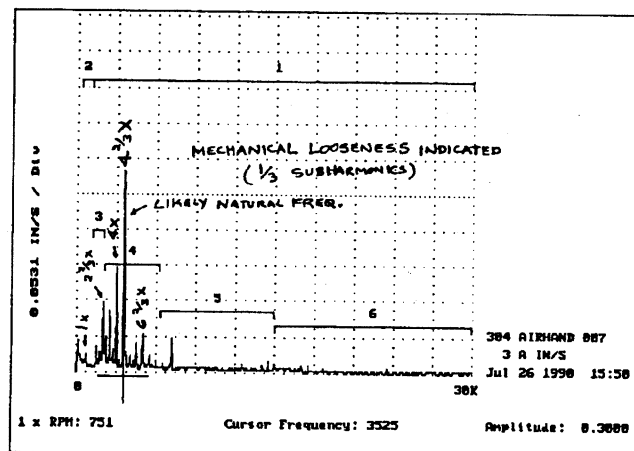
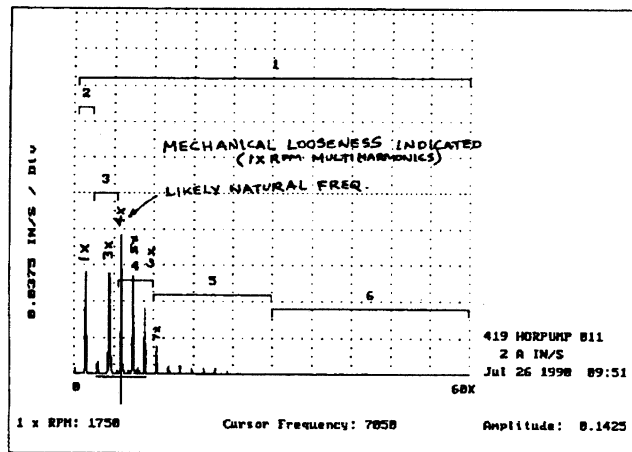
2-1-89

$$002\% \times 0.5000 = 0.01000 \quad 1 \times \text{RPM: } 1750.00$$

Machine	Pos	Dir	Units	Date
344 HOPUMP 001	1	A	IN/S	Aug 03 1988 07:27

Frequency	1 x RPM	Amplitude
300	0.17	0.01398
3000	1.71	0.01096
7200	4.11	0.01096
14100	8.06	0.01597
21600	12.34	0.01247

FIGURE 6.06F
ADVANCING PROBLEM WITH BEARING LOOSE ON SHAFT
(PARTIALLY RESOLVED)



$$0.02\% \times 0.500 = 0.0100 \quad 1 \times \text{RPM:} \quad 751.00$$

Machine	Pos	Dir	Units	Date	
304 AIRHAND 007	3	A	IN/S	Jul 26 1990 15:50	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
150	0.20	0.0450	5550	7.39	0.0225 = 7 1/2 x
750	1.00	0.0275 = 1 x	5700	7.59	0.0100
975	1.30	0.0100 = 1 1/2 x	6000	7.99	0.0150 = 2 x
1125	1.50	0.0100 = 1 1/2 x	6525	8.69	0.0125 = 2 1/4 x
1500	2.00	0.0350 = 2 x	7200	9.59	0.0500 = 7 1/2 x
1800	2.40	0.0300	12525	16.68	0.0125 = 4 1/4 x
2025	2.70	0.1025 = 2 3/4 x	14025	18.68	0.0100 = 1 1/2 x
2250	3.00	0.0525 = 3 x	14550	19.37	0.0150 = 1 1/4 x
2475	3.30	0.0900 = 3 1/2 x	15000	19.97	0.0100 = 2 x
2775	3.70	0.0325 = 3 1/2 x	17025	22.67	0.0100
3000	3.99	0.1550 = 4 x	17550	23.37	0.0100
3225	4.29	0.0150 = 4 1/2 x			
3525	4.69	0.3000 = 4 3/4 x			
4050	5.39	0.0225 = 5 1/2 x			
4275	5.69	0.0150 = 5 3/4 x			
4500	5.99	0.0425 = 6 x			
5025	6.69	0.0575 = 6 3/4 x			

T - Threshold adjust Q - Quit

FIGURE 6.06G
2 DIFFERENT TYPES OF SPECTRA INDICATING
TYPE C MECHANICAL LOOSENESS

6.07 TRACKING OF ROLLING ELEMENT BEARING FAILURE STAGES USING VIBRATION AND HIGH FREQUENCY ENVELOPING AND DEMODULATED SPECTRAL TECHNIQUES

ABSTRACT:

Throughout the last decade, much research along with in-plant testing has been conducted in an attempt to accurately evaluate the condition of rolling element bearings. This has included studies to approximate remaining bearing life. This life is very dependent upon the vibration to which the bearing is subjected. The following paper will present a number of failure senors which have been identified to date for tracking rolling element bearing failure stages using both vibration and spike energy spectral analysis. These results have been directly correlated with studies which have been rigorously conducted in laboratories. Such studies have included evaluation of many types of rolling element bearings including deep groove ball, angular contact ball, needle, cylindrical roller, spherical roller and tapered roller bearings. It is the expressed purpose of this paper to provide the reader with solid tools with which he can not only evaluate the current health of specific rolling element bearings, but also can assist him in predicting remaining life and/or taking proactive steps immediately required to noticeably extend the life of the bearings.

INTRODUCTION:

A tremendous cross section of today's process and utility machinery is outfitted with rolling element bearings. In most all cases, these bearings are the most precise components within the machine, generally held to tolerances only 1/10th those of many of the remaining machine components. Yet, only about 10 to 20% of bearings achieve their design life due to a variety of factors. These primarily include lubrication inadequacies, use of the wrong lubricant, contamination with dirt and other foreign particles, improper storage outside their shipping packages, introduction of moisture, false brinelling during shipment or when standing idle, misapplication of the wrong bearing for the job, improper installation of bearings, etc.

Figure 6.07A shows the components of a rolling element bearing. Note the location of the accelerometer relative to the bearing's outer race, inner race, rolling elements and cage. When rolling element bearings wear, the vibration signal most readily travels from defects on the outer race to the accelerometer. Such flaws will normally appear on two or more of these components prior to eventual failure. Figure 6.07B illustrates various types of rolling element bearings.

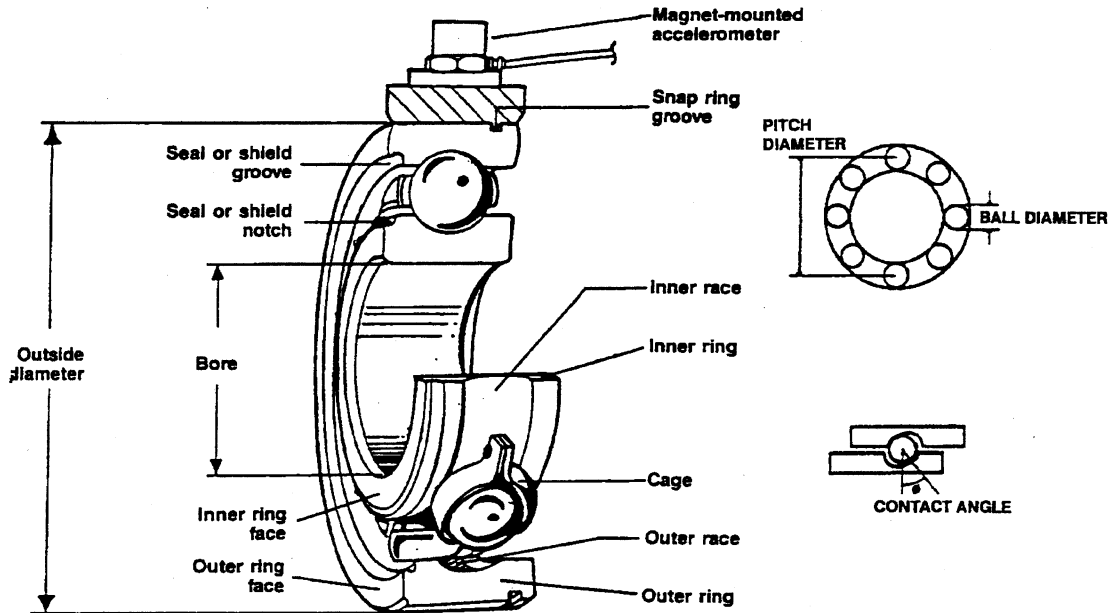
One of the leading contributors to premature rolling element bearing failure is excessive vibration and the high dynamic loads that it can transmit into bearings. Following below is the design formula used in calculating theoretical ball bearing life which will show why it is so critical to bearing life to minimize the dynamic loads imposed upon them from vibration:

$$L_{10} \text{ Life} = \left(\frac{16,666}{\text{RPM}} \right) \left(\frac{\text{RATING}_B}{\text{LOAD}_E} \right)^3 \text{ HOURS}$$

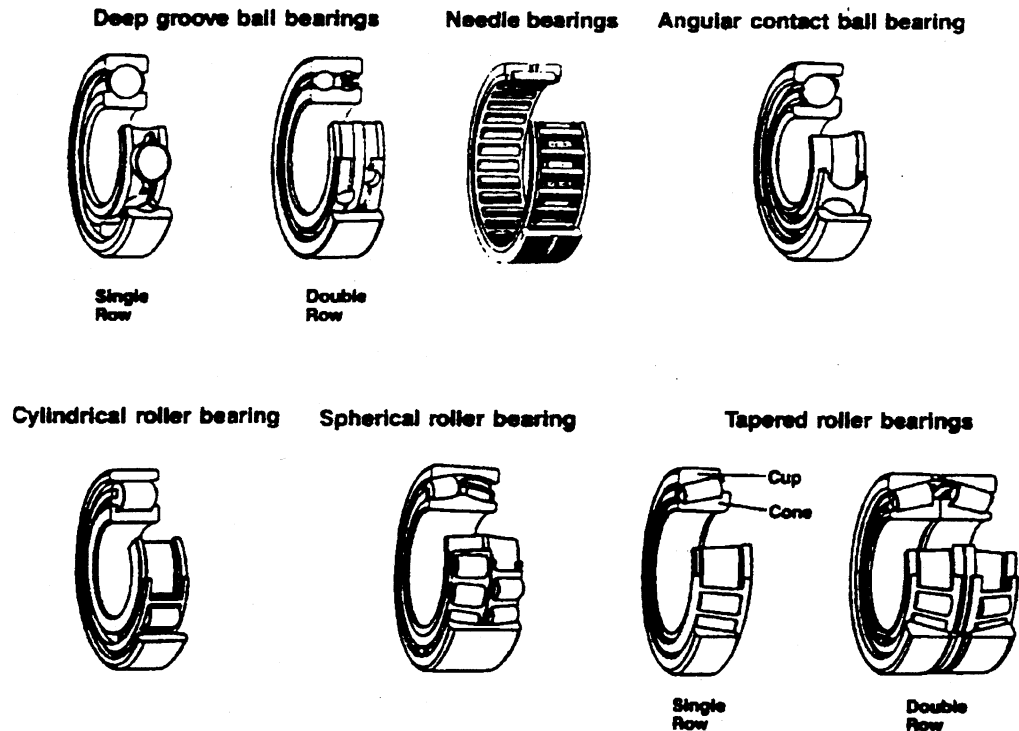
where:

- | | | |
|-----------------------|---|---|
| $L_{10} \text{ Life}$ | = | No. of Hours that 90% of a group of bearings should attain or exceed prior to onset of fatigue failure. |
| RATING_B | = | Basic Dynamic Load Rating for a given bearing (lb) |
| LOAD_E | = | Equivalent Radial Load taken by a bearing - including Radial and Axial Loads (lb) |

**FIGURE 6.07A
ROLLING ELEMENT BEARING TERMINOLOGY**



**FIGURE 6.07B
ROLLING ELEMENT BEARING TYPES**



This formula shows that the greater the speed, the lower the anticipated life. However, of even greater significance is that the theoretical ball bearing life varies with the 3rd power of the load to which it is subjected (and to the 3.33 power in the case of roller bearings). Therefore, if the designer only considered the static loads on the bearing as well as those from other components such as belt tension, he may be surprised to learn of the magnitude of the dynamic forces which can be introduced from vibration.

The key point is that rolling element bearings in truth rarely fail due to defective workmanship themselves. In most all cases, outside influences act on the bearing to bring about its premature failure. If outside influences such as unbalance, misalignment, belt drive problems, soft foot, inadequate lubrication and improper installation can be taken care of, the bearings themselves should have satisfactory life.

To bring home the point, if one considers only the load from unbalance, this alone can generate significant dynamic loads. For example, consider a 2000 lb rotor turning at 6000 RPM with 1 oz of unbalance on a 3 foot diameter (18" radius). The amount of centrifugal force from the unbalance alone can be calculated as follows:

$$F_c = \frac{mr\omega^2}{g_c} = \left(\frac{Wr}{(386)(16)} \right) \left(\frac{2\pi n}{60} \right)^2$$

$$F_c = .000001775 Un^2 = .00002841 Wrn^2$$

where:

- F_c = Centrifugal Force (lb)
- U = Unbalance of Rotating Part (oz-in)
- W = Weight of Rotating Part (lb)
- r = eccentricity of the rotor (in)
- n = Rotating Speed (RPM)

Now substituting for the sample rotor with a 1 oz unbalance at an 18" radius ($U = 18$ oz-in) turning 6000 RPM,

$$F_c = (.000001775)(18 \text{ oz-in})(6000 \text{ RPM})^2$$

$$F_c = 1150 \text{ lbs (from centrifugal force due to unbalance alone)}$$

That is, only a 1 oz unbalance on a 3 foot diameter wheel turning 6000 RPM would introduce a centrifugal force of 1150 lbs which would have to be supported by the bearings in addition to the 2000 lb static rotor weight. Therefore, if the designer had only anticipated supporting 2000 lbs by the bearings, but in fact had to withstand 3150 lbs, his design life calculation would be off by a factor of:

$$\text{Corrected } L_{10} \text{ Life} = (\text{Initial Life}) \left(\frac{2000}{3150} \right)^3$$

$$\text{Corrected } L_{10} \text{ Life} = .25 \times \text{Initial Life Calculation (only 25\% of design life)}$$

Therefore, the actual theoretical life would only be 25% of his initial design life if this wheel were subjected only to unbalance, not to mention other dynamic forces introduced from vibration due to misalignment, looseness, cavitation or any other problems.

Of great importance is the ability to "track" the condition of rolling element bearings and to know when they will need replacement, right from the beginning when initial baseline signatures are

acquired. Much has been learned during the past decade on how vibration signature analysis can contribute to this capability. Following in Section 6.073 will be a separate illustrated discussion on how this can be accomplished using not only vibration signature analysis, but also some of the newer high frequency enveloping signal processing techniques as well. This discussion will follow sections on "Which vibration parameter should normally be used?" (Section 6.071) and "What types of vibration frequencies are generated by defective rolling element bearings?" (Section 6.072).

6.071 Optimum Vibration Parameter For Rolling Element Bearing Condition Evaluation (Acceleration, Velocity Or Displacement)?

Particularly during this last decade, we have learned that we certainly can no longer depend on overall vibration alone to accurately evaluate the health of rolling element bearings. Also, we have found we cannot depend only on the measurement of ultrasonic frequency broadband measurements (Spike Energy, HFD, Shock Pulse, etc.) which make measurements in approximately the 5000 Hz to 60,000 Hz region. What we have learned is that these ultrasonic measurements are only an indicator, not the indicator of bearing health. They likewise are most effectively used in conjunction with vibration signature analysis to best evaluate bearing condition.

Of course, when employing vibration signature analysis, either of 3 vibration parameters can be used - acceleration (g), velocity (in/sec) or displacement (mils). Figure 6.07C shows how each of these parameters vary with frequency in terms of severity. Following below are comments on the attributes of each of these vibration parameters when specifically evaluating rolling element bearing health:

- A. Displacement** - Unfortunately, displacement spectra miss a great deal of bearing health information. Since displacement is low frequency intensive, it tends to suppress or almost eliminate much of the spectral content available that indicates bearing defect problems. In fact, one of the real problems with employing displacement on low-speed machines less than 200 RPM is the fact that while the spectrum may successfully display 1X RPM, it most often will almost completely miss bearing frequencies until the problem is quite severe.
- B. Acceleration** - Unlike displacement, acceleration tends to overemphasize much of the high frequency content generated by the rolling element bearing defects. As a result, if one is not greatly familiar with working with it, acceleration spectra might "cry wolf" far too often. For plant programs trying to establish themselves, this can do great harm to credibility. Although acceleration itself is probably a better indicator in the very early stages of bearing problems, it quickly gives way to vibration velocity which more accurately and clearly tells the "true story" of current bearing health.
- C. Velocity** - Velocity spectra should be one of the best parameters for evaluating most bearing problems, even on low-speed machines (for example, even if the speed was only about 60 RPM, much of the bearing frequency content would be above 500 CPM). For common rotational speeds ranging from 1200 to 3600 RPM, most of the spectral vibration content containing bearing defect information will be below 2000 Hz (120,000 CPM). In general, depending on the type of transducer employed, velocity will remain somewhat "flat" in frequencies ranging from 600 CPM up to 120,000 CPM (10 Hz - 2000 Hz). This means that somewhat "equal weight" can be assessed to a bearing defect frequency occurring at either 6000 CPM or at 60,000 CPM, whereas this could not be done with acceleration or displacement which are highly frequency dependent (see Figure 6.07C). When rolling element bearing machines operate at speeds above approximately 10,000 RPM, acceleration would then likely be the best rolling element bearing health indicator.

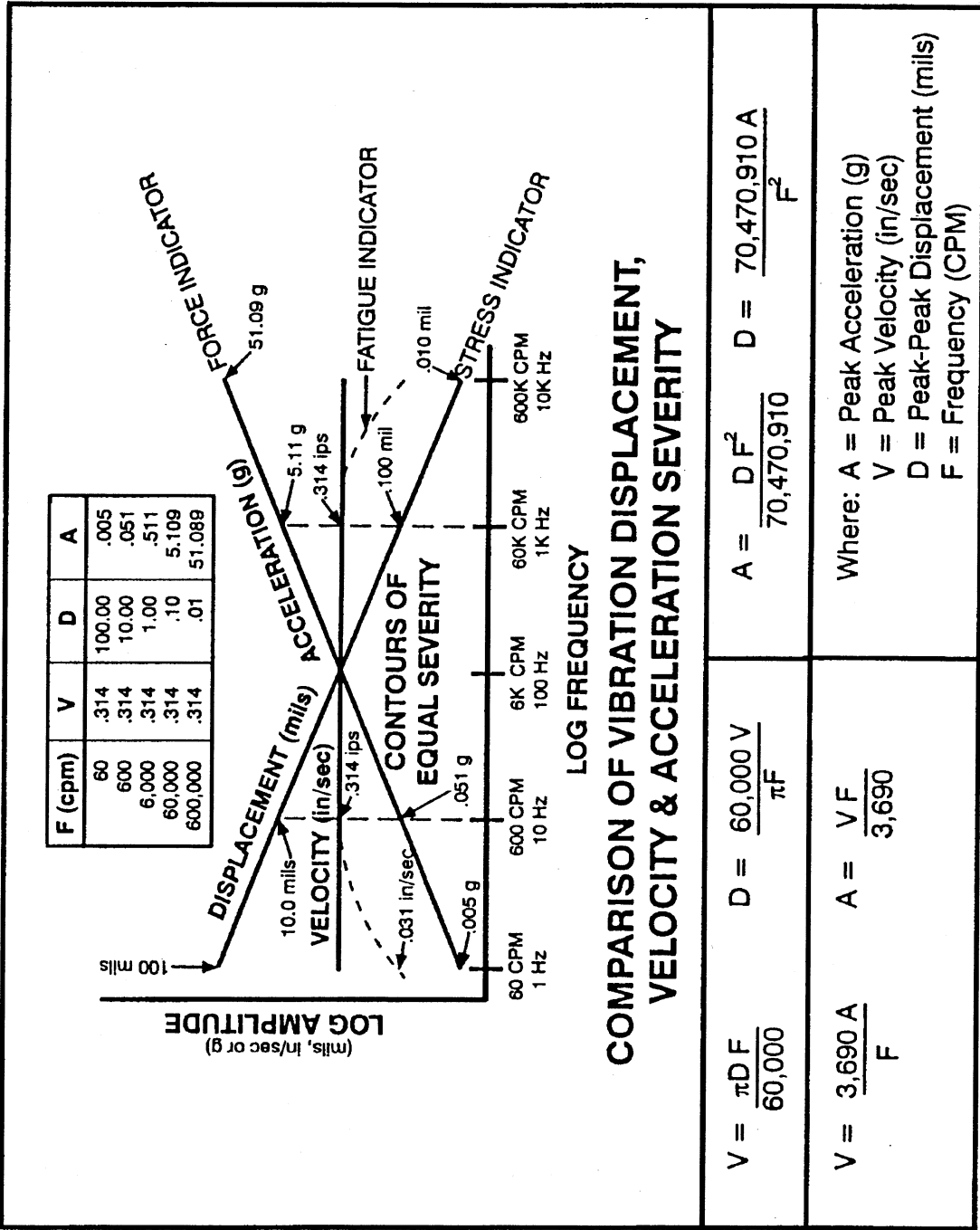


FIGURE 6.07C

The remainder of this section on rolling element bearing health will assume velocity spectra. However, if one takes into account how the other parameters such as acceleration vary with frequency, he can apply many of these same techniques. Acceleration spectra may detect such bearing problems earlier than will velocity spectra, particularly on high-speed machinery. In addition, high frequency envelope demodulated spectra will likely provide even earlier warning of bearing wear and lubrication problems, and should be employed in addition to vibration spectra alone on critical machinery (this will be discussed in Section 6.073 of the paper).

6.072 Types Of Vibration Spectra Caused By Defective Rolling Element Bearings:

Defective rolling element bearings generate each of 4 types of frequencies once they begin to develop defects. These frequencies include (a) random, ultrasonic frequencies; (b) natural frequencies of bearing components; (c) rotational defect frequencies; and (d) sum and difference frequencies. Following below will be a discussion on each of these types of frequencies and their significance:

(a) Random, Ultrasonic Frequencies:

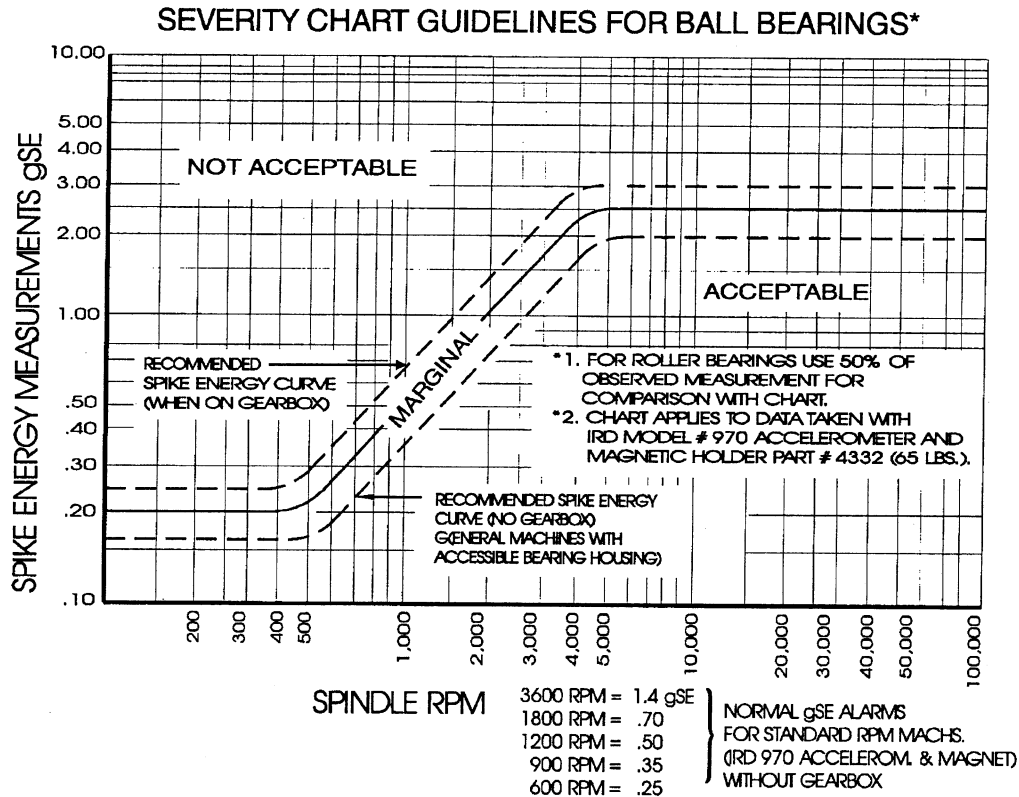
Measurements in this ultrasonic frequency region ranging from approximately 5000 Hz to 60,000 Hz are made by a variety of instruments and employ a similar variety of techniques. These include spike energy, HFD high frequency acceleration, shock pulse measurement and others. Each of these are meant to be incipient failure detection parameters that can track bearing health from its installation until just prior to or including eventual failure. Each of them have their own strengths and weaknesses, not only concerning their bearing health evaluation accuracy, but also how well each of them can be trended with time. In general, the overall number they provide gives just one more piece of information to be considered when evaluating bearing health. However, the information contributed by the vibration spectral data should be given significantly more weight.

Figure 6.07D provides a severity chart for specifying alarm levels of Spike Energy. This chart shows that machine speed must be taken into account when evaluating Spike Energy amplitudes. Similarly, Figure 6.07E provides a comparable severity chart for "HFD" and Shock Pulse (SPM), comparing the relative amplitudes of these parameters with Spike Energy levels in a study conducted by Mr. Charles Berggren (References 16 & 19).

(b) Natural Frequencies of Bearing Components (when installed):

Reference 11 documents that the natural frequencies of installed rolling element bearing components range from approximately 500 to 2000 Hz (30,000 to 120,000 CPM). Like every other member, these bearing components "ring" at these natural frequencies when they are impacted. In the case of rolling element bearings, intermittent impacts of the rolling elements striking flaws on the raceways ring their natural frequencies. Actually, there are several bearing component natural frequencies in the region of 30,000 to 120,000 CPM, but some are much more predominant than others. Therefore, when defects progress beyond microscopic size, they begin to excite these natural frequencies making them the "second line of detection" (Failure Stage 2, discussed in Section 6.073). As the defects worsen, they can cause greater impacts which cause greater response from the natural frequency peaks. Eventually, when wear progresses, more frequencies around these resonances appear, many of which will be 1X RPM sidebands of these natural frequencies (often, such modulating peaks will be spaced at bearing defect frequencies rather than 1X RPM sidebands).

FIGURE 6.07D
RECOMMENDED SPIKE ENERGY SEVERITY CHART (IRD SPIKE ENERGY)
 (Ref. 25)



SPIKE ENERGY™ MEASUREMENTS

Energy is generated by repetitive transient mechanical impacts. Such impacts typically occur as a result of surface flaws in rolling-element bearings or gear teeth.

This energy is conducted from its source through various paths to the outer surface of the machine structure, and is seen as a small-amplitude vibration at the surface. Accelerometers coupled to the surface generate corresponding electrical signal.

The accelerometer signals processed by unique filtering and detection circuitry to produce a single "figure of merit" related to the intensity of the original impacts. This figure of merit is expressed in "gSE" units.

SPIKE ENERGY™ gSE readings are measurements which can with experience, be correlated with the severity of the casual surface flaws. Even though gSE readings are affected by the nature of the conductive path between the impact source and the accelerometer, similar machine structures will provide a reasonable basis for comparison between the structures.

The gSE figure of merit has proven to be effective in detecting mechanical defects in meshing gears and rolling element bearings. The gSE measurement, when used in conjunction with conventional measurement of vibration velocity and acceleration, provides early indications of mechanical deterioration.

** When used with magnetic holders, accelerometers must be installed with a light coating of silicone grease and tightened to 40 in-lb. torque.

FIGURE 6.07E
MAINTENANCE DIAGNOSTIC VIBRATION AND HIGH FREQUENCY
GENERAL TOLERANCE CHART FOR PROCESS MACHINERY WITH
ROLLING ELEMENT BEARINGS^①

Unfiltered Vibration Velocity	Shock Pulse & Acoustic Emission	Microlog HFD	IRD Spike Energy ^②	Severity Level and Alarm Status
Inches/Second Peak	Db	g	gSE	
1.5+	50+	5.0+	3.0+	Danger (Shutdown)
.75 - 1.49	40 - 49	3.0 - 4.99	1.50 - 2.99	Very Rough (Alert)
.40 - .74	30 - 39	1.50 - 2.99	.80 - 1.49	Rough (Alert)
.20 - .39	20 - 29	.75 - 1.49	.40 - .79	Fair (Acceptable)
.10 - .19	10 - 19	.30 - .74	.20 - .39	Good (Acceptable)
.01 - .09	1 - 9	.01 - .29	.01 - .19	Smooth (Acceptable)

① Provide by Charles Berggren of Monsanto

② Spike Energy Amplitudes measured using an IRD 970 accelerometer outfitted with IRD's 2-pole, 65 lb. magnet.

One important point about these bearing component natural frequencies is the fact that they are independent of running speed. That is, whether the shaft is turning at low or at very high speeds, the natural frequencies remain at the same frequency location. However, their response amplitude is proportional to the impact velocity, which means that with greater rotational speeds, they normally will respond at greater amplitudes.

(c) Rotational Defect Frequencies:

Through the years, a series of formulas have been developed which can help detect specific defects within rolling element bearings. They can separately detect faults on the inner race, outer race, cage or rolling elements themselves. They are based on the bearing geometry, the number of rolling elements and the bearing rotational speed. Figure 6.07F provides the formulas for each of these four rolling element bearing defect frequencies.

The power of these equations is that if one knows the design parameters of his bearings (pitch diameter, rolling element diameter, number of rolling elements & contact angle), he is able to detect problems which occur on the races, cage or rolling elements, and he is enabled to track these problems as deterioration continues. In many cases, the analyst may not know all the parameters to insert for a particular bearing in the equations, but he might know the bearing manufacturer and model number. In these cases, there are several publications and software offerings which tabulate each of the 4 defect frequencies for each of the bearing model numbers. Figure 6.07G is an example of one of the better known publications listing these defect frequencies (as per Reference 12). Note on Figure 6.07G that this sheet provides the number of rolling elements (N_b), rolling element diameter (B_d), bearing pitch diameter (P_d), contact angle

($\emptyset$), outer race defect frequency (BPFO), inner race defect frequency (BPFI), cage defect frequency (FTF) and rolling element defect frequency (BSF). Note that Figure 6.07G also provides an example assuming the SKF N220 bearing. Note that each one of the bearing frequencies are given in terms of running speed orders. Therefore, the frequencies for the SKF N220 bearing show, for example, that the inner race frequency (BPFI) is 9.13X RPM. Thus, if this bearing were turning at 1000 RPM, and a frequency was detected at 9130 CPM, he could conclude that this peak is the bearing inner race defect frequency and that a fault is developing there. On the other hand, if a frequency of 6860 CPM occurred on this same unit, he would know this is the outer race defect frequency of this SKF N220 bearing (since BPFO = 6.86X RPM).

The great advantage of knowing these bearing defect frequencies is that, for example, an analyst can separately evaluate the outboard and inboard bearings on the same machine, particularly if they are different model numbers. For example, assume that the outboard bearing of our example was an SKF N220 and that its inboard bearing was an SKF N228 bearing. Note from Figure 6.07G that they would have an entirely different set of defect frequencies (for example, the inner race frequency on the N228 would be 10.19X RPM as compared to 9.13X RPM on the N220).

A number of interesting facts can be stated about these bearing defect frequencies as follows:

1. How Bearing Frequencies Differ From Other Frequencies (Defect Frequencies):

One thing setting rolling element bearing defect frequencies apart from other vibration sources is the fact that they are defect frequencies. In other words, bearing defect frequencies should not be present. When they are present, they signal at least an incipient problem. On the other hand, other common frequencies such as 1X RPM are always present whether or not there is satisfactory or unsatisfactory balance or alignment; pumps and vanes always show vibration at some amplitude for blade pass frequencies; gears cause vibration at the number of teeth X RPM. However, the presence of these other frequencies does not mean there is necessarily a defect or problem. The appearance of bearing defect frequencies sends a message to the analyst to "pay attention". However, it is also important to point out the presence of such defect frequencies do not necessarily mean there are defects within the bearing. They also will appear if there is insufficient lubricant allowing metal-to-metal contact, or if the bearings are improperly loaded (excessive press fit, excessive thrust on a bearing not necessarily designed to take thrust, or if a thrust bearing is installed backwards, etc.).

2. Bearing Defect Frequencies are Noninteger Multiples of Operating Speed:

Referring back to Figure 6.07G, note that each of the bearing defect frequencies are noninteger multiples. That is, they are one of the few machinery vibration sources that do not generate integer multiples of rotational speed (GMF = #teeth X RPM; BPF = #blades X RPM). This is helpful when the manufacturer and model number of the bearing is unknown. Look for real number RPM multiples (such as 5.78X, or 7.14X, etc.).

3. Sum of BPFO Multiplier and BPFI Multiplier = Number of Balls:

Figure 6.07F shows an interesting relationship between the outer and inner race frequency multipliers and the number of rolling elements. Note that the sum of the number of balls equals the sum of the outer and inner race frequencies (BPFO + BPFI). For years, many have looked for the appearance of frequencies at the number of balls X RPM similar to what they have experienced with gears (#teeth X RPM), blade pass frequency (#Blades X RPM) and so forth. However, the author has rarely seen the appearance of a frequency at "bearing ball pass frequency", but has commonly seen each of the race frequencies themselves which are on the other side of the equation.

FIGURE 6.07F ROLLING ELEMENT BEARING DEFECT FREQUENCIES

These equations assume the inner race is rotating with the shaft while the outer race is fixed (stationary):

$$\text{INNER RACE} = \text{BPFI} = \frac{N_b}{2} \left(1 + \frac{B_d}{P_d} \cos \theta \right) \times \text{RPM}$$

$$\text{OUTER RACE} = \text{BPFO} = \frac{N_b}{2} \left(1 - \frac{B_d}{P_d} \cos \theta \right) \times \text{RPM} = N_b \times \text{FTF}$$

$$\text{BALL (OR ROLLER)} = \text{BSF} = \frac{P_d}{2B_d} \left[1 - \left(\frac{B_d}{P_d} \cos \theta \right)^2 \right] \times \text{RPM}$$

$$\text{CAGE} = \text{FTF} = \frac{1}{2} \left(1 - \frac{B_d}{P_d} \cos \theta \right) \times \text{RPM} = .35 - .45 \times \text{RPM}$$

NOTE:

If the inner ring is fixed (stationary) while the outer race rotates with the shaft, the minus sign must be changed to a plus sign within the parentheses of the Cage Frequency Equation. In this case, $N_b \times \text{FTF}$ will now equal BPFI rather than BPFO; and $\text{FTF} \approx .55 - .65 \times \text{RPM}$. However, BPFO, BPFI and BSF will remain the same.

(NOTE THAT $\text{BPFI} + \text{BPFO} = N_b$)

where:

- N_b = Number of Balls or Rollers
- B_d = Ball or Roller Diameter (in. or mm)
- P_d = Bearing Pitch Diameter (in. or mm)
- θ = Contact Angle (degrees)

APPROXIMATIONS (If Bearing Dimensions are Unknown, but do know # of Balls or Rollers) (Reference 18):

$$\text{Approx. BPFI} \approx \left(\frac{N_b}{2} + 1.2 \right) \times \text{RPM}$$

$$\text{Approx. BPFO} \approx \left(\frac{N_b}{2} - 1.2 \right) \times \text{RPM}$$

$$\text{Approx. BSF} \approx \frac{1}{2} \left(\frac{N_b}{2} - \frac{1.2}{N_b} \right) \times \text{RPM}$$

$$\text{Approx. FTF} \approx \left(\frac{1}{2} - \frac{1.2}{N_b} \right) \times \text{RPM}$$

FIGURE 6.07G
EXAMPLE TABULATION OF ROLLING ELEMENT BEARING DEFECT
FREQUENCIES (REFERENCE 12)

SKF

SINGLE ROW CYLINDRICAL ROLLER BEARINGS

<u>BEARING NUMBER</u>	<u>Nb</u>	<u>Bd</u>	<u>Pd</u>	<u>Ø</u>	<u>BPFO</u>	<u>BPFI</u>	<u>FTF</u>	<u>BSF</u>
N205	12	.256	1.516	0	4.99	7.01	.41	2.87
N206	12	.295	1.811	0	5.00	6.94	.41	2.99
N207	12	.354	2.106	0	4.97	6.98	.41	2.89
N208	13	.394	2.362	0	5.42	7.58	.41	2.92
N209	14	.394	2.539	0	5.97	8.07	.42	3.17
N210	15	.394	2.968	0	6.50	8.48	.43	3.70
N211	15	.433	3.051	0	6.43	8.56	.43	3.45
N212	16	.472	3.476	0	6.92	9.08	.43	3.61
N213	16	.512	3.641	0	6.88	9.11	.43	3.49
N214	16	.512	3.738	0	6.93	9.06	.43	3.68
N215	17	.551	4.035	0	7.34	9.65	.43	3.59
N216	16	.590	4.331	0	6.91	9.08	.43	3.60
N217	16	.620	4.626	0	6.91	9.08	.43	3.60
N218	16	.709	4.921	0	6.84	9.15	.43	3.40
N219	16	.748	5.216	0	6.85	9.14	.43	3.41
N220	16	.787	5.512	0	6.86	9.13	.43	3.43
N222	16	.905	6.102	0	6.81	9.18	.42	5.59
N224	16	.945	6.594	0	6.85	9.14	.42	3.41
N225	18	.945	7.087	0	7.91	10.07	.43	3.32
N228	18	1.024	7.677	0	7.76	10.19	.43	3.68
N305	10	.354	1.713	0	4.01	5.99	.40	2.32
N306	11	.394	2.007	0	4.42	6.58	.40	2.45
N307	11	.433	2.264	0	4.45	6.53	.40	2.51
N308	12	.472	2.559	0	4.88	7.07	.41	2.62
N309	11	.551	2.854	0	4.43	6.56	.40	2.49
N310	12	.590	3.150	0	4.85	7.09	.40	2.57
N311	11	.669	3.445	0	4.42	6.56	.40	2.48
N312	12	.709	3.740	0	4.85	7.10	.41	2.54
N313	12	.748	4.035	0	4.87	7.08	.41	2.60
N314	12	.787	4.330	0	4.89	7.06	.41	2.66
N315	12	.866	4.626	0	4.85	7.09	.41	2.57
N316	13	.866	4.921	0	5.36	7.63	.41	2.75
N317	12	.945	5.216	0	4.89	7.06	.41	2.67
N318	13	.984	5.512	0	5.34	7.65	.41	2.71
N319	14	1.024	5.807	0	5.77	8.22	.41	2.74
N320	14	1.102	6.201	0	5.75	8.24	.41	2.68
N321	14	1.181	6.496	0	5.72	8.27	.41	2.66
N322	14	1.260	6.890	0	5.72	8.27	.41	2.64

EXAMPLE:

FOR SKF N220 BEARING, BPFI = 9.13 X RPM (Inner Race)
BPFO = 6.86 X RPM (Outer Race)
BSF = 3.43 X RPM (Rolling Elements)
FTF = .43 X RPM (Cage Freq.)

4. Description of How Defect Frequencies are Generated Within Rolling Element Bearings:

Figure 6.07F provides the formulas required to calculate the four defect frequencies for each rolling element bearing (BPFI, BPFO, BSF and FTF). Figure 6.07H illustrates how such defect frequencies are generated within the bearings. For example, if there is a defect on the outer race on the bottom of the bearing within the load zone as shown in Figure 6.07H, note that an impulse occurs in the time waveform at each instant when a rolling element passes over and impacts this defect. On the other hand, if the inner race had a defect, an impulse would occur in the time domain when the inner race rotated past each rolling element (assuming the inner race is press fit on the shaft). An important fact shown by Figure 6.07H is that the amount of response from the rolling elements striking the inner race defect will depend on where the inner race is positioned at that particular instant of time when the impact occurs (that is, if the inner race defect is positioned within the load zone, it will have significantly more response than it would if the impact occurred with the same inner race defect positioned 180° away, clearly out of the load zone). This explains why inner race defect frequencies are often surrounded by sidebands spaced at 1X RPM since their amplitude is modulated at the rate of once per revolution as shown by Figure 6.07H. On the other hand, since the outer race does not rotate in this instance, the amplitude response in the time domain should remain near constant. Therefore, 1X RPM sidebands surrounding outer race frequencies (BPFO) are much more serious than those surrounding the inner race frequency (BPFI), again assuming the inner race is press fit on the shaft (if the inner race is stationary while the outer race rotates, the reverse with respect to sidebands would occur). The presence of 1X RPM sidebands surrounding outer race (BPFO) frequencies normally means the problem is sufficiently serious to cause the bearing to actually impede the motion of the shaft. As Section 6.073 will later point out, when the inner race frequency (BPFI) becomes surrounded by several "families" of 1X RPM sidebands, this can likewise indicate a more serious problem.

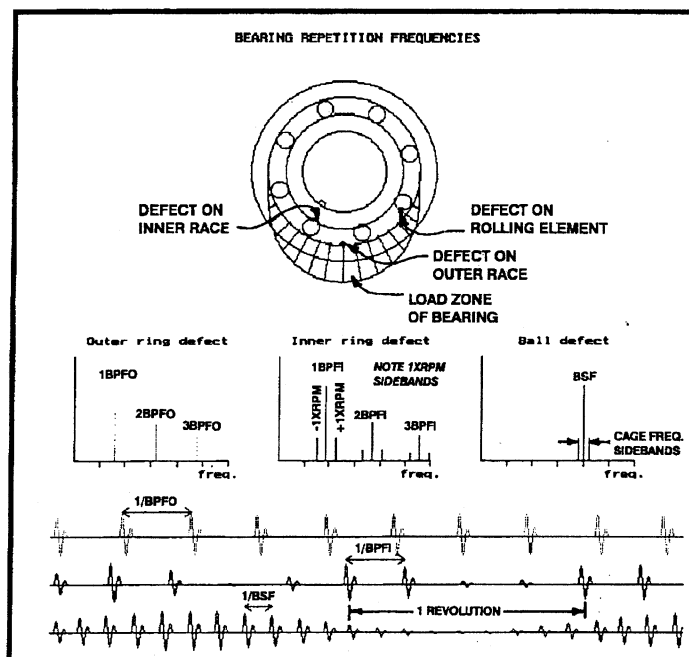


FIGURE 6.07H
ILLUSTRATION OF HOW DEFECT FREQUENCIES ARE
GENERATED WITH ROLLING ELEMENT BEARINGS

5. Relative Outer and Inner Race Amplitudes:

Normally, the amplitude of the outer race frequency is higher than that of the inner race frequency. This is likely due to the fact that the transducer itself is much closer to the outer race (see Figure 6.07A). In addition, the vibration signal from the inner race frequency (or harmonics) must pass through several more interfaces including the constantly rotating rolling elements on its path out to the vibration transducer.

6. Usual Order of Appearance by Defect Frequencies:

Normally, defects will first appear on the races before the problem propagates to the rolling elements and cage. Thus, outer and inner race frequencies are normally first established before the appearance of a ball spin frequency. Later, the cage frequency normally can appear either as a fundamental or as a sideband to another frequency. Likewise, the ball spin frequency will sometimes appear as a sideband above and below an outer or inner race defect frequency.

7. Where Cage Frequency Normally Appears:

Although rolling element bearing problems are classically thought of as high frequency problems, fundamental cage frequencies will always be subsynchronous, ranging from approximately .33X RPM up to .48X RPM with the majority falling between .35X and .45X RPM. However, the cage frequency will not normally appear at its fundamental frequency. Instead, it most often will appear as a sideband around ball spin frequency (BSF), or around one of the race frequencies (BPFO or BPFI) with the sideband difference frequency equal to the cage frequency (FTF). For example, see Figure 6.09I which shows an outer race frequency BPFO at 10,260 CPM (5.78X RPM) with cage frequency sidebands (FTF) spaced at 720 CPM (.41 X running speed which is 1775 RPM). This is how a cage frequency most often will appear.

8. Frequencies Generated by Faults on Balls or Rollers:

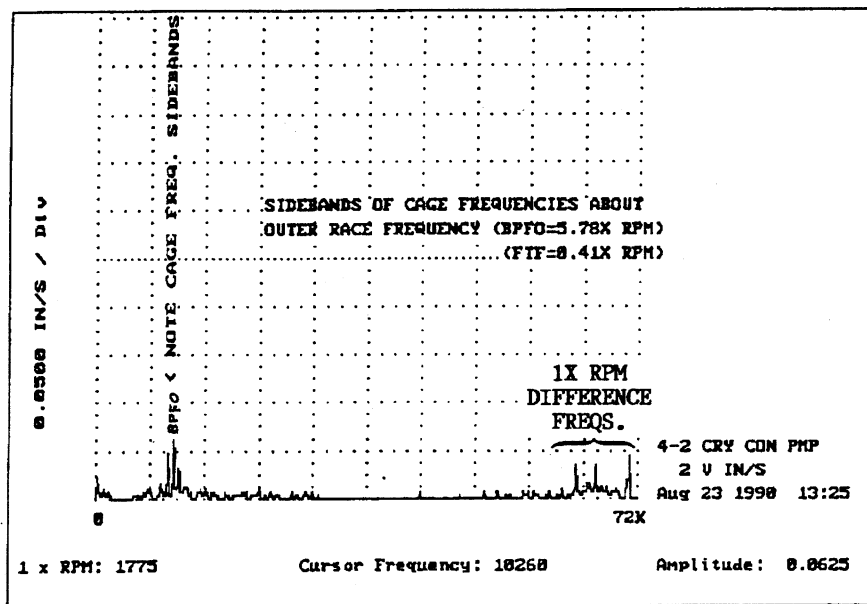
When defects occur on rolling elements themselves, they will often generate a frequency not only at the ball spin frequency (BSF), but also at the cage frequency (FTF, also known as fundamental train frequency).

9. Ball Spin Frequency May Appear if Cage is Broken:

The appearance of a ball spin frequency does not always necessarily mean there is a defect on the rolling elements. However, it still means that there is a problem present. In this case, it can indicate that a cage is broken at a rivet and if the balls are thrusting hard against the cage (as per Reference 12).

10. Frequency Generated if More than One Rolling Element has Faults:

If more than one rolling element has defects, a frequency equal to the number of balls having defects X the ball spin frequency will be generated. In other words, if defects are present on 5 balls or rollers, a frequency at 5X Ball Spin Frequency would most often appear.



002% x 0.500 = 0.0100 1 x RPM: 1775.00

Machine	Pos	Dir	Units	Date
4-2 CRY CON PMP	2	V	IN/S	Aug 23 1990 13:25

Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
180	0.10	0.0225	67140	37.83	0.0125
1080	0.61	0.0100	67860	38.23	0.0125
6840	3.85	0.0100	69120	38.94	0.0100
7200	4.06	0.0125	70920	39.95	0.0425
8640	4.87	0.0150			
9540	5.37	0.0475 ← -FTF (.41x)			
10260	5.78	0.0625 ← BPFO			
10980	6.19	0.0325 ← +FTF (.41x)			
11880	6.69	0.0125			
13860	7.81	0.0100			
14580	8.21	0.0100			
21600	12.17	0.0100			
61920	34.88	0.0100			
63720	35.90	0.0350			
65160	36.71	0.0150			
65520	36.91	0.0150			
66420	37.42	0.0350			

FIGURE 6.07I
HOW THE CAGE FREQUENCY (FTF) MOST OFTEN APPEARS
WITHIN A VIBRATION SPECTRUM

11. Allowable Vibration at Bearing Defect Frequencies:

It is very difficult to assign definite vibration severity levels to bearing defect frequencies similar to that which is commonly done for unbalance amplitude at 1X RPM. For one, there are a variety of rolling element bearings in a variety of different machines, each of which may provide different paths for the vibration signal to the transducer. However, one can make a statement which will apply almost uniformly for all the various types and combinations - the allowable vibration for unbalance at 1X RPM is much greater than that allowed for a defect frequency for a rolling element bearing.

Much research has proven that no absolute answer can be given to allowable vibration amplitudes at bearing defect frequencies. Not only does it depend on which particular machine a bearing is installed and what the operating speed is, it also depends greatly on which bearing failure scenario path it will travel. For example, on rare occasions, the author has personally witnessed bearings which still did not have significant damage even with an amplitude of .30 in/sec at a fundamental BPFO when this damage was concentrated at one particular location on the outer race. On the other hand, considerable damage has been discovered in other bearings when no one defect frequency had an amplitude greater than only .03 in/sec in everyday machinery like pumps and blowers running at common speeds such as 1780 RPM. In fact, extensive damage has been found in large dryer roll bearings on paper machines running less than 100 RPM with bearing frequency amplitudes that ranged from only .003 to .006 in/sec.

The key point common to each of the latter two scenarios having low vibration but significant bearing damage was that in each case, not just one bearing frequency was present in the spectra; instead, a number of bearing frequency harmonics were present (a "number of bearing frequencies present" means for example that either 3 to 5 BPFO or BPFI harmonics might simultaneously be present; or that 2 or more harmonics of BPFO might be present along with 2 or more harmonics of BPFI). In addition, further investigation has shown that when these bearing frequency harmonics were surrounded by sideband frequencies spaced at 1X RPM of the problem bearing, even more damage is indicated (particularly if these 1X RPM sidebands surround BPFO harmonics, assuming the bearing is press fit on the shaft). It should also be emphasized that these sidebands may be spaced at bearing frequencies themselves rather than at 1X RPM (i.e., 4 or 5 BPFO harmonics may all be present, each with sidebands of FTF or BSF above and below them which would likewise indicate a potentially serious problem).

Therefore, the most important thing to look for indicating significant bearing wear is the presence of a number of bearing defect frequency harmonics, particularly if they are surrounded by sidebands spaced at either 1X RPM or sidebands spaced at other defect frequencies of the bearing - independent of amplitude. If these are present in a spectrum, replace the bearing as soon as possible.

12. Evaluating Bearings on Low-Speed Machinery (less than 250 RPM):

Special precautions must be taken when making measurements on low-speed machinery. Rolling element bearings have been successfully evaluated at speeds as low as 1.5 RPM. However, one must be particularly aware of the low frequency limitations of both his analyzer and his transducer. If he uses one of the computer-loaded data collectors popular today, he should be aware that many of these instruments are outfitted with high-pass filters that begin filtering out signals below fixed frequency ranges at rates of from approximately 12 to 24 dB/octave. This cutoff frequency may be close to DC (0 Hz) on some data collectors, but may be as high as 8 Hz (480 CPM) on other models. Thus, in the latter case, a machine may be

generating 1.00 in/sec of vibration at 240 RPM, but the instrument may display only approximately .25 in/sec at this frequency if it has such a high-pass filter. Obviously, if one attempted to make measurements at frequencies below 120 CPM with this instrument, the data collector would detect and display only a small percentage of this peak. And, if one wanted to single or double integrate the signal from acceleration to velocity or displacement, respectively, there would likely be even greater suppression of the amplitude in the data collector (also, it is important to point out that some data collectors do not integrate acceleration signals to velocity at frequencies below approximately 120 CPM; data below this cutoff frequency is left nonintegrated). In addition to filtering, the low frequency response of the analyzer is also effected by its signal conditioning and input circuitry components which also needs to be considered.

Not only does one have to be concerned about the analyzer during low frequency measurements, he also must be aware of the frequency response of the transducer whether it is an accelerometer, a velocity pickup, or a proximity probe. Typically, most general purpose accelerometers in use with data collectors today are flat within 5% between approximately 5 Hz and 10,000 Hz (300 - 600,000 CPM). However, special seismic piezoelectric accelerometers can be obtained which will extend the $\pm 5\%$ flat response down to as low as approximately 0.1 Hz (6 CPM). These transducers will typically have exceptionally high sensitivities ranging from 1000 to 10,000 mV/g (as compared to only 10 to 100 mV/g for the standard accelerometer) and will normally have much greater weight on the order of 400 to 1000 grams (as compared to only 10 to 50 grams for the standard unit). They also will typically have much longer discharge time constants of 20 seconds or more (compared to only 0.5 second for standard accelerometers) and will work best when connected to analyzers having input impedances of approximately 1,000,000 ohms. Finally, the low frequency performance of these transducers can be further enhanced by connecting them to special power supplies and signal conditioning equipment.

In addition to the instrument limitations discussed, it should be pointed out that both 1X RPM and bearing frequency amplitudes themselves will inherently be much lower on large, low-speed machinery. In the case of 1X RPM, there will be minimal unbalance forces since these vary with the square of speed. Thus, if one attempts to specify spectral alarm bands for this machinery, he will have to spec them at much lower alarm amplitudes than for machinery rotating above 1200 RPM. The best way to specify them will be to capture actual data and perform statistical analysis on the overall levels and those of individual frequency bands.

Fortunately, even though many of today's standard data collectors and transducers may not be capable of evaluating 1X RPM and 2X RPM vibration on much low-speed machinery, these same instruments might still be able to satisfactorily evaluate the health of their bearings. Please refer to Figure 6.07G which shows some typical rolling element bearing frequencies. Even though the fundamental cage (FTF) and ball spin frequencies (BSF) may still be below the "reach" of the system, the more common outer race (BPFO) and inner race frequencies (BPFI) will most often range from 4 to 12X RPM. Therefore, a measurement system which could not "see" vibration at, for example, 100 RPM speed very likely could detect fundamental outer and inner race frequencies which would probably range from about 500 to 1000 CPM on this machine with little or no loss of signals. And, of course, harmonics of these bearing frequencies could easily be evaluated.

Each of these items will be further discussed in more detail in the seminar entitled "Analysis II". For now, the most important point is that one can successfully evaluate bearing health on low-speed machinery if he takes into account the frequency response performance characteristics of his instrumentation system and takes the necessary provisions outlined in this section.

13. Where Transducer Should be Placed to Properly Detect Bearing Frequencies:

It is most important to place the vibration transducer as close as possible to the load zone of the bearing, particularly if the bearing only supports radial load. Reference 13 points out that this is of critical importance for a spherical roller bearing in which he claims the vibration signal strength can be affected by as much as 100% if the transducer is not placed in the load zone, giving the analyst a false sense of security when he actually does have a problem.

14. Approximate Bearing Defect Frequencies if only the Number of Rolling Elements is Known:

Figure 6.07F provides 4 formulas which approximate the bearing defect frequencies if only the number of rolling elements (N_b) is known (as per Reference 18). These equations have been proven to be considerably more accurate than older approximations which assumed BPFO to equal approximately $.4 \times N_b \times \text{RPM}$ and BPFI to equal approximately $.6 \times N_b \times \text{RPM}$. Still, optimum precision can only be realized if one knows each of the other bearing design parameters (B_d , P_d and $\emptyset$), particularly in the case of the ball spin frequency (BSF).

15. Sensing of Improper Bearing Load or Installation:

Not only can bearing frequencies be used to detect defects within the bearings, but also they can be used to detect when bearings are improperly loaded or installed (that is, they do not always indicate that a defect is present within the bearing). For example, even when a new bearing is installed, if there is excessive interference of the bearing seat on the bearing housing causing it to be "jammed into" the seat, it can result in take-up of all the internal clearances forcing the rolling elements against the races. If this occurs, the bearing will immediately generate ball pass frequencies of the outer and/or inner race upon startup. In addition, the author has taken data several times when thrust bearings were installed backwards. When this has occurred, the improperly installed thrust bearings have generated excessively high amplitudes at race defect frequencies, sometimes on the order of 1.0 in/sec, or greater. Excessive press fit of a bearing onto a shaft can also immediately generate a pronounced outer race or inner race defect frequency (BPFO or BPFI), letting the analyst know excessive and/or improper load has been placed on the bearing. And, even though no real wear might yet have occurred, if the problem is not detected and corrected, the bearing will likely fail long before its predicted life. In all improperly loaded situations, if the same bearings were reinstalled properly or loaded properly, the defect frequencies were significantly reduced or disappeared altogether. This has happened on a number of occasions. In fact, one client manufacturing textile machinery uses this technique to detect assembly problems in its quality assurance program.

16. Frequencies Generated by Inadequate Bearing Lubrication:

Reference 12 states that "unique signatures generated by inadequately lubricated bearings are characterized by 3 or 4 peaks in the frequency range of 900 to 1600 Hz (54,000 to 96,000 CPM). The difference frequency between the peaks ranges from 80 to 130 Hz (4800 to 7800 CPM). Some signatures of properly lubricated bearings contain these frequency components; however, the amplitude is very low - about 0.05 in/sec or less. The amplitude increases to as much as 0.10 or 0.20 in/sec when the lubrication is inadequate". He adds that "empirical evidence indicates that frequencies from 900 to 1600 Hz are natural frequencies of the installed bearing." With this in mind, if frequencies within this range (approximately 50,000 to 100,000 CPM) do occur with difference frequencies on the order of 5000 to 8000 CPM, it may be a good idea to check lubrication, particularly if high spike energy (or equivalent) levels are also measured on this same bearing housing. Also, even

greater evidence of lubrication problems would occur if neither the high spike energy nor a spectrum having these components was present on the other bearing that is supporting this same shaft.

In addition, some research has shown that inadequately lubricated bearings which allow metal-to-metal contact can also generate bearing defect frequencies. In these cases, one might go ahead and add lubricant while monitoring both overall spike energy (or equivalent) and vibration FFT spectra to determine if the ultrasonic levels drop considerably, and to find if the bearing defect frequencies disappear from the spectrum. If they do, he should return to the machine 12 to 24 hours later and see if either have reoccurred. If not, lubrication was likely the problem. If they do return, the bearing is likely suffering a wear problem as described in Section 6.073.

(d) Sum and Difference Frequencies:

When a single defect has developed on a bearing, it will generate a defect frequency and will begin deteriorating. When the defect grows, it can contribute to the development of other defects in the bearing. In so doing, other frequencies will be generated and a number of patterns might develop. Some frequencies will add to and subtract from others. In fact, the fundamental frequency for a particular defect might never occur. When it appears, this defect frequency might act as a sideband to other frequencies which are already present. For example, a cage frequency itself will not normally occur at its fundamental frequency of approximately .35X to .45X RPM. Instead, the cage frequency itself normally will sideband the race frequencies (BPFO or BPFI) or the ball spin frequency of the bearing (see Figure 6.07I). In some cases, bearing defect frequencies can even modulate frequencies generated by sources other than bearings. Following below are some of the more important facts concerning sum and difference frequencies and their behavior:

1. It is not uncommon for the fundamental ball spin frequency or harmonics never to appear even if significant faults are present on the rolling elements. In this case, BSF will appear, but as a sideband of other frequencies.
2. If only a single fault is present on either the outer or inner race, only a single race frequency will appear. However, when faults begin to appear around the periphery, a number of harmonics of these race frequencies will appear.
3. When defects grow on raceways, the amplitude of the race frequencies themselves will often increase somewhat. However, even greater indicators of deterioration are the number of bearing frequency harmonics, as well as the appearance of 1X RPM sideband frequencies above and below the race frequencies (particularly BPFO). In effect, the unbalance forces at shaft speed will tend to modulate the frequency components of the races, generating sidebands. In the case of BPFI, 1X RPM sidebands are often created around it since its amplitude is modulated with much greater response when the inner race defect impacts rolling elements within the load zone than those generated outside the load zone. The rate of this modulation in amplitude is at 1X RPM, therefore generating the 1X RPM sidebands (see Figure 6.07H).
4. In rolling element bearing frequency analysis, the emphasis is not on amplitude, but on content of the spectra. In fact, amplitudes of bearing frequencies themselves often begin to drop as condition worsens, particularly in the case where faults begin to propagate around the periphery of the outer or inner race. In this case, serious weight should be placed on the fact that a multiple number of fault frequencies are appearing and that many of these fault frequencies are sidebanded by vibration at bearing RPM.

-
5. As deterioration continues, modulation effects can continue to have more and more influence until eventually, the spectrum can become a series of nothing but 1X RPM harmonics.
 6. Reference 14 shows that double row bearings having staggered rolling elements can generate 2 sets of bearing defect frequencies - (a) If a defect is confined to only one side of the raceway, the bearing defect frequencies generated should be calculated using the number of rolling elements only in that single row; however, (b) if a defect occurs on both sides of the raceway, the frequencies should be calculated using the total number of rolling elements. Here again, either of these frequencies can act as a sideband around other frequencies with the sideband difference depending on whether the fault has occurred on one or both raceways. Importantly, this number of balls (either those for one row or the total number) must be entered into the outer race, inner race and ball spin frequencies.
 7. A single defect on the inner race of a rolling element can be difficult to diagnose from frequency spectra alone since amplitudes will most often be low, and discrete spectral lines might not be seen at calculated defect frequencies. This particularly applies to bearings with inner race rotation mounted with an interference fit between the inner race and the shaft. In these cases, the load zone might be noticeably less than 180° which explains why discrete peaks at defect frequencies may not appear. Reference 15 points out the real problem occurs if a crack appears on the inner race. In these cases, the shaft fit can be relieved and internal clearances of the bearing lost. He points out that such a situation is extremely dangerous which can bring about rapid seizure of the bearing causing the inner race to spin on the shaft or the outer race to spin in the housing resulting in catastrophic failure. In these cases, since the inner race frequency itself or harmonics may not show, but act as a sideband on other frequencies, he recommends referring to the time domain (or time waveform) in addition to the vibration spectra themselves.
 8. In rolling element bearings, when multiple running speed harmonics are present, they can signal either looseness of the bearing on the shaft or within its seat or, much more importantly, they can likewise signal a bearing turning on the shaft or in the housing. The author has been involved in several situations where a number of very low amplitude running speed harmonics (less than .04 in/sec) were present which resulted in catastrophic failures, even with only the low amplitudes. In many of these cases, there were in fact no real defects within the bearings themselves, but the bearing turning on the shaft resulted in severe shaft damage (in one case, over .25 inch diameter was lost before eventual failure).

Thus, there is great information present within the signatures of rolling element bearings that will help identify their current condition and assist the analyst in recommending possible corrective actions. Following in Section 6.073 will be a presentation of a series of typical spectra which can be used by the analyst to track the condition of rolling element bearings under some of the most common failure scenarios which have occurred in the experience of the author.

6.073 Typical Spectra For Tracking Failure Stages Through Which Rolling Element Bearings Pass:

During the last decade, concentrated research and experimental investigations have been conducted on how to best evaluate rolling element bearing condition using vibration analysis and high frequency enveloping techniques. Some of the better papers which have been written include References 11, 12 and 16. In addition, Technical Associates has been deeply involved in further developing rolling element bearing diagnostic capabilities. We have consulted with a variety of clientele on a wide array of machinery ranging from massive, low-speed machines outfitted with large, expensive rolling element bearings to very small, high speed rolling element bearing machinery.

Experiments have been conducted purposely running rolling element bearings to failure while carefully tracking them with vibration signatures (Reference 11). At various intervals, the bearings were broken open and closely inspected for the extent of damage, the type of deterioration and the probable cause for the onset of faults within the bearings. These results would then be closely correlated with the vibration spectra, along with various ultrasonic frequency measurements (i.e., Spike Energy, Shock Pulse and HFD). The bearings were purposely opened up and examined when such data indicated them to be in a wide range of condition (from those with supposedly no faults to those indicated to have extensive damage). Some only had increases in ultrasonic measurements (not in vibration); others showed increases only in high frequency regions; while clearly defined bearing defect frequencies were within vibration spectra of others (some were allowed to catastrophically fail while vibration and ultrasonic responses were closely tracked). Such experiments were performed on a variety of rolling element bearings at different speeds and conditions. This work has enabled the development of fairly well proven diagnostic methods for the tracking of rolling element bearing condition using vibration signature analysis, along with time waveform and ultrasonic analysis.

It was found that the majority of rolling element bearings followed a fairly predictable failure path from the very onset of deterioration through eventual catastrophic failure. This failure path is graphically portrayed in Figure 6.07J which plots bearing damage versus time. Note importantly that bearing damage typically will increase exponentially during the final 10% to 20% of its life. It is here where intensive research and field investigations have found how to use vibration analysis and high frequency enveloping tools to identify "failure stages". From this, a classic 4 stage failure scenario has been developed which will apply to approximately 80% of rolling element bearing failures. This will be presented in Figure 6.07K as Scenario A entitled "4 Primary Failure Stages Through Which Most Rolling Element Bearings Pass". Text describing each of the 4 failure stages will accompany Figure 6.07K.

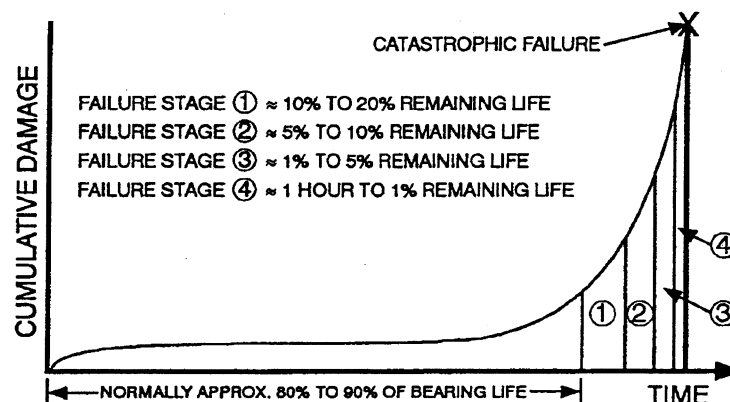


FIGURE 6.07J
TYPICAL FAILURE PATH TAKEN BY ROLLING ELEMENT BEARINGS

As was expected, not all rolling element bearings follow a failure path identical to that which will be outlined for Scenario A. At least five (5) other failure scenarios have been identified. However, due to considerations limiting article length, only this first scenario encompassing the normal failure path taken by approximately 80% of rolling element bearings will be covered (Scenario A). The complete paper referenced in notes accompanying the article covers the remaining five failure scenarios and gives real-world examples of each (Reference 26).

SCENARIO A. 4 PRIMARY FAILURE STAGES THROUGH WHICH MOST ROLLING ELEMENT BEARINGS PASS (FIGURE 6.07P):

Figure 6.07K presents typical velocity spectra for each of the 4 failure stages for most rolling element bearings. These spectra follow the bearing from the very onset of bearing problems in Stage 1 through imminent failure of the bearing in Stage 4 (see Figure 6.07J). Note that the overall spike energy (or HFD) amplitudes given to the right of each spectrum are meant to be rough approximations only. Also, note that documentation below the title of each one of the stages refers to an approximate " L_{10} Life Remaining" which corresponds to the approximate remaining anticipated life of the bearing based on a 90% confidence level (L_{10} Life is documented and its formula is given in the introductory section of this Section 6.07). Today, in general, the machine designer normally will attempt to provide a design life of approximately 5 to 10 years for most common machinery. Therefore, when a percentage on the order of 10% of L_{10} Life is quoted below, this will generally mean from 6 months to 1 year remaining life anticipated (depending on the type of machine and its intended application). Of course, this can vary widely with the machine type, with particular design parameters and whether or not the bearings are provided with the proper lubrication, operating temperature and subjected to acceptable vibration levels.

It is important to point out that bearings do not follow a linear deterioration path, but instead tend to fail exponentially; that is, when the bearing for example enters Stage 3 outlined below, it may still have a fair remaining life, or it may fail rapidly. Once the bearing enters Stage 3, the failure can progress rapidly, particularly towards the end of this stage. Generally, low speed bearings may still have fairly predictable remaining life (unless exposed to high dynamic loads - see L_{10} Life Formula discussion in the introduction to this Section 6.07 which shows that bearing life is inversely proportional to the 3rd power of imposed loading).

Table 6.07A compiled by Charles Berggren in Reference 16 (reprinted here with his permission) roughly adheres to the 4 stages illustrated in Figure 6.07K and documented below:

Stage 1 (Approximately 10% to 20% L_{10} Life Remaining):

The appearance of Spike Energy (or HFD, Shock Pulse, etc.) from very low levels (below .02 to .04 gSE) up to .15 to .25 gSE is the event which defines the onset of Failure Stage 1. For example, Stage 1 shows a normal spectrum indicating a healthy bearing and has only the normal first 3 running speed harmonics in the velocity spectrum. The only evidence of possible bearing problems is that Spike Energy has grown from near 0 to approximately .20 gSE (this is an example amplitude only; actual levels depend on the particular bearing and how close the measurement is to the bearing housing). Note that the Spike Energy itself is not meant to be part of the spectrum in Figure 6.07K but only an overall level (Spike Energy spectra are also available and can be used to confirm that a bearing problem is the cause for increase of this Spike Energy overall level. The Spike Energy response itself responds in the ultrasonic frequency ranges in the region of 2,000,000 CPM (approximately 35,000 Hz). HFD is also a high frequency overall level extending as high as 1,800,000 CPM (30,000 Hz).

During Stage 1, no sound will be detectable by the human ear indicating bearing damage and no change in bearing temperature would be anticipated at this point. Table 6.07A shows that no defects detectable by the human eye would be expected in this stage since such defects would be near microscopic at this stage. Even though physical examination by the naked eye and hand would not normally indicate wear at this stage, a metallurgical analysis would likely reveal damage within the outer .004 to .006 inch (.10 to .15 mm) surface layer of the races and/or rolling elements.

FIGURE 6.07K
4 PRIMARY FAILURE STAGES THRU WHICH MOST ROLLING ELEMENT BEARINGS PASS (VELOCITY SPECTRA)

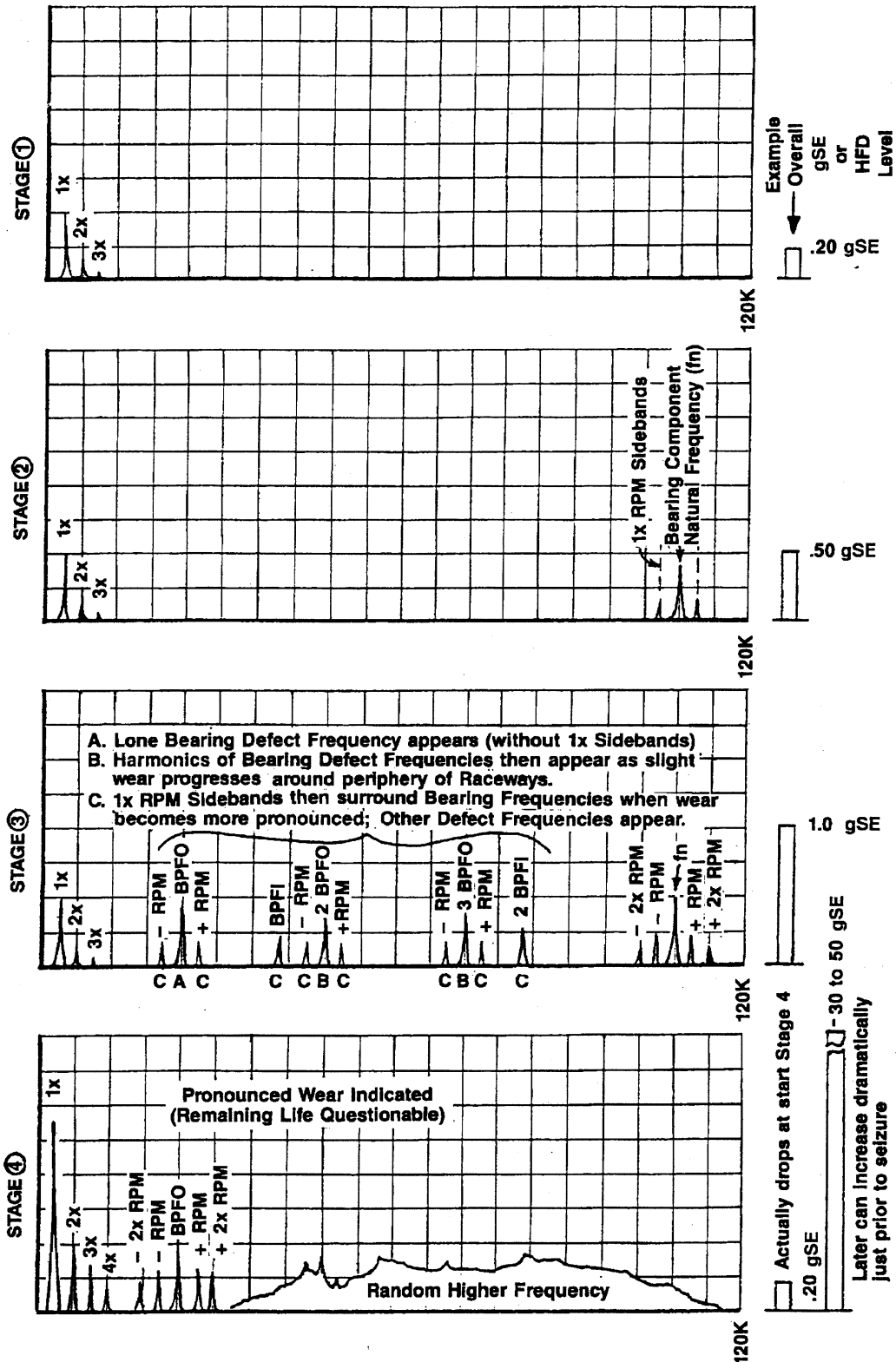


TABLE 6.07A
4 ROLLING ELEMENT FAILURE STAGES (Ref. 16)

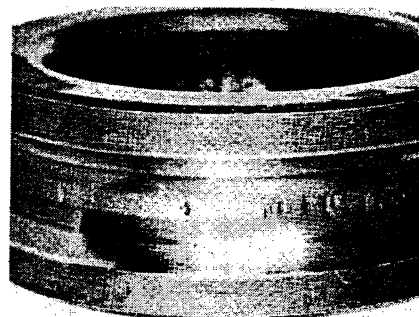
- noise level normal
- temperature normal
- measurable increase in ultrasonic sound, acoustic emission, spike energy, and outer race deflection
- overall vibration low; no discrete spikes at bearing
- frequencies remaining life more than 10% of B-10 rating*

a. Initial Phase.



- slight increase in noise level
- temperature normal
- large increase in ultrasonic sound, acoustic emission, spike energy, and outer race deflection
- slight increase in overall vibration acceleration and velocity
- bearing frequencies clearly visible on log scale, barely visible on linear scale of vibration spectrum; noticeable rise in noise floor
- remaining life less than 5% B-10 rating

b. Second Phase.



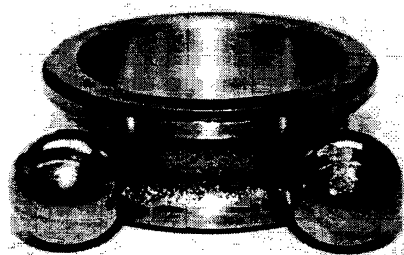
- noise level quite audible
- slight increase in temperature
- very high ultrasonic sound, acoustic emission, spike energy, and outer race deflection
- large increase in overall vibration acceleration and velocity
- bearing frequencies with harmonics and sidebands clearly visible on linear scale
- of vibration spectrum; noticeable rise in noise floor
- remaining life less than 1% B-10 rating*

c. Third Phase.



- change in pitch of noise level
- significant temperature increase
- gradual decline followed by rapid increase in ultrasonic sound, acoustic emission, spike energy, and outer race deflection immediately prior to failure
- significant increase in overall vibration displacement and velocity; decrease in acceleration

d. Final Phase.



Stage 2 (Approximately 5% to 10% L_{10} Life Remaining):

Slight bearing defects begin to "ring" natural frequencies of the installed bearing components (see Figure 6.07K). These natural frequencies are concentrated in the 30,000 to 120,000 CPM range (500 to 2000 Hz). Natural frequencies of most bearings lie between 30,000 and 90,000 CPM (500 - 1500 Hz). These are natural frequencies of the assembled rolling element bearings themselves which do not change in frequency with a change in operating speed (however, they normally will show higher amplitudes with increasing speed due to greater impact velocity). These natural frequencies are excited by the momentary impact between the defective rolling elements and bearing races which not only excite the bearing natural frequencies, but also increase Spike Energy or HFD response (for example, roughly doubling in many cases). It has been the experience of the author that during initial Stage 2, only one or more frequencies appear in these higher frequency regions. Later, towards the end of Stage 2, these frequencies will not only grow in amplitude, but also become modulated with the running speed when wear progresses (that is, 1X RPM sidebands will later appear above and below these natural frequencies). Although modulation of these bearing component natural frequencies most often occurs at 1X RPM, it should be pointed out that such sidebands can also be spaced at bearing defect frequencies (BPFO or BPFI) about the bearing natural frequencies.

Note that the defects themselves may not yet be readily visible to the naked eye in Stage 2. There should be only a slight increase in bearing noise and its temperature should still be roughly normal. Notice that bearing defect frequencies will not likely yet be visible in the velocity spectrum. However, acceleration spectra may now begin to pick up harmonics of such bearing defect frequencies, particularly if a log amplitude scale is employed. Still, at this stage, bearing defect frequency response will normally be erratic.

Stage 3 (Approximately 1% to 5% L_{10} Life Remaining):

Note that each of 3 progressive events are documented for Stage 3 in Figure 6.07K (shown as A, B and C). The letters shown below the horizontal axis of the Stage 3 spectrum correspond to peaks which appear during these 3 progressive events. For the first time, bearing defect frequencies associated with faults in the inner race (BPFI), outer race (BPFO), rolling elements (BSF) and cage (FTF) appear in the velocity spectrum (Event A). Later on in Stage 3, harmonics of these bearing defect frequencies then will appear as slight wear progresses around the periphery of the raceways and/or faults appear on more than one rolling element (Event B). Normally, outer race (BPFO) frequencies themselves will not be modulated by running speed when they first appear. However, when deterioration progresses, 1X RPM sidebands will surround the outer race frequencies. In the case of inner race frequencies (BPFI), 1X RPM sidebands will often appear as soon as do the inner race frequencies (or their harmonics) since inner race defects will have much more response when they are impacted by rolling elements and are within the load zone (they will see very little response when such impacts occur with the inner race defect outside the load zone). Therefore the 1X RPM sidebands often appear with BPFI (and harmonics) due to what is known as "amplitude modulation" (which is covered in depth in the "Analysis II" seminar).

Spike Energy and HFD will continue to grow, doubling or tripling in amplitude. At the end of Stage 3, not only will more and more 1X RPM sidebands appear around bearing defect frequencies, but more sideband families will also usually appear around the bearing component natural frequencies (Event C).

A word of caution should be taken at this point. When the bearings approach the conclusion of Stage 3, the rate of wear becomes highly unpredictable. In any case, it will be well into the exponential part of the failure curve. How much longer the bearing lasts will largely depend on its lubrication, temperature, cleanliness and dynamic loads being imposed upon it by vibration forces from unbalance, misalignment and so forth. At this point, there will be a noticeable change in sound level and frequency and a slight increase in bearing housing temperature. Addition of lubricant at this point may temporarily lower Spike Energy and possibly have some effect in reducing vibration. However, since wear is the problem at this point, both vibration and Spike Energy will return, normally within 12 to 24 hours (the lubricant addition at this point could actually hasten failure if the bearings already were over-lubricated).

Figures 6.07L and 6.07M show a spectrum and trend plot taken from a pump bearing with a classic Stage 3 bearing failure underway. Notice that the Spike Energy increased from .271 to .944 gSE between May 24 and August 23 measurements exceeding the alarm of .750 gSE (Figure 6.07M). Then, looking at the velocity spectrum of Figure 6.07L, each of 4 harmonics of inner race frequency are present ($\text{BFPI} = 7.44 \times \text{RPM} = 13,200 \text{ CPM}$). Note the harmonics at 26400, 39600 and 52800 CPM. Also note the multiple 1X RPM sidebands which have appeared around 2X BPFI and 3X BPFI. Looking at both the velocity spectral data and Spike Energy overall level, the plant would be well advised to replace this bearing right away even though amplitudes of all bearing frequencies are below .10 in/sec.

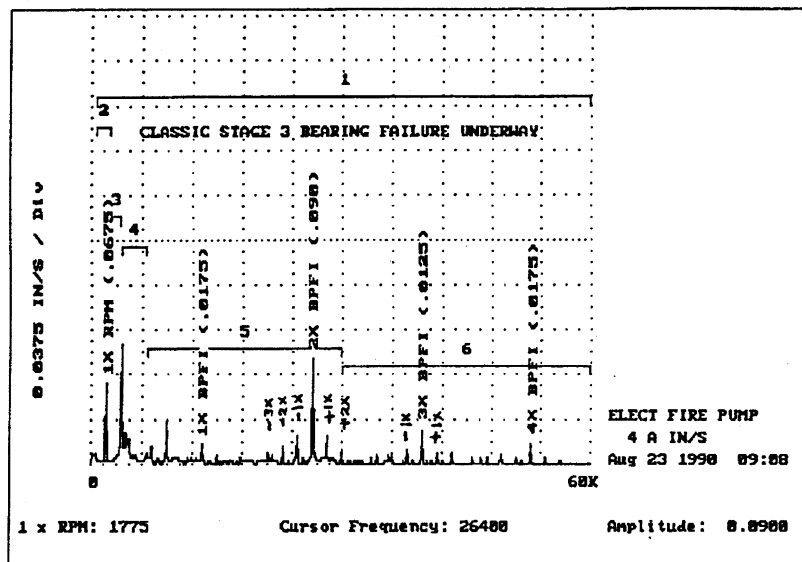
Stage 4 - Approaching Catastrophic Failure (Approximately 1 Hour to 1% L_{10} Life Remains):

The level of 1X RPM normally begins to grow for the first time throughout the bearing failure process in Stage 4, along with harmonics at 2X and 3X RPM. Spike Energy and HFD levels actually begin to drop and amplitudes of the higher bearing frequency harmonics and natural frequencies also will normally drop. Many 1X RPM sidebands appear around bearing defect frequencies (indicating pronounced wear throughout the periphery of the bearing and/or load zone). There will now be a noticeable change in pitch of the bearing noise and likely a significant increase in bearing housing temperature.

Later on in Stage 4, discernible bearing defect component natural frequencies actually begin to disappear and are replaced by a random high frequency "noise floor" which can extend far down into the spectrum obliterating discrete frequency peaks.

There may now be one week of remaining bearing life, on the other hand, the bearing may fail within one hour! The point is that no one knows! Finally, at the end of Stage 4, Spike Energy and HFD normally will decline again (as the bearing itself "disappears"); but, just prior to failure, Spike Energy levels can grow dramatically up to 50 to 100 gSE just before final seizure. Thus, a bearing should never be allowed to continue to operate in Stage 4, for no one knows when it will catastrophically fail.

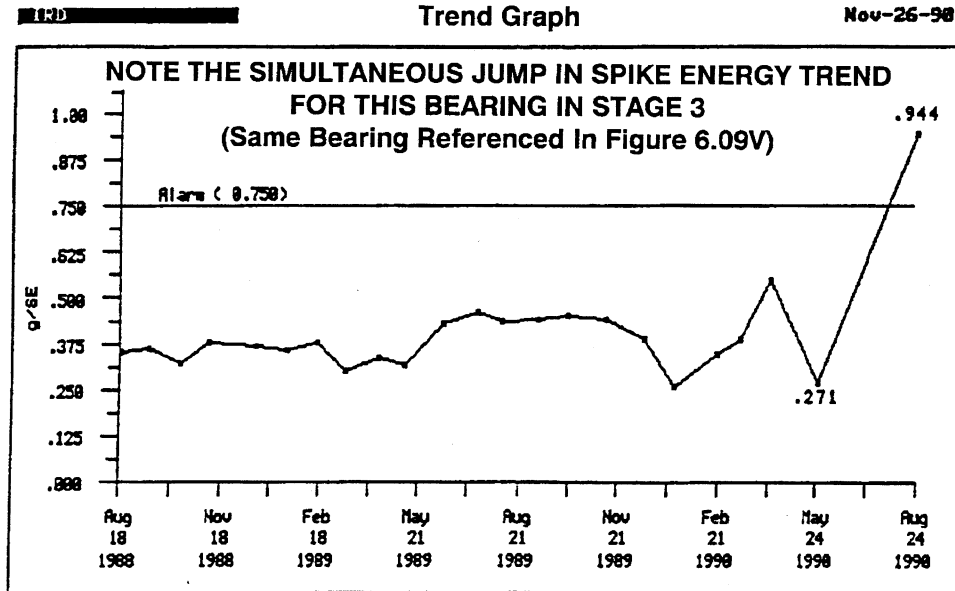
FIGURE 6.07L



2 x 0.500 = 0.0100 1 x RPM: 1775.00

Machine		Pos	Dir	Units	Date	
ELECT FIRE PUMP		4	A	IN/S	Aug 23 1990 09:08	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude	
150	0.08	0.0100	39600	22.31	0.0275	3x BPFI
600	0.34	0.0100	41400	23.32	0.0100	
1800	1.01	0.0675	43200	24.34	0.0100	
3600	2.03	0.1000	52800	29.75	0.0175	4x BPFI
4050	2.28	0.0250				
4500	2.54	0.0200				
6600	3.72	0.0100				
7200	4.06	0.0150				
8850	4.99	0.0350				
13200	7.44	0.0175				BPFI
21000	11.83	0.0100				-3x RPM
22800	12.85	0.0150				-2x RPM
24600	13.86	0.0225				-1x RPM
26400	14.87	0.0900				2x BPFI
28200	15.89	0.0225				+1x RPM
30000	16.90	0.0125				+2x RPM
37800	21.30	0.0125				

FIGURE 6.07M



ELECT FIRE PUMP 4 H .750 g/SE

IRD Mechanalysis Data Analysis System

Machine	ELECT FIRE PUMP	Pos	4	Dir	H	Units	g/SE	Alarm	.750
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
Code	mm/dd/yy			Code	mm/dd/yy				
.353	0	08-18-88	9:25	.442	0	09-07-89	8:51		
.361	0	09-13-88	0:07	.449	0	10-05-89	8:50		
.320	0	10-11-88	9:18	.442	0	11-09-89	10:00		
.378	0	11-07-88	9:19	.390	0	12-14-89	15:38		
.368	0	12-20-88	10:34	.261	0	01-11-90	8:52		
.355	0	01-17-89	9:10	.347	0	02-19-90	8:40		
.380	0	02-14-89	8:51	.388	0	03-12-90	8:58		
.302	0	03-14-89	8:45	.549	0	04-09-90	7:54		
.338	0	04-13-89	8:03	.271	0	05-24-90	7:51		
.316	0	05-09-89	9:18	.944	0	08-23-90	9:08		
.429	0	06-13-89	9:17						
.460	0	07-13-89	9:20						
.434	0	08-03-89	9:25						

Page 1

Machine Report (Latest Data) for: ELECT FIRE PUMP

Date	Time	Pos	Dir	Ampl	Units	Change	Alarm	Inspection
Aug-23-90	09:06	1	A	.150	IN/S	-6%	.300	
Aug-23-90	09:06	1	H	.129	IN/S	48%	.300	
Aug-23-90	09:06	1	H	.329	g/SE	186%	.600	
Aug-23-90	09:06	1	O	.117	IN/S	-----	.300	
Aug-23-90	09:06	1	V	.077	IN/S	8%	.300	
Aug-23-90	09:07	2	A	.157	IN/S	-16%	.300	
Aug-23-90	09:07	2	H	.127	IN/S	14%	.300	
Aug-23-90	09:07	2	H	.332	g/SE	-1%	.600	
Aug-23-90	09:07	2	O	.129	IN/S	-----	.300	
Aug-23-90	09:07	2	V	.061	IN/S	-65%	.300	
Aug-23-90	09:07	3	A	.211	IN/S	41%	.300	
Aug-23-90	09:07	3	H	.180	IN/S	10%	.300	
Aug-23-90	09:07	3	H	.337	g/SE	42%	.600	
Aug-23-90	09:07	3	V	.189	IN/S	26%	.300	
Aug-23-90	09:08	4	A	.204	IN/S	-19%	.300	
Aug-23-90	09:08	4	H	.204	IN/S	2%	.300	
Aug-23-90	09:08	4	H	.944	g/SE	248%	.750	
Aug-23-90	09:09	4	V	.121	IN/S	-15%	.300	

18 lines this report

Other Rolling Element Bearing Failure Scenarios Which Have Been Identified

While Scenario A outlined in Figure 6.07K probably encompasses the failure paths followed by approximately 80% of rolling element bearings, other scenarios have been identified. To date, five (5) other failure scenarios have been documented. Due to limitations on the length of this article, it is not feasible to provide all the text, spectrum diagrams and example illustrations as was done with Scenario A for these other scenarios (complete documentation on each of the remaining failure scenarios is included within the "Analysis II" seminar text, Reference 26). However, following below in Table 6.07B is a tabulation of each of these remaining failure scenarios:

**TABLE 6.07B
DESCRIPTION OF OTHER IDENTIFIED ROLLING
ELEMENT BEARING FAILURE SCENARIOS**

FAILURE SCENARIO	SCENARIO DESCRIPTION
B.	"One Pronounced Localized Fault on One Raceway Exciting One Dominant Bearing Defect Frequency"
C.	"Continual Wear Throughout the Periphery of a Raceway"
D.	"Development of a Serious Fault Frequency Acting as a Sideband Rather than a Fundamental Frequency"
E.	"Condition Deterioration Ending Either With Severe Mechanical Looseness or the Bearing Turning on the Shaft"
F.	"Development of Excessive 1X RPM Modulation About Race Frequencies Ending up with Excessive Fatigue Spalling on Raceways"

6.08 INTRODUCTION TO GEAR PROBLEM DETECTION

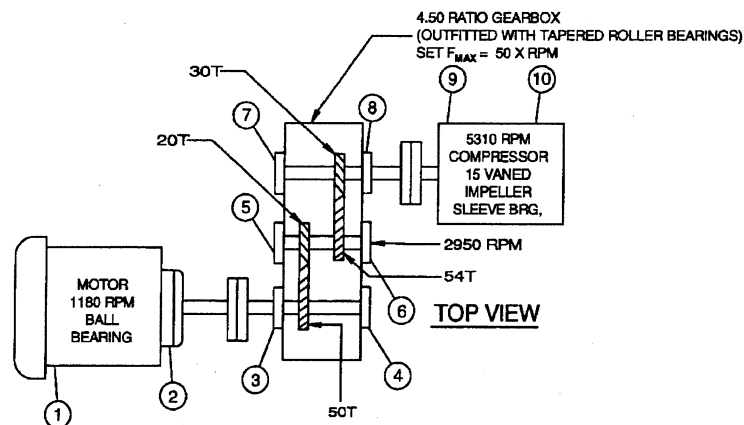
Following in this paper will be an introduction on how problems are detected with gears using vibration signature analysis (occasionally with the assistance of ultrasonic spectral analysis such as spike energy spectra). This paper will cover the fundamentals of gear analysis, particularly concentrating on the wear of gear teeth. The focus of this paper will be on how wear of gear teeth is detected while the emphasis of papers in higher level seminar texts will be on how to detect cracked or broken gear teeth, gear eccentricity and/or excessive backlash, gear assembly phase problems, hunting tooth problems, etc. However, it is of great importance that the analyst who is beginning to evaluate gears understand the significance of gear mesh frequencies (#teeth X RPM) and of sideband frequencies surrounding the gear mesh frequencies (spaced at 1X RPM on either side of gear mesh frequencies).

Figure 6.08A will help illustrate how this frequency important to gear analysis known as the gear mesh frequency (GMF) is calculated. As the figure shows, $GMF = \# \text{Gear Teeth} \times \text{Gear RPM}$. In this case, notice there are two gear mesh frequencies in this 2-stage gearbox (GMF_1 and GMF_2). The first mesh involves a 50 tooth gear operating at 1180 RPM meshing with a 20 tooth gear at 2950 RPM. The next mesh (GMF_2) involves a 54 tooth gear operating at the same 2950 RPM meshing with a 30 tooth gear running at 5310 RPM. Note just below the drawing the calculations of GMF_1 and GMF_2 . Notice that the number of teeth and RPM on either gear can be used to calculate the gear mesh frequency. For example, in the case of GMF_1 , it can be calculated by multiplying 50T X 1180 RPM, which is the same as 20T X 2950 RPM, or 59,000 CPM.

6.081 Specification of Spectral Setup for Detecting Gear Wear:

Later, it will be pointed out that one should always use a frequency span up to a minimum of 3.25 X the gear mesh frequency when evaluating gears (if GMF is unknown, it is probably a good idea to specify the maximum frequency (F_{MAX}) at about 200X RPM and to employ 1600 FFT Lines to allow detection of low-speed bullgear sideband frequencies as well as higher speed pinion sidebands). This is due to the fact that many gear wear problems do not necessarily occur at the fundamental gear mesh frequency (1X GMF), but even at 2X GMF or 3X GMF. In fact, one of the most common frequencies at which gear wear is first detected is at 3X GMF. This is likely due to the fact that each mesh event includes three separate events - a sliding action as one tooth enters mesh with another; a rolling action as they approach the root of the mating gear; and another sliding action as the teeth disengage. Any interruption to this meshing action can therefore generate frequencies at three pulses per mesh (or 3X GMF). Since taking data up to 3.25X GMF will often require high frequency measurements, it will often mandate acceleration spectra if F_{MAX} exceeds approximately 300,000 CPM (or 5000 Hz). In addition, use of 1600 line FFT spectra is necessary in order to even detect lower speed bullgear RPM sidebands at these high frequencies. Using 1600 lines of FFT resolution will separate them from the GMF fundamental or harmonics.

Because gear analysis requires measurements up to 3.25X GMF as a minimum, it may sometimes require the use of two separate transducers on each gearbox position. The first transducer might be used to measure a span extending up to only about 40X or 50X RPM which would detect such problems as unbalance, misalignment, eccentricity, bearing wear, etc. The second measurement at the same gear locations (and likely with a much higher frequency span) would be used to evaluate potential gear wear. Therefore, each individual location may require two separate measurements having two different frequency spans for complete evaluation of gear health during each survey.



$$GMF_1 = (50T)(1180 \text{ RPM}) = (20T)(2950 \text{ RPM})$$

$$GMF_1 = 59,000 \text{ CPM}$$

$$\text{POS. 3HI } F_{MAX} = GMF \times 3.25 \\ = 191,750 \text{ CPM}$$

$$\text{USE POS 3HI } F_{MAX} = 200,00 \text{ CPM}$$

$$GMF_2 = 54T \times 2950 \text{ RPM} = (30T)(5310 \text{ RPM})$$

$$GMF_2 = 159,300 \text{ CPM}$$

$$\text{POS. 6HI } F_{MAX} = 3.25 \times 159,300 \\ = 517,725 \text{ CPM}$$

$$\text{USE POS 6HI } F_{MAX} = 540,00 \text{ CPM}$$

(Must Measure Acceleration Here)

NOTE:

Spectral Band Alarm Setup for the 2-Stage speed increaser shown in Figure 6.08A is shown for all motor, gearbox and compressor positions in Figure 3 of Chapter 7.

FIGURE 6.08A
CENTRIFUGAL COMPRESSOR DRIVEN BY A 2-STAGE GEARBOX

For example, in the case of the gearbox in Figure 6.08A, a good frequency range for gearbox input shaft positions #3 and #4 would be a frequency span of about 60X the 1180 RPM, or a maximum frequency (F_{MAX}) up to approximately 72,000 CPM. Comparing this to the required high frequency gear wear frequency span of 200,000 CPM shown in Figure 6.08A for evaluation up to 3.25X GMF_1 , a second measurement (Positions 3HI and 4HI) should also be specified, at least in the horizontal and axial directions up to this high F_{MAX} .

Note that helical gears (like those shown in Figure 6.08A) and bevel gears tend to cause much vibration in the axial direction when wear develops; while worn spur gears normally cause greatest vibration in radial directions (horizontal or vertical).

In addition, the two different frequency spans at each of the gearbox positions may require two separate transducers, and even two separate units of measure (velocity and acceleration). Looking at Figure 6.08A, note that Positions 6HI and 8HI will require analysis up to 540,000 CPM (9000 Hz); whereas the 2950 RPM of this shaft would require analysis only up to about 150,000 CPM (for detection of unbalance, misalignment, looseness, etc.). Therefore, the 150,000 CPM could be accurately evaluated by a magnet mounted accelerometer measuring velocity; whereas the high frequency acceleration measurements up to 540,000 CPM would likely require either a stud mounted or adhesive mounted accelerometer in order to give repeatable, accurate data up to such a high frequency. This is very common with such multi-stage speed increasers.

Figure 6.08B pictures a large bullgear which is meshing with each of four pinions mounted on the impeller assembly shafts of centrifugal air compressors. Note that even though the bullgear itself meshes with each of the four pinions, there will only be one gear mesh frequency (GMF) since the one common gear (bullgear) meshes with all four other gears.

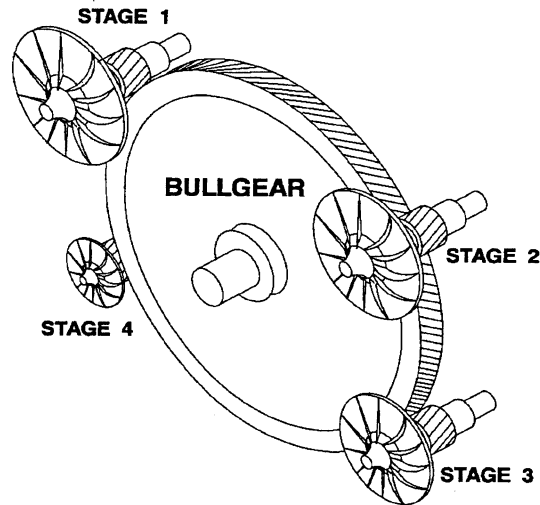


FIGURE 6.08B
PERSPECTIVE VIEW OF AN INGERSOLL-RAND CENTAC CENTRIFUGAL
AIR COMPRESSOR BULLGEAR MESHING WITH 4 IMPELLER ASSEMBLY
PINIONS (MODEL C-21 2100 ICFM)

COMPONENT	# TEETH	RPM	Hz
Stage 4	25T	49,270	821.13
Stage 3	27T	45,620	760.3
Stage 2	32T	38,490	641.51
Stage 1	42T	29,325	488.75
Bullgear	344T	3,580	59.68

GMF = #Bullgear Teeth X Bullgear RPM = # Stage 1 Teeth X Stage 1 RPM

GMF = 344T X 3580 RPM = 29,325 RPM X 42T

GMF = 1,231,680 CPM = 20,528 Hz

6.082 Indications of Gear Tooth Wear:

A fairly large number of indications of potential gear problems exist and are summarized in the **Illustrated Vibration Diagnostics Chart** found in the beginning of Chapter 6. Here, we will concentrate on gear tooth wear itself, leaving some of the more exotic analyses to the texts of higher level seminars.

Figure 6.08C will be used to illustrate how gear wear is detected (note that Figure 6.08C pictures a 2-Stage rotary screw compressor including its driving bullgear, timing gears on the low and high pressure stages, and the drive gear arrangement for the gear lubrication pump). Following below will be many of the comments which should be made concerning detection of gear tooth wear:

1. One of the first indications of gear tooth wear is a pronounced increase in amplitude of the fundamental gear mesh frequency (1X GMF) and/or its harmonics (2X GMF, 3X GMF, etc.). However, one of the problems with watching only the amplitude of gear mesh frequencies is that gear mesh frequencies can sometimes significantly increase or decrease, depending on load (particularly in the case of the fundamental gear mesh frequency which may see an amplitude increase of a factor of 2X or 3X, depending on the machine and gear type).
2. Of even greater concern will be watching for increases in both the amplitude and the number of sideband frequencies surrounding gear mesh frequencies. Such sideband frequencies will be spaced at the RPM of each mating gear. For example, please refer to the plot shown in Figure 6.08D. Here, each of three gear mesh harmonics for the gear pump itself exist and are labeled "OPMF", "2X OPMF" and "3X OPMF". Note particularly the number of sideband frequencies surrounding 2X OPMF. There are four families of sidebands surrounding 2X OPMF with a spacing of approximately the 2719 RPM speed of the pump shaft (F_{op} in Figure 6.08C).
3. Normally, when gears are in good condition and well aligned with respect to one another, the amplitude of the fundamental gear mesh frequency (1X GMF) will be highest with levels of 2X GMF and 3X GMF falling off. In addition, they will usually only show about one family of sidebands which will be spaced at the RPM of each gear. For example, referring to Figure 6.08C, note the meshing of the 55 tooth gear operating at 1780 RPM and the 36 tooth gear operating at 2719 RPM oil pump shaft speed. In this case, the gear mesh frequency (GMF_{op}) is 97,900 CPM (55T X 1780 RPM). Therefore, if this mesh is in good shape and well aligned, normally a low amplitude in this case would occur at 97,900 CPM (GMF_{op}) and its multiples; as well as lower amplitude sideband frequencies spaced in this example at 1780 RPM input speed and at 2719 RPM output speed. Such sidebands would occur on either side of this gear mesh frequency.

In the case of the spectrum shown in Figure 6.08D, note that each of four sideband families were found to be surrounding 2X oil pump mesh frequency (2X OPMF), each of which were spaced at the oil pump shaft speed of 2719 RPM. Therefore, this indicated a potential problem. In this case, since amplitudes at both the fundamental oil pump mesh frequency (OPMF) and 3X OPMF were low, and since there were so many 2719 RPM sidebands surrounding 2X OPMF, it was concluded that the oil pump gears were possibly misaligned with respect to one another. Subsequent inspection found this was the case as evidenced by wear patterns showing that the load was only being carried by about 25% to 30% of the face width of each gear tooth. Corrective measures were taken to better align these gears, resulting in significantly increased life of the oil pump.

For comparison purposes, Figure 6.08E was taken on a nearby sister machine (Compressor #1) which showed much lower vibration at 2X OPMF (59,565 CPM) with data taken at the identical location (Position 9A in Figure 6.08C). In the case of Figure 6.11E, note that narrowband envelope alarms were specified for each compressor and were used by the PMP software to detect any alarm violations like that which occurred with the #3 rotary screw compressor with plots shown in Figures 6.08D and 6.08E.

4. Often, initial tooth wear will be detected at 3X gear mesh frequency due to the sliding - rolling - sliding action which takes place during each mesh as described above. Here again, in this case, look not only for significant amplitude increases in 3X GMF, but even greater amplitude increases in sidebands surrounding it; as well as more and more sideband families appearing during subsequent surveys.

5. If more than one gear in a mesh has worn teeth, sidebands will occur around gear mesh frequencies with a spacing at the RPM of each of the gears having worn teeth. For example, referring back to Figure 6.08C, if both the 32 tooth and 48 tooth timing gears on the high pressure stage become worn, sidebands spaced at both the 32 tooth gear speed of 11,443 RPM, as well as at the 48 tooth gear speed of 7629 RPM would appear around the high pressure gear mesh frequency (GMF_{HP}) of 366,171 CPM and/or its harmonics.

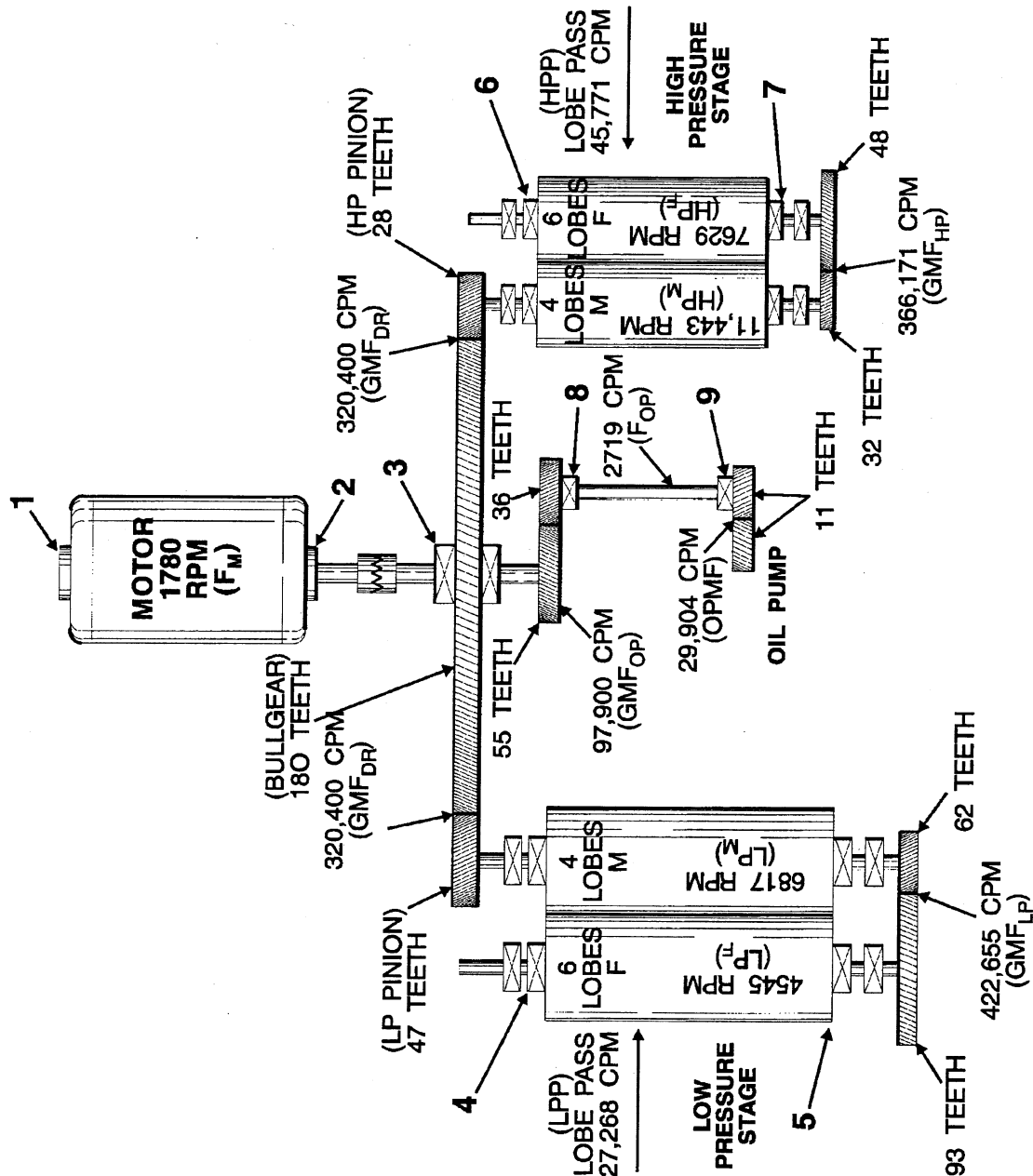


FIGURE 6.08C
ACCELEROMETER LOCATIONS ON A 2-STAGE ROTARY SCREW
AIR COMPRESSOR

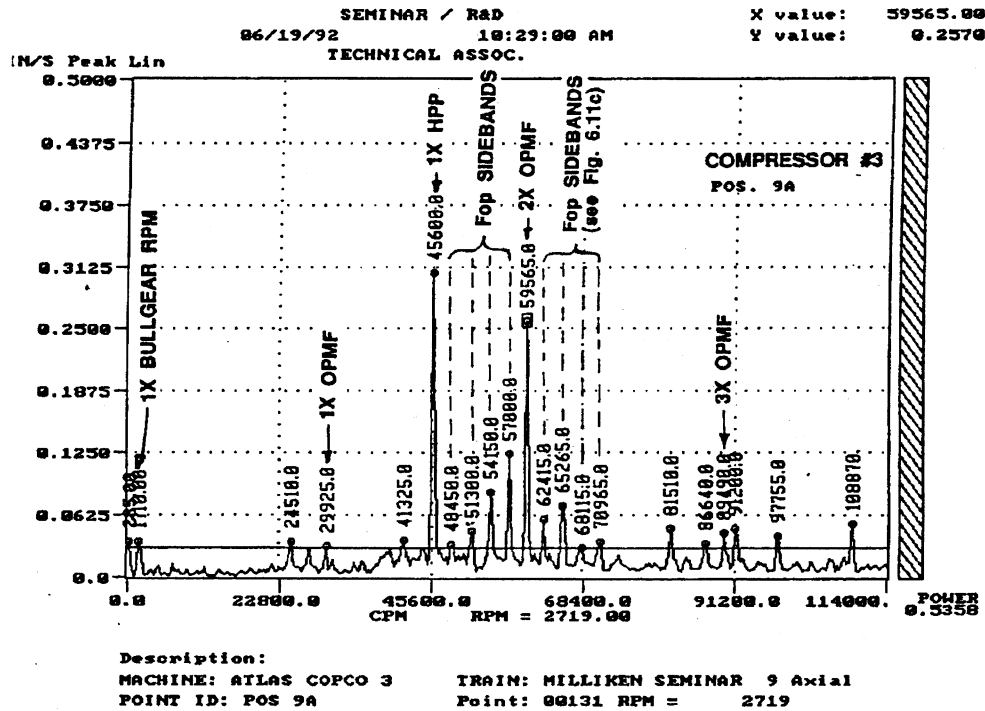


FIGURE 6.08D
THRESHOLD PLOT

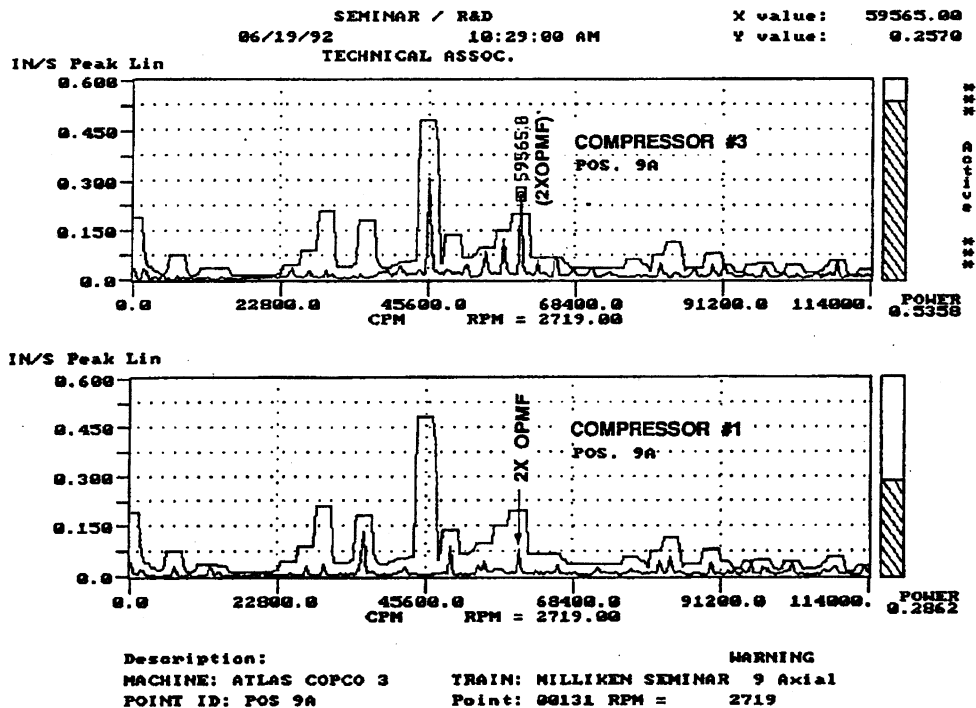


FIGURE 6.08E
NARROWBAND ALARM PLOT COMPARISON

6.09 INTRODUCTION TO ELECTRICAL PROBLEM DETECTION

Many find it surprising that you can detect not only mechanical problems, but also electrical problems using vibration signature analysis. However, remember that the magnetic fields within a motor create flux which induces electromagnetic forces, and that these, along with forces from mechanically induced problems must all be supported by the bearings. These forces shown in Figure 6.09A can then be measured directly by force transducers placed on the bearing housings, or indirectly by vibration transducers such as accelerometers, velocity pickups or noncontact displacement probes. Accelerometers directly measure the quotient of force divided by mass (f/m), which is acceleration (according to Newton's Law). Again, it does not matter whether these forces themselves are mechanically or electrically induced.

Figures 6.09B and 6.09C picture the stator and rotor of an induction motor, along with a close-up view of a rotor, rotor bars and shorting rings. Note some important formulas that are included in Figure 6.09B. These include the calculation of synchronous speed (N_s) which is dependent on the line frequency (F_L) supplied and the number of poles (P). Also shown is a formula for determining the slip frequency (F_s) which is just the difference in the synchronous speed and actual RPM (for example, in a two pole motor turning 3580 RPM, the slip frequency would be $3600 - 3580$, or 20 CPM). Also shown is a formula for calculating what is known as the pole pass frequency (F_p) which equals the slip frequency X the number of poles (in the example given above having 2 poles, the slip frequency would be $20 \text{ CPM} \times 2 \text{ poles} = 40 \text{ CPM}$). Finally, another formula is given for calculating the rotor bar pass frequency (RBPF) which equals the number of rotor bars X the rotor RPM.

Please refer to Figure 6.09C which shows the three components making up the rotor including the rotor bars, rotor laminations (windings) and shorting rings on either end (also sometimes known as end rings).

The **Illustrated Vibration Diagnostics Chart** given in the beginning of Chapter 6 shows that a large variety of electrical problems can be detected using vibration analysis. In this introductory paper on electrical problem detection, most of the emphasis will be given on how to calculate problems within the stator, rotor problems and variable air gap between the rotor and stator. Note that both stator problems and variable air gap problems are detected by high amplitudes at 2X electrical line frequency ($2F_L$). In fact, probably the majority of electrical problems occur at this same frequency ($2F_L$). The remaining electrical problems included in the **Illustrated Vibration Diagnostics Chart** will be discussed in texts given in higher seminar levels.

6.091 Why many electrical problems occur at 2X line frequency:

Many electrical problems are detected due to "higher than normal" amplitudes at 2X electrical line frequency (also known as synchronous frequency). In the United States, the line frequency is normally set at 60 Hz (3600 CPM), whereas in Europe and other locations, it is at 50 Hz (3000 CPM). Therefore, with respect to the United States, a frequency of great importance when detecting electrical problems will be that at 120 Hz (or 7200 CPM).

Figure 6.09A helps explain why so many electrical problems involve twice line frequency ($2F_L$) rather than the fundamental line frequency. Referring to Figure 6.09A, note that during one rotation of the 3600 CPM stator field of this two-pole motor, the magnetic pull towards the closest pole rises from zero to a maximum 2X per revolution in this rotor which is displaced off center (eccentrically) within the stator. Since the field itself revolves at 3600 revolutions per minute, the magnetic pull reaches a maximum at a frequency of 7200 times per minute (or 7200 CPM). Another way of looking at it is that because the close side of the rotor will first be attracted to the north, and then to the south pole, the force itself will vary at 2X line frequency of the magnetic

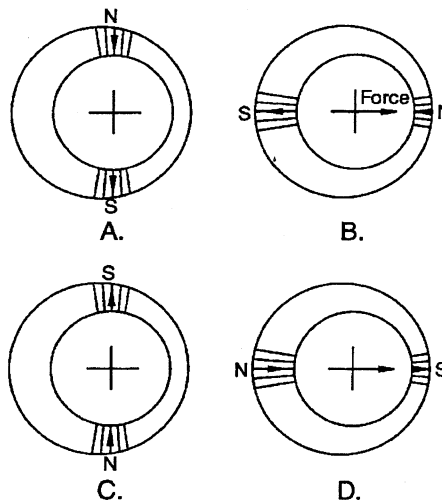


FIGURE 6.09A
UNEVEN AIR GAP BETWEEN MOTOR AND STATOR

field relative to the eccentricity. Therefore, when the rotor is not centered within the stator causing a variable air gap between the rotor and stator, it will always affect 2X line frequency vibration. This fact is true whether you have a two-pole motor or not. That is, if one had a four-pole motor energized with 1800 CPM synchronous line frequency, each of the four poles would pulse as they passed by the small air gap pictured in illustration B of Figure 6.09A generating four pulses per revolution. Then, given the 1800 CPM synchronous frequency of the four-pole motor, this same 7200 CPM frequency will be affected. This same phenomenon would occur with a six-pole motor supplied with 1200 CPM synchronous frequency, still producing 7200 CPM twice line frequency.

Following below will be a discussion on how electrical problems concerning the stator and variable air gap are detected:

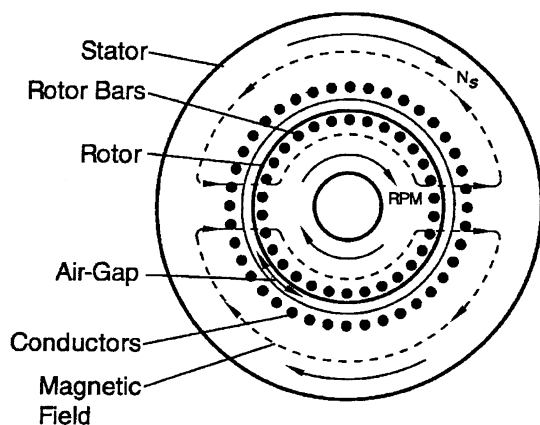
6.092 Stator Problems:

Stator problems detectable by vibration analysis include each of the following:

- Stator Eccentricity (an eccentric stator producing a stationary differential air gap between the rotor and stator)
- Shorted Laminations (insulation problems with lamination layers which can cause localized heating)
- Loose Iron (any looseness or weakness in the stator)

These problems exhibit the following characteristics:

1. All such stator problems generate high vibration at 2X line frequency. However, they do not necessarily generate pole pass frequency sidebands since they originate within the stator, and are not therefore modulated by either running speed or slip frequency. Figures 6.09D and 6.09E indicate a serious electrical problem. Figure 6.12D is a spectrum captured by a data collector during a regularly scheduled PMP route. Note the high amplitude of .230 in/sec at 7200 CPM which exceeded Band 3. A real-time analyzer was then employed on the



$$N_s = \frac{120 F_L}{P} = \text{SYNCHRON. SPEED}$$

$$F_s = N_s - \text{RPM} = \text{SLIP FREQ.}$$

$$F_p = (F_s)(P) = \text{POLE PASS FREQ.}$$

$$\text{RBPF} = \# \text{ ROTOR BARS} \times \text{RPM}$$

WHERE:

F_L = ELECTRICAL LINE FREQ. (often 60 Hz)

RPM = ROTOR SPEED

N_s = SYNCHRONOUS SPEED

F_s = SLIP FREQUENCY ($N_s - \text{RPM}$)

F_p = POLE PASS FREQUENCY

P = # POLES

RBPF = ROTOR BAR PASS FREQUENCY

FIGURE 6.09B
DIAGRAM OF AN INDUCTION MOTOR ALONG WITH ITS STATOR, ROTOR, ROTOR BARS, AIR GAP AND MAGNETIC FIELD

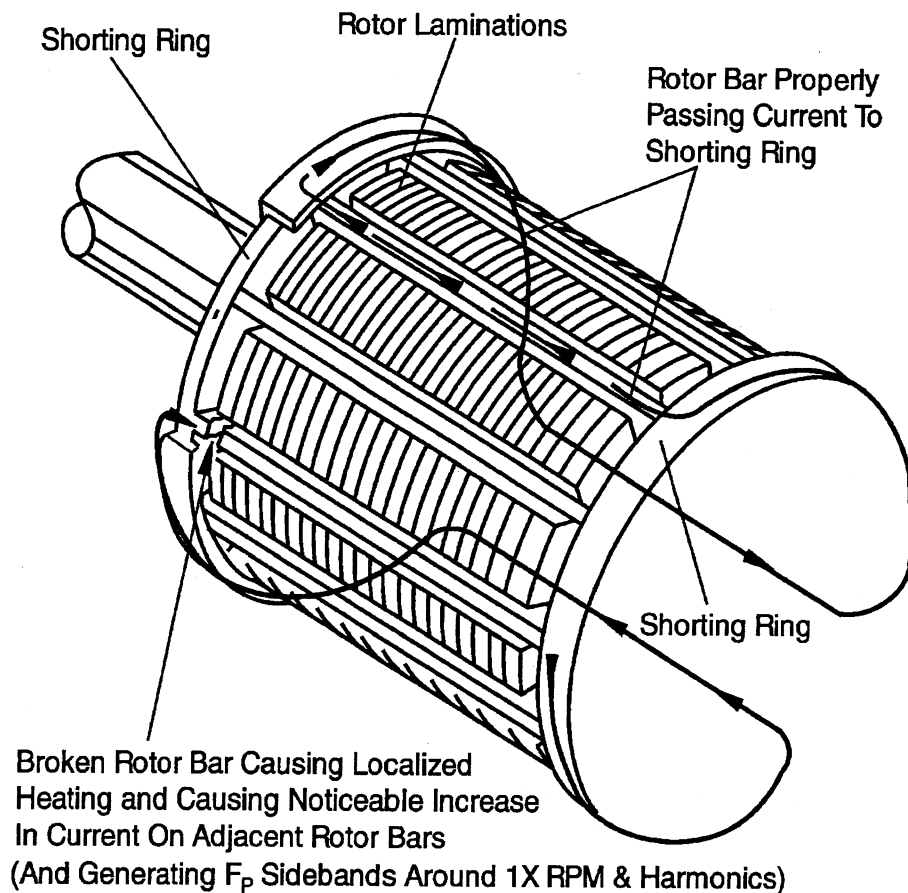


FIGURE 6.09C
ISOMETRIC VIEW OF A ROTOR INCLUDING ITS ROTOR BARS, SHORTING RINGS AND ROTOR LAMINATIONS

same day to better define this problem which is shown by the zoom spectrum of Figure 6.09E. This spectrum showed a high level of .228 in/sec at 7200 CPM ($2F_L$) and only .0044 in/sec at 2X RPM. Please also note the absence of any pole pass sidebands about 2X line frequency which indicated a stator problem. Upon inspection, a stator eccentricity problem was found.

2. Concern should likely be given motors whose vibration under load exceeds .025 in/sec peak at 2X line frequency on new or rebuilt motors, or .070 in/sec peak on in-service motors having a stator problem (applies to motors in general ranging from 50 HP to 1000 HP). This amplitude applies specifically to the peak at 7200.0 CPM itself ($2F_L$). However, if this motor is directly driving a precision machine tool spindle, $2F_L$ levels must be much lower, on the order of .015 in/sec or less (Note importantly that these amplitudes apply to induction motors running under at least 60 to 70% load).
3. Stator eccentricity produces an uneven stationary air gap between the rotor and stator which results in highly directional vibration, depending on the largest gap differential. The largest magnetic forces occur at a minimum rotor/stator gap. Therefore, the electro-magnetic forces themselves go from a minimum to a maximum each revolution producing vibration at twice line frequency (7200 CPM).
4. Differential air gaps should not exceed 5% for induction motors and 10% for synchronous motors. If the vibration amplitude at $2F_L$ grows over time, the motor should be inspected (if physically possible) by marking a point both on the rotor and the stator. Then, measuring the air gap at the point where the marks align, rotate the rotor in 45° increments and measure the air gap at the point where the stator is marked. If the variation exceeds approximately 5%, the air gap difference is likely due to an eccentric rotor. The next step is to rotate the rotor again in 45° increments, this time measuring the gap at the point where the rotor is marked. If this gap varies more than 5% for these measurements, an eccentric stator is indicated.
5. Loose iron is due to localized stator support weakness or looseness.
6. Shorted stator laminations can cause uneven, localized heating which can actually bow a motor shaft. This produces thermally-induced bow which can significantly grow with operating time, sometimes causing the stator to contact the rotor which can be catastrophic.

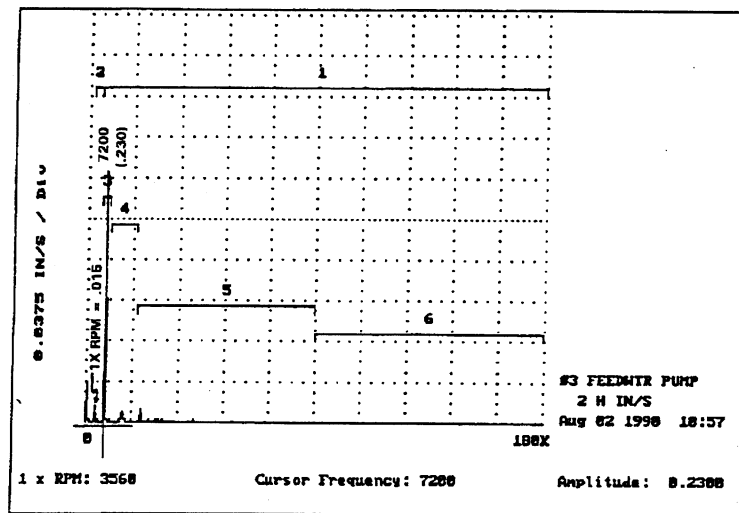


FIGURE 6.09D
NORMAL PMP ROUTE SPECTRUM WITH $F_{MAX} = 50 \text{ X RPM}$

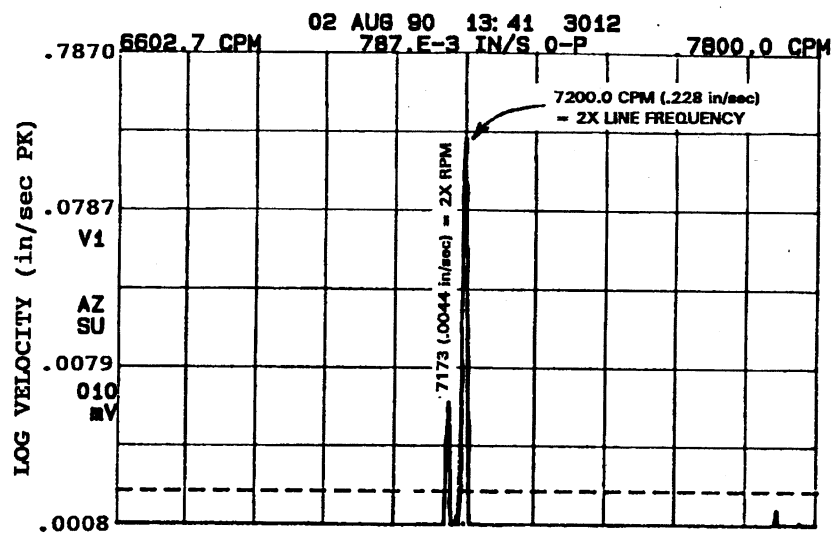


FIGURE 6.09E
ZOOM SPECTRUM INDICATING PROBLEM WITH THE STATOR
(NOTICE THE ABSENCE OF POLE PASS SIDEBANDS)

6.093 Eccentric Rotor and Variable Air Gap:

In the case of an eccentric rotor, the rotor itself is not concentric with its centerline (see Figure 6.09F). Therefore, an eccentric rotor produces an uneven air gap between the rotor and stator which will rotate with the rotor (as opposed to a stationary air gap in the case of an eccentric stator). An eccentric rotor can be caused by shorted rotor laminations resulting in localized heating and inducing a bowed rotor, or a simple out-of-round rotor.

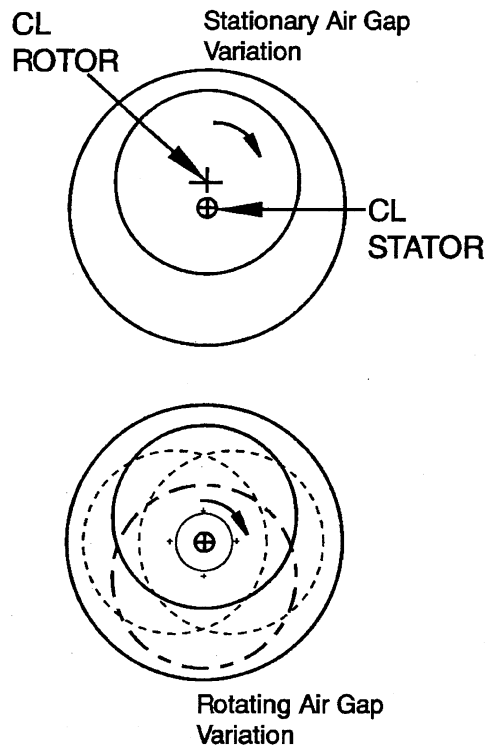


FIGURE 6.09F
STATIONARY & ROTATING AIR GAP VARIATIONS

An eccentric rotor will exhibit the following characteristics:

1. An eccentric rotor most often will produce a high vibration at twice line frequency ($2F_L$) accompanied by sidebands spaced at pole pass frequencies ($F_p = \# \text{ Poles} \times \text{slip frequency}$). That is, for a two-pole, 3600 RPM motor, the sidebands will be at twice slip frequency; while for a four-pole motor, sidebands would be at 4X slip frequency. Figures 6.09G and 6.09H show a spectrum indicative of an eccentric rotor problem producing a variable air gap. First, the wideband spectrum of Figure 6.09G showed a high overall of .295 in/sec with .162 in/sec at a "so-called" 7200 CPM frequency. The zoom spectrum of Figure 6.09H revealed a high level of .166 in/sec at 7200 CPM, along with well-formed pole pass sidebands indicative of an eccentric rotor.

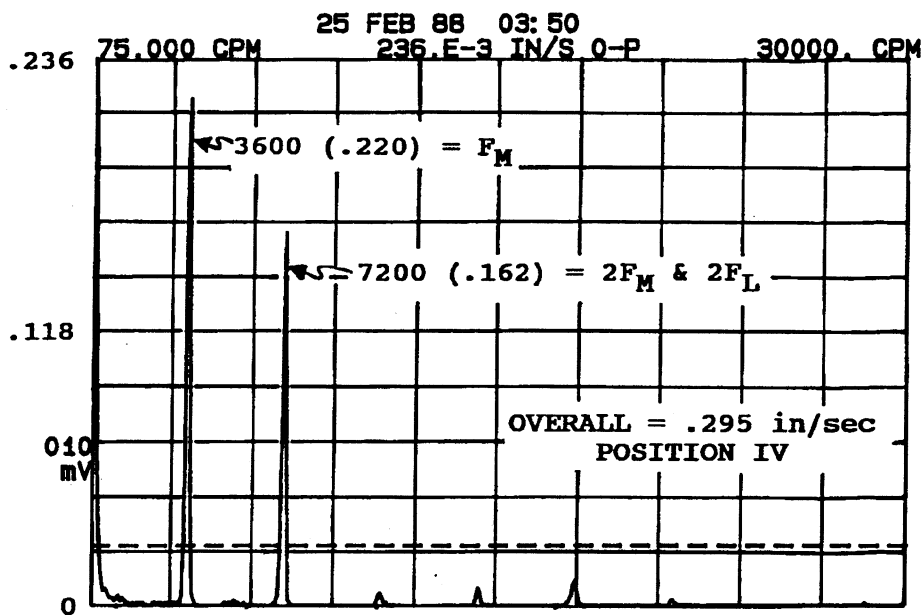


FIGURE 6.09G
30,00 CPM SPECTRUM FOR A CIRCULATING WATER PUMP MOTOR

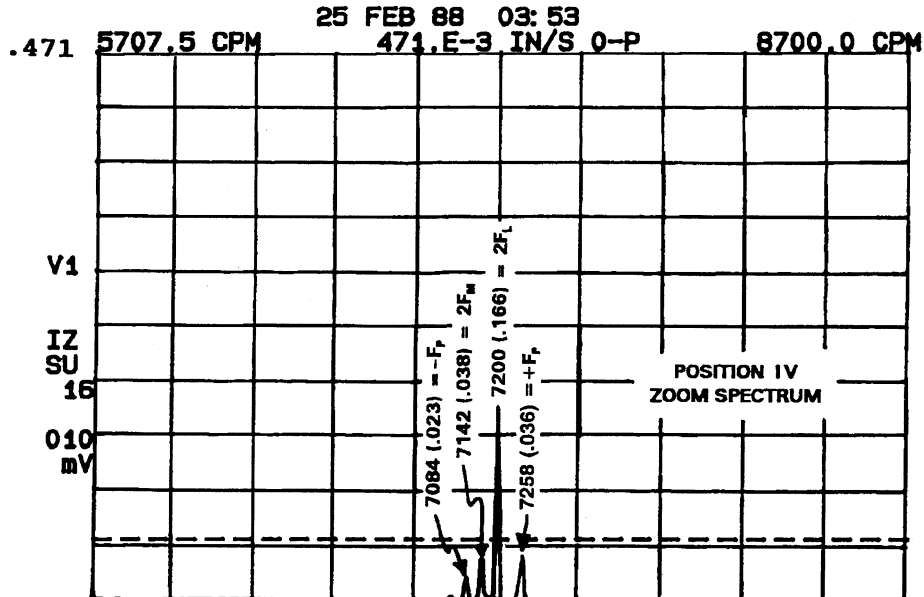


FIGURE 6.09H
ZOOM SPECTRUM INDICATING AN ECCENTRIC ROTOR
(NOT MECHANICAL LOOSENESS OR MISALIGNMENT)

2. Concern should likely be given motors running under load whose amplitude at twice line frequency (7200 CPM) exceeds approximately .025 in/sec for new or rebuilt motors; or .070 in/sec peak for in-service motors assuming these are general purpose motors. If serving a precision machine tool spindle, the allowable at $2F_L$ would be only approximately .015 in/sec. This would apply to most induction motors ranging in size from approximately 50 HP to 1000 HP. It is important to clarify that this amplitude applies specifically to 2X line frequency itself. Note that these amplitudes apply to induction motors running under at least 60 to 70% load.
3. An eccentric rotor may require adjustment of bearing housings themselves and/or machining of the rotor journals in order to provide an air gap within tolerance all around the periphery. Importantly, an eccentric rotor can also be produced by mechanical problems such as severe misalignment.
4. In a predictive maintenance program, when an eccentric rotor is indicated by pole pass frequency sidebands around $2F_L$, they should be closely trended in future surveys. For example, when a peak at $2F_L$ exceeds roughly .070 in/sec ("Alarm 1"), it should be closely trended in future surveys. If its amplitude noticeably increases above roughly .100 in/sec ("Alarm 2"), and if pole pass sideband amplitudes likewise increase, much greater concern should be given, particularly if even more pole pass frequency sidebands appear above and below $2F_L$ (normally 7200 CPM). On the other hand, if the amplitudes at 7200 CPM and sidebands remain stable over several surveys, no further damage is likely being done to the motor - even if these amplitudes are as much as .175 in/sec at 7200 CPM. In these cases, it will likely be satisfactory just to continue trending. However, the motor itself will likely have a lowered life expectancy.
5. Note that a motor having an eccentric rotor will often experience higher and higher vibration as it comes up to temperature. For example, when such a motor is first started, it may have a level of only .10 in/sec. Then, after about 10 minutes operating time, the amplitude might increase to, for example, .14 in/sec. Next, after about 20 minutes, it may continue to grow up to about .18 in/sec. Finally, after 30 minutes, it may experience possibly .25 in/sec or more. This can be caused by nonuniform heating of the rotor on one side relative to the other. Left uncorrected, it can result in catastrophic failure if the rotor bows sufficiently to cause the rotor to contact the stator.

6.094 Rotor Problems:

Rotor problems detectable by vibration analysis include each of the following:

- Broken/Cracked Rotor Bars or Shorting Rings (see Figure 6.09I)
- Bad High Resistance Joints between Rotor Bars & Shorting Rings
- Shorted Rotor Laminations
- Loose/Open Rotor Bars not making good contact with End Rings

These problems exhibit the following characteristics:

1. Probably the key area of concern for broken or cracked rotor bars, is the presence of pole pass frequency sidebands around 1X RPM and running speed harmonics (in the case of 2-pole motors, these sidebands will be at 2X slip frequency while at 4X slip frequency for 4-pole motors). Figure 6.09I shows spectra typical for a 2-pole motor having serious rotor

problems. Initially, the 30,000 CPM wideband spectrum gave little hint of a serious problem, seemingly showing mechanical looseness as evidenced by several running speed harmonics (Plot A of Figure 6.09I). However, upon closer examination, the zoom spectra of Plots B thru D of the figure revealed a series of well-formed pole pass sidebands around 1X, 2X and 3X RPM, respectively. These spectra all indicated cracked or broken rotor bars, shorting ring problems or shorted rotor laminations.

2. As suggested by Figure 6.09I, in addition to the pole pass frequency sidebands around 1X RPM, broken or cracked rotor bars and/or high resistance joints can produce pole pass sidebands around higher running speed harmonics up to and including the 2nd, 3rd, 4th and 5th running speed harmonics. In this case, more than 1 rotor bar is often found cracked or broken since there is more than 1 pulse event per revolution. In fact, when zoom spectra were made around 4X RPM up thru 8X RPM in this figure, they also showed multiple pole pass (F_p) sidebands indicating serious rotor problems. Subsequent checks found this motor to have 4 cracked rotor bars and scoring of each shorting ring (likely due to excessive heat).
3. The key area of concern for loose/open rotor bars is vibration at much higher frequencies at rotor bar pass frequency (RBPF) and also harmonics of this frequency:

$$\text{RBPF} = \# \text{ Rotor Bars} \times \text{RPM}$$

4. Here, the concern are amplitudes exceeding approximately .06 in/sec at either rotor bar pass frequency (RBPF) or approximately .03 in/sec at 2X RBPF assuming motors are under at least 60 to 70% load. In addition, the sideband spacing around RBPF and its harmonics will be at exactly twice line frequency ($2 F_L$). The reader is cautioned that while RBPF itself may be acceptable, if he extends the frequency range to encompass 2X RBPF, he may find amplitudes 10 or more times higher than those at RBPF (particularly if arching is occurring between rotor bars and end rings).

For example, Figure 6.109J shows a spectrum for a motor confirmed to have two or more open rotor bars. This motor had 57 rotor bars and operated at a speed of 1793 RPM, giving the fundamental RBPF at about 102,200 CPM. The wide band spectrum of Figure 6.09J showed an amplitude of only .008 in/sec at RBPF. However, the story out at 2X RBPF was completely different. The zoom spectrum of Figure 6.09K showed an excessive .340 in/sec at 204,380 CPM, or 2X RBPF (over 28 times higher amplitude than at 1X RBPF). Importantly, if a maximum frequency high enough only to capture the fundamental RBPF was taken, this problem would have been missed entirely. Again, the key indicators were the excessively high level at 2X RBPF, which was accompanied by sidebands at exactly 7200 CPM ($2F_L$).

5. Sometimes the amplitudes of RBPF or its harmonic will not be the highest amplitude. Instead, one of the sidebands spaced at difference frequencies of 2X line frequency (usually 7200 CPM with 60 Hz F_L) might be highest. This "array" of frequencies will still include RBPF and exactly $2F_L$ sidebands, and will still indicate loose or open rotor bars (assuming 60 Hz F_L) and/or variable air gap.
6. Even though the amplitude of concern in the area of 1X rotor bar pass frequency is about .06 in/sec for most motors, the important task will be first to detect these problems, and then to trend them before making any rash decisions about overhauling motors. If trending of several sets of spectra do not show any real increases, substantial damage is likely not continuing even with RBPF levels of approximately .10 to .15 in/sec. On the other hand, if the rate of change shows substantial increases from one survey to the next, it does indicate rapidly deteriorating condition which does require corrective action.

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7. In Condition Monitoring programs, it is a must to specify each of the following two special points on a route for each motor to detect electrical problems:

a. Low Frequency Motor Electrical Point:

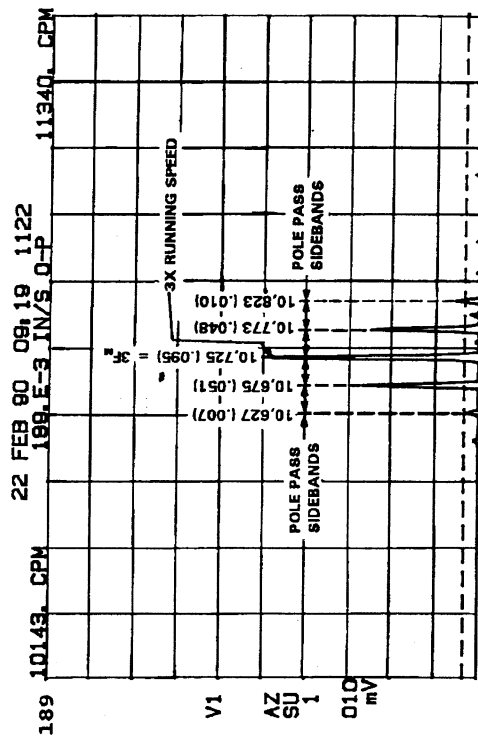
$F_{MAX} = 24,000$ CPM; 6400 FFT lines; 2 Ave and 50% Overlap Processing. This will allow one to separate the true amplitude at $2F_L$ and at motor running speed harmonics. It will also allow one to see if pole pass frequencies (F_p) surround 1X RPM or its harmonics, which could signal significant rotor bar problems (a 400 line spectrum, even with a frequency span of only 24,000 CPM will almost never even show these pole pass frequencies; instead it will simply show only one peak at these running speed harmonics instead of the 3 or more peaks which are truly present at what "appeared" to be only one peak at each running speed harmonic when only a 400 line spectrum was taken.

b. High Frequency Rotor Bar Pass Evaluation Point:

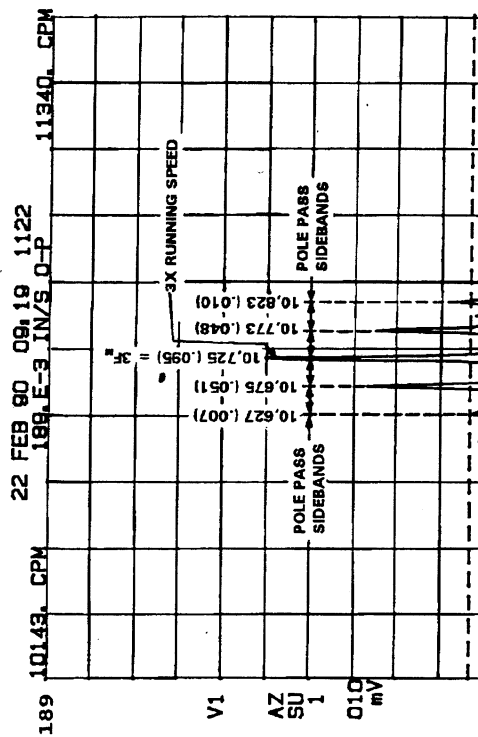
$F_{MAX} = 360,000$ CPM; 1600 FFT lines; 8 Ave. (for motors with more than two poles, an F_{MAX} of 240,000 CPM will likely be sufficient to pick up both 1X and 2X RBPF.) This will allow the detection of potential problems at RBPF and its multiples. Look for difference frequencies spaced at exactly $2F_L$ (usually 7200 CPM), even if the number of rotor bars is unknown. Figure 6.12L is a good example. Note the high amplitude peak at RBPF which penetrated both "Alarm 1" (.060 in/sec) and "Alarm 2" (.100 in/sec) narrowband envelopes. Also, notice the 7200 CPM ($2F_L$) sidebands surrounding RBPF. The reader is cautioned to carefully mount his transducer and to employ one with sufficient frequency response out to 360,000 CPM (6000 Hz) if he must make these higher frequency measurements on 2-pole motors.

6.095 Important Closing Comments on Electrical Measurements:

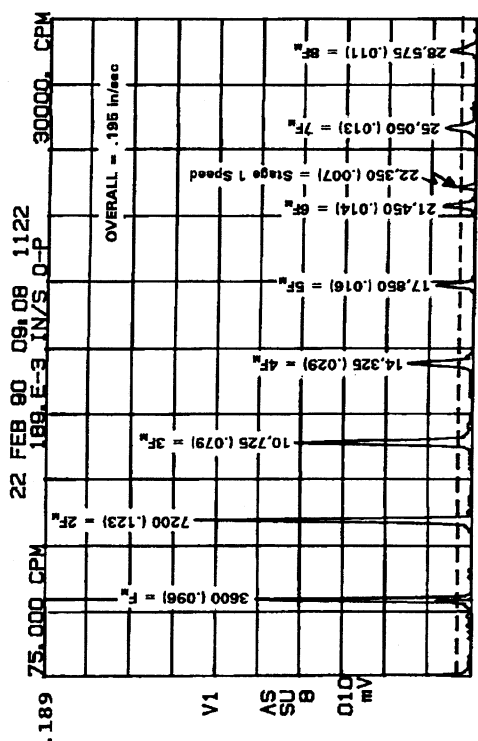
Experience has shown that these high resolution measurements are normally best taken in the horizontal direction. It is important that these measurements be taken with the motor under load (since many of the electrical problems will not even appear if operated unloaded). Remember that these two spectral measurements are in addition to the standard condition monitoring route points on each motor. Experience has shown that taking these two additional points on each motor during a route will require approximately 34 more seconds per motor. There is a cost associated with this additional survey time. However, if one detects only one motor with cracked or broken rotor bars, and prevents catastrophic failure, he can easily save many times this amount not only due to a much lower repair bill, but also due to probable far less downtime. Again experience has shown one very likely will detect such problems if (and possibly only if) he adds these high resolution points to his route. He cannot depend on "catching them with his usual 400 line spectra and THEN going out to perform them. The plain fact is HE WILL NOT EVEN LIKELY DETECT THESE PROBLEMS WITHOUT THEM FOR HE WILL NOT EVEN "SEE" THE POLE PASS FREQUENCY SIDE BANDS!



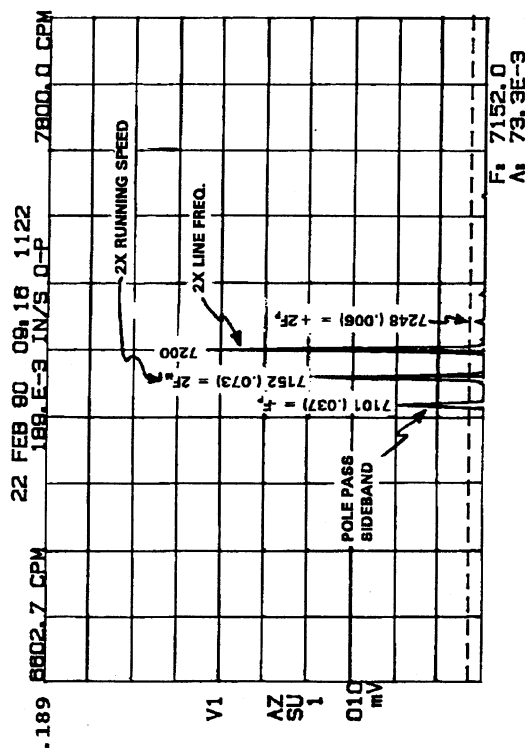
PLOT B
ZOOM SPECTRUM AROUND 1X RPM SHOWING MULTIPLE POLE PASS SIDE BANDS (BROKEN OR CRACKED ROTOR BARS/SHORTING RING PROBLEMS INDICATED)



PLOT D
ZOOM SPECTRUM AROUND 3X RPM
(NOTE POLE PASS SIDE BANDS HERE ALSO)



PLOT A
30,000 CPM SPECTRUM ON A CENTAC MOTOR (POS. 2H)



PLOT C
ZOOM SPECTRUM AROUND 2X RPM
(POLE PASS SIDE BANDS ALSO HERE)

FIGURE 6.091
WIDE BAND & ZOOM SPECTRA FOR A MOTOR HAVING CRACKED OR BROKEN ROTOR BARS OR SHORTING RING PROBLEMS

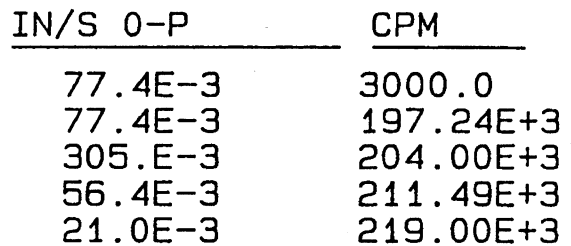
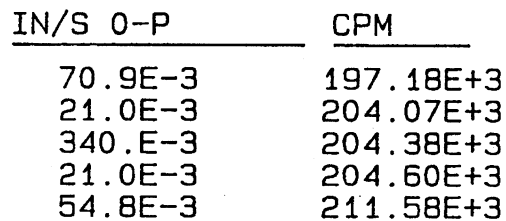
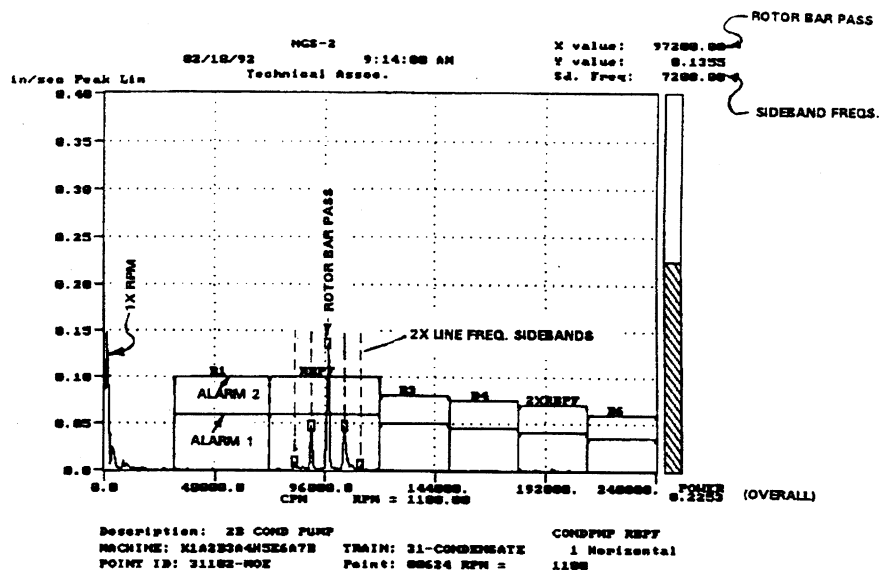


FIGURE 6.09J
SERIOUS ROTOR BAR PROBLEM DETECTED AT 2X RBPF, BUT
COMPLETELY MISSED BY 1X ROTOR BAR PASS FREQUENCY



NOTE: 6-10-90 G.E. Repair Shop phoned to confirm this motor had 2 or more open rotor bars which will likely result in motor replacement.

FIGURE 6.09K
SERIOUS ROTOR BAR PROBLEM DETECTED AT 2X ROTOR BAR
PASS FREQUENCY



02/18/92
9:14:00 AM

Threshold Report

PLANT: NGS-2
Description: 2B COND PUMP
MACHINE: K1A2B3A4H5E6A7B
POINT ID: 31102-MOE
Units: in/sec Peak Lin
Threshold: 0.0050

TRAIN: 31-CONDENSATE
Point: 00624
RPM: 1180.00

Frequency Hz	CPM	Order	Spect Amp	% Exceeded
18.95	1.14E+3	0.96	0.1481	2861.08
61.43	3.69E+3	3.12	0.0237	374.55
138.02	8281	7.02	0.0098	95.67
1383.46	8.30E+4 - 4F _L	70.35	0.0084	68.81
1503.55	90213 - 2F _L	76.45	0.0468	835.97
1623.67	97420 = RBPF	82.56	0.1355	2610.03
1743.41	1.05E+5 + 2F _L	88.65	0.0468	835.97
1863.95	1.12E+5 + 4F _L	94.78	0.0053	5.24
Power	Sync	Subsync	Non-Sync	
0.2253	0.2138	0.0	0.0710	

FIGURE 6.09L
TYPICAL SETUP OF 6 SPECTRAL ALARM BANDS FOR A 6-POLE
CONDENSATE PUMP MOTOR (1180 RPM NAMEPLATE SPEED)

6.10 BELT DRIVE PROBLEMS

There are a whole variety of belt-driven machines ranging from those with inherently high vibration such as reciprocating air compressors to those requiring very low levels such as machine tool spindles. However, if the proper precautions are taken, the great majority of such machines should have low vibration. Probably 3 of the greatest factors affecting how much vibration a belt-driven machine will experience are:

- Alignment of Sheaves
- Sheave Eccentricity
- Sheave Construction and Attachment Method

If each of these factors are carefully considered, there is no reason to have inherently high vibration in belt-driven machines.

Before discussing belt drive problems detectable by vibration analysis in particular, general statements concerning belt drives should be made:

1. It is best to take radial measurements in line with belt direction as shown in Figure 6.10A.

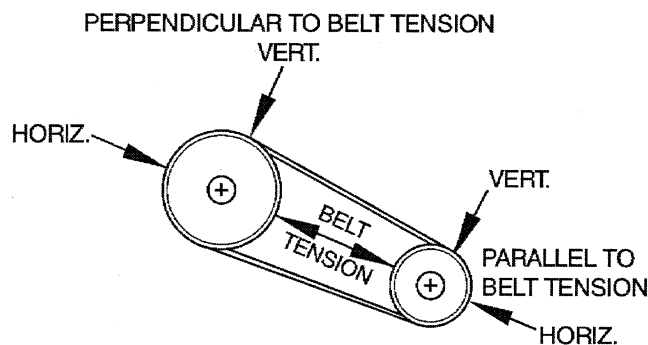


FIGURE 6.10A
PROPER MEASUREMENTS ON A BELT DRIVE

2. Adjustable V-belt sheaves create undue vibration and premature belt and sheave deterioration. These devices have inherent vibration problems since it is not possible to keep sheave faces parallel with one another which allows belts to ride up and down in the grooves with each revolution. As a result, this creates belt tension variation which generates high vibration and accelerates belt and sheave wear.
3. Another critical factor in belt drives is the amount of sheave eccentricity (i.e., runout). Unfortunately, sheaves which are purchased for general utility machinery almost always have inherently high eccentricity much greater than that of other components which are used in general rotating machinery. As a result, as soon as these are assembled to the machine, they themselves can generate high vibration and even cause noticeable variations in belt length and tension with each revolution. It is up to industry to demand tighter concentricity tolerance sheaves and to enforce these specifications right at machine acceptance.

-
4. V-belt drives are often blamed as the source of high vibration when, in fact, they are simply reacting to other problems such as unbalance, misalignment, mechanical looseness, etc. When these other problems are present, they can cause high vibration in the belts which themselves are not the source of the problem.

Following below are discussions on many of the belt drive problems which can be detected by vibration analysis:

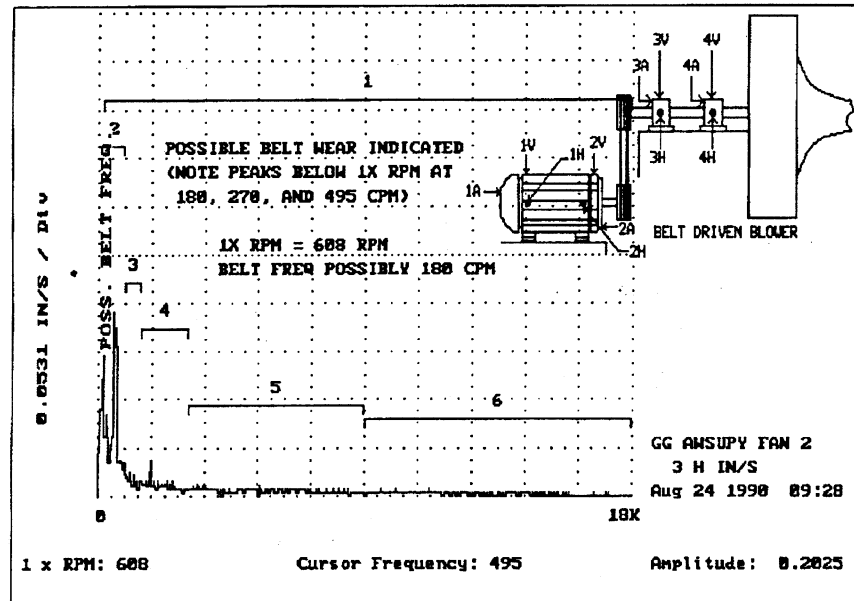
6.101 Worn, Loose or Mismatched Belts:

A typical spectrum indicating worn belts is given on page 4 of the **Illustrated Vibration Diagnostic Chart**. Note the belt frequency harmonics, all of which happen to be below both the driver and driven RPM in this example. The belt frequency (or belt RPM) is calculated as follows:

$$\text{Belt Freq.} = \frac{(3.142)(\text{Pulley RPM})(\text{Pulley Pitch Dia.})}{\text{Belt Length}}$$

When using the above equation, it is important to enter both the RPM and pitch diameter of the same pulley. It does not matter which pulley is used as long as the variables both come from the same pulley. Note that in all cases the belt RPM will be less than either the driver or driven RPM. However, belt frequency harmonics often will be higher than one or both of these. Worn, loose or mismatched belts display the following characteristics:

1. Worn belt defects detectable by vibration analysis include cracks, broken-off pieces of belt, hard and soft spots, lumps on belt faces, and also a crooked belt which has taken a set deformed shape during packing and storage.
2. When the problem is a worn belt(s), they will normally generate 3 to 4 multiples of belt frequency. Often, the 2X belt frequency peak may be dominant; in other cases, the fundamental belt frequency peak itself may not even show. In addition, worn belts sometimes lift the baseline of the spectrum throughout the subsynchronous frequency region, and just beyond driver and driven speeds as demonstrated in Figure 6.10B. In each case, worn belts normally cause unsteady amplitudes that sometimes pulsate with either the driver or driven RPM if any of these harmonics are close to either the driver or driven speed.
3. Belt defects usually show higher amplitudes in the direction parallel to belt tension. To get an idea how much vibration is caused by the belt defects themselves, compare amplitudes for the belt RPM frequencies themselves in a direction parallel with belt tension versus that in a direction perpendicular to belt tension.
4. Other belt specific problems which show up at belt RPM harmonics include belt width variations which cause the belts to ride up and down pulley grooves, creating vibration due to belt tension variations.
5. A loose cog belt is indicated by high vibration at the #cogs X RPM, and/or high vibration at the cog belt frequency itself which is calculated using the above equation.
6. Multiple V-belt drives can generate high vibration in the axial direction if they are unequally tensioned. This can result in excessive thrust bearing wear. These problems are sometimes solved by replacing several individual belts with one multi-belt which has been molded into one piece. If these are used, they place even greater importance on carefully aligning the sheaves.



2% x 0.500 = 0.0100 1 x RPM: 608.00

Machine			Date		
GG AWSUPY FAN 2			Aug 24 1990 09:28		
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
180	0.296	0.1550	2520	4.145	0.0175
270	0.444	0.0900	2610	4.293	0.0100
495	0.814	0.2025	3240	5.329	0.0100
585	0.962	0.1850 = 1x RPM	3375	5.551	0.0125
765	1.258	0.0400	3600	5.921	0.0100
900	1.480	0.0375	5490	9.030	0.0100
1080	1.776	0.0250			
1215	1.998	0.0225			
1305	2.146	0.0150			
1440	2.368	0.0175			
1620	2.664	0.0175			
1800	2.961	0.0400			
1890	3.109	0.0175			
2025	3.331	0.0150			
2160	3.553	0.0125			
2340	3.849	0.0175			
2430	3.997	0.0125			

FIGURE 6.10B

-
7. Worn, loose or mismatched belts normally generate highest vibration in the radial direction, particularly in line with belt tension.

6.102 Belt/Sheave Misalignment:

Probably one of the greatest sources of belt drive vibration is misalignment of driver and driven sheaves. It is often amazing to see how much vibration can be reduced simply by employing a chalk line between sheaves. Misaligned sheaves display the following characteristics:

1. Axial vibration is most always generated by sheave misalignment which can highly accelerate the rate of wear of thrust bearings.
2. Misaligned sheaves produce high vibration at 1X RPM, predominately in the axial direction. Dominant vibration is quite often at driver RPM, but occasionally at driven RPM. The ratio of amplitudes of driver to driven RPM depends exactly on where the data is taken as well as on relative mass and frame stiffness.
3. Often with sheave misalignment, the highest axial vibration on the motor will be at fan RPM, while the highest axial when measuring on the fan will be at motor RPM. However, this is not always the case.

6.103 Eccentric Sheaves:

Eccentric sheaves are one of the greatest contributors to high vibration in belt-driven machines today, often due to a lack of emphasis on specifying good concentricity in purchase specs. Sheave eccentricity displays the following characteristics:

1. Eccentric and/or unbalanced sheaves cause high vibration at 1X RPM of the eccentric sheave.
2. Highest amplitude is normally in line with the belts and should show up on both the driver and driven sheaves.
3. Unlike unbalance, reaction forces caused by an eccentric pulley are not equally applied throughout the entire 360° rotation of the pulley. The force instead is concentrated in the direction of belt tension along a line passing through the centerline of the 2 shafts. As a result, this highly directional vibration will show comparative horizontal and vertical phase readings which either are identical or 180° opposite one another, depending on which side of the bearing the transducer is located for the measurement. In any case, both phase readings show that the bearing is moving in one line.
4. Since the forces are so directional in nature, the resultant vibration cannot be totally corrected by balancing via attaching washers to taperlock bolts. Even if balanced, the sheave eccentricity will still induce vibration in the belt due to belt length and tension variations, and will result in premature accelerated wear of belts and/or sheaves along with the driver and driven bearings.

6.104 Belt Resonance:

Just like everything else in nature, belts themselves have a natural frequency which corresponds to the stiffness of the belt, the belt mass, and how much deflection is induced in the belt during operation. The natural frequency of a belt can be determined simply by pulling on the belt, releasing it and measuring the response. Belt resonance will display the following characteristics:

1. A spectrum indicating a belt resonance problem is shown on page 5 of the Table 6.0 **Illustrated Vibration Diagnostic Chart**. In this example, the belt natural frequency lies very close to 1X RPM of either the driver or driven machine. If this occurs, this will cause great flapping of the belt, particularly on the tension side at a frequency corresponding to the belt natural frequency.
2. Not only can a pulley speed excite a belt resonance, but also this can occur if a belt RPM harmonic should line up with its natural frequency.
3. Both the amplitude and phase of vibration at the belt resonant frequency will be unsteady.
4. The belt natural frequency itself can be changed either by altering the belt tension, belt length, sheave center distance, adding an idler pulley, etc..

6.105 Excessive Motor Vibration At Fan Speed Due to Motor Frame/Foundation Resonance:

Often, in the case of belt-driven machines, there will be excessive vibration on the motor. However, when a spectrum analysis is taken, it will show low vibration on the motor at motor speed, but high vibration on it at fan RPM. This can occur in either radial or axial directions. This will be evident by an excessive vibration at 1X RPM of the driven unit, particularly in one direction on the motor. Often, when this occurs, the problem source is excitation of a motor frame or foundation natural frequency by the incoming fan speed vibration. This can be confirmed by simple impulse natural frequency tests explained in Section 6.05. In this case, the solution will normally be stiffening of the frame or foundation by addition of bracing or addition of concrete on the base if it is resonant.

6.106 Loose Pulley or Fan Hub:

Excessive vibration sometimes occurs in belt-driven machines due to looseness either of a pulley or a fan hub itself. This is evidenced by each of the following characteristics:

1. There will be excessive vibration particularly at 1X RPM, but also at several running speeds harmonics.
2. One of the best indicators will be unstable phase. If operating properly, the phase in horizontal, vertical and axial directions should be steady. If a fan hub held on by set screws is loose on the shaft, it may show a difference in both phase and amplitude each time the unit is started up. In these cases, balancing would be only a temporary solution. It may help for a "couple of hours", but later when the fan hub or pulley rotates slightly on the shaft, it will upset the whole balance possibly dramatically changing the phase and amplitude. This same thing can occur if either the pulley or fan hub has a taper fit, and is not properly pulled up and fastened to the shaft.
3. The solution to each of these cases is ensuring that all rotors attached to the shaft are securely fastened, and properly oriented in the case of a taper fit.

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